

Workbook

Chapters 0–5

Chapters 0–2: minimal theory taken from the book's own definitions and strategies, followed immediately by the book's own exercises, every one of them worked in the text or solved in Appendix S. Chapters 3–5: theory, worked examples and stretch exercises for the same sections, to be rebuilt from the book once its pages are uploaded.

- **Source.** C. Newstead, *An Infinite Descent into Pure Mathematics*, Utrecht adaptation (version of 29 July 2026). In Chapters 0–2 all numbers (Definition 0.36, Strategy 1.1.28, Exercise 2.2.26, ...) are the book's numbers.
- **Built from the book:** Chapter 0 (pages 1–25, including 0.E), Chapter 1 (pages 29–75, including 1.E), Section 2.1 (pages 78–84) and Section 2.2 up to Definition 2.2.46 (pages 85–93), plus book page 176 in Chapter 5.
- **Not yet from the book:** the rest of §2.2 (tuples, Cartesian products), 2.E, and Chapters 3–5. The uploaded files stop at page 93. Those chapters are included in their earlier form (see the red notice at the start of Chapter 3) so that this is one document for the whole exam range; upload pages 94–180 and they will be rebuilt like Chapters 0–2.

How this workbook works

Every section is a sequence of short cycles: **one piece of theory, then the exercises that use it.**

Definition n.m

The book's definition, condensed. Learn it word for word: every proof starts by unfolding one of these.

Theorem / Proposition / Axiom

A result you may quote by number in a proof.

Strategy n.m

The book's proof strategies. Each one is a rule for what to write next.

Exercise n.m badge

A book exercise worked in full, in the style the book uses. Cover the solution and try first.

NOW YOU

The remaining book exercises for that piece of theory. Do them; every one is solved in Appendix S.

Watch out

The book's own warnings ("Common error", asides) and the places where marks are lost.

Example n Self Tutor

Chapters 3–5 only: a worked example at quiz level (not a book exercise).

Example n Book level

Chapters 3–5 only: a worked example at the level of the hardest planning exercises, with its invention step named.

STRETCH EXERCISES

Chapters 3–5 only: unsolved problems with hints; solutions in Appendix T. Appendix U is a one-page proof toolbox.

Flags from the course planning

★ = **essential** exercise in the planning ☆ = **recommended** ○ = in the book but not in the planning (the polynomial and complex-number part of Chapter 0, and the questions that use it, are outside the course; they are still solved here so that you can do *any* question in the book).

Quiz 1 (23 Sep): numbers and §1.1. Quiz 2 (7 Oct): sets and functions. Exam 4 Nov.

Conventions of this book

- $\mathbb{N} = \{0, 1, 2, \dots\}$: 0 is a natural number (Definition 0.6). $[n] = \{k \in \mathbb{N} \mid k < n\}$ (Definition 0.25), so $[0] = \emptyset$ and $[3] = \{0, 1, 2\}$.
- Truth values in truth tables are ✓ (true) and × (false), as in the book.
- “ b divides a ” means $a = qb$ for some $q \in \mathbb{Z}$ (Definition 0.36); it is about multiplication, never about the fraction a/b .
- “Odd” means “not even” (Definition 0.41); that a is odd iff $a = 2b + 1$ is a *proposition* (1.1.58), not the definition.
- □ ends a proof. {grey text in braces} names the definition, strategy or rule used for a step.

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Chapter 0

Getting started

Contents

- A Propositions, proofs, number sets
- B Sets, notation, intervals
- C Number bases
- D Division of integers
- E Irrational numbers
- F Polynomials and complex numbers ◦
- G Chapter 0 exercises (0.E)

Planning

Planning (Mon 7 Sep): Chapter 0 up to and including irrational numbers.

★ 0.2, 0.21, 0.24, 0.40; 0.E.3, 4, 22, 27, 28

★ 0.14, 0.30, 0.38, 0.45, 0.51; 0.E.17, 18, 29, 30

Section F and the 0.E questions on polynomials, complex and algebraic numbers are not in the planning.

A PROPOSITIONS, PROOFS, NUMBER SETS

Definitions 0.1 and 0.3 · proposition, proof

A **proposition** is a statement to which it makes sense to assign a truth value (true or false, not both). A **proof** of a true proposition is an argument that demonstrates its truth: it starts from what is known or assumed, uses agreed logical steps, and can be verified by its intended reader. Results are called propositions, theorems (key results), lemmas (stepping stones) or corollaries (immediate consequences).

NOW YOU 0.2 ★ · solutions in Appendix S

0.2 Give (a) a true proposition, (b) a false one, (c) one whose truth value you don't know.

Definitions 0.4–0.10 · the number line and the number types

Real numbers are the points of the number line (0.4). **Natural numbers** $0, 1, 2, \dots$ are reached from 0 by unit steps to the right (0.6); **integers** by unit steps in either direction (0.8).

Closure (Assumptions 0.5, 0.7, 0.9): $\mathbb{R}$ is closed under $+$, $-$, $\times$ and $\div$ by non-zero; $\mathbb{N}$ under $+$ and $\times$ only; $\mathbb{Z}$ under $+$, $-$, $\times$. Between an integer n and $n + 1$ there is no integer.

Floor and ceiling (0.10): for $x \in \mathbb{R}$, $\lfloor x \rfloor$ is the integer n with $n \leq x < n + 1$; $\lceil x \rceil$ is the integer n with $n - 1 < x \leq n$.

Proposition 0.12

Let n be an integer. Then $\lfloor n \rfloor = n = \lceil n \rceil$. *Proof idea:* n itself satisfies $n \leq n < n + 1$, and the floor is the (only) integer with that property.

The first proofs in the book all have the same shape: *name the integer that the definition asks for, then check the two inequalities.*

Exercise 0.13

from the book

Suppose that x is a real number that is not an integer. Prove that $\lceil x \rceil = \lfloor x \rfloor + 1$.

Let $n = \lfloor x \rfloor$. By Definition 0.10, n is an integer with $n \leq x < n + 1$. Since x is not an integer, $x \neq n$, so in fact $n < x < n + 1$.

Now $n + 1$ is an integer, and $(n + 1) - 1 < x \leq n + 1$. By Definition 0.10, $\lceil x \rceil$ is the integer m with $m - 1 < x \leq m$; so $\lceil x \rceil = n + 1 = \lfloor x \rfloor + 1$. $\square$

Exercise 0.14

recommended

Suppose that x is a real number. Prove that $\lfloor -x \rfloor = -\lceil x \rceil$ and $\lceil -x \rceil = -\lfloor x \rfloor$.

Let $n = \lceil x \rceil$, so $n \in \mathbb{Z}$ and $n - 1 < x \leq n$ (Definition 0.10). Multiplying through by -1 reverses the inequalities: $-n \leq -x < -n + 1 = (-n) + 1$. So $-n$ is an integer m with $m \leq -x < m + 1$, which is exactly the defining property of $\lfloor -x \rfloor$. Hence $\lfloor -x \rfloor = -n = -\lceil x \rceil$.

Similarly let $n = \lfloor x \rfloor$, so $n \leq x < n + 1$. Then $-n - 1 < -x \leq -n$, i.e. $(-n) - 1 < -x \leq -n$, so $\lceil -x \rceil = -n = -\lfloor x \rfloor$. $\square$

Definition 0.15 · rational number

$x \in \mathbb{R}$ is **rational** if there are integers a, b with $b \neq 0$ and $x = \frac{a}{b}$.

Theorem 0.16 · closure of $\mathbb{Q}$

If $x, y \in \mathbb{Q}$ then $x + y, x - y, xy \in \mathbb{Q}$, and $x/y \in \mathbb{Q}$ if $y \neq 0$.

Proof pattern (the book proves $x + y$): write $x = \frac{a}{b}, y = \frac{c}{d}$ with $b, d \neq 0$; compute; check that numerator and denominator are integers and the denominator is non-zero.

Exercise 0.17

from the book

Complete the proof of Theorem 0.16.

Let $x = \frac{a}{b}$ and $y = \frac{c}{d}$ with $a, b, c, d \in \mathbb{Z}$ and $b, d \neq 0$ (Definition 0.15).

$x - y = \frac{ad - bc}{bd}$: here $ad - bc \in \mathbb{Z}$ and $bd \in \mathbb{Z}$ by closure of $\mathbb{Z}$ (Assumption 0.9), and $bd \neq 0$ since $b \neq 0$ and $d \neq 0$. So $x - y \in \mathbb{Q}$.

$xy = \frac{ac}{bd}$: $ac, bd \in \mathbb{Z}$ and $bd \neq 0$. So $xy \in \mathbb{Q}$.

If $y \neq 0$ then $c \neq 0$ (otherwise $y = 0$), and $\frac{x}{y} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}$ with $ad, bc \in \mathbb{Z}$ and $bc \neq 0$. So $x/y \in \mathbb{Q}$. $\square$

B SETS, NOTATION, INTERVALS**Definitions 0.18, 0.19 and the two notations**

A **set** is a collection of objects, its **elements**; $x \in X$ asserts that x is an element of X (revised in §2.1). The **number sets** are $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ (and $\mathbb{C}$, 0.62).

List notation $\{0, 1, 2, 3\}$, implied lists $\{0, 1, 4, 9, \dots\}$.

Set-builder notation. $a \in \{x \mid p(x)\}$ iff $p(a)$ is true. Two variants:

- $\{x \in X \mid p(x)\} = \{x \mid x \in X \text{ and } p(x)\}$: the elements of X satisfying p ;

- $\{\text{expr}(x) \mid p(x)\}$: all objects of the form $\text{expr}(x)$ for some x satisfying p . E.g. $\{n^2 \mid n \in \mathbb{N}\} = \{0, 1, 4, 9, \dots\}$.

Exercise 0.21**essential**

Determine the truth values: (a) $2 \in \mathbb{N}$; (b) $-2 \in \mathbb{N}$; (c) $-2 \in \mathbb{Q}$; (d) $\frac{2}{5} \in \mathbb{Z}$; (e) $\frac{\sqrt{8} + \sqrt{2}}{\sqrt{8} - \sqrt{2}} \in \mathbb{Q}$; (f) $\pi^\pi \in \mathbb{Q}$; (g) $\sqrt{-7} \in \mathbb{R}$; (h) $\mathbb{N} \in \mathbb{Z}$.

(a) True: 2 is a non-negative whole number. (b) False: $-2 < 0$. (c) True: $-2 = \frac{-2}{1}$ with $-2, 1 \in \mathbb{Z}$, $1 \neq 0$. (d) False: $0 < \frac{2}{5} < 1$ and there is no integer strictly between 0 and 1. (e) True: $\sqrt{8} = 2\sqrt{2}$, so the number is $\frac{3\sqrt{2}}{\sqrt{2}} = 3 = \frac{3}{1} \in \mathbb{Q}$. (f) This is a proposition, but its truth value is not known: nobody has proved π^π rational or irrational. (g) False: $\sqrt{-7} = \sqrt{7}i$ is not a point of the real number line (0.62). (h) False: the elements of $\mathbb{Z}$ are numbers, and $\mathbb{N}$ is a set, not a number. $\square$

Exercise 0.24**essential**

A dyadic rational is a rational number that can be expressed as an integer divided by a power of 2. Express the set of all dyadic rationals in set-builder notation in at least two ways.

Filter form: $\{x \in \mathbb{Q} \mid \exists a \in \mathbb{Z}, \exists n \in \mathbb{N}, x = \frac{a}{2^n}\}$. Expression form: $\{\frac{a}{2^n} \mid a \in \mathbb{Z}, n \in \mathbb{N}\}$.

A third: $\{x \in \mathbb{R} \mid 2^n x \in \mathbb{Z} \text{ for some } n \in \mathbb{N}\}$. $\square$

Definitions 0.25 and 0.28 · $[n]$ and intervals

For $n \in \mathbb{N}$, $[n] = \{k \in \mathbb{N} \mid k < n\}$, so $[0] = \emptyset$, $[1] = \{0\}$, $[3] = \{0, 1, 2\}$: $[n]$ has n elements.

For $a < b$: $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$, $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$, $[a, b)$, $(a, b]$ likewise; $(-\infty, b) = \{x \in \mathbb{R} \mid x < b\}$, $[a, \infty) = \{x \in \mathbb{R} \mid x \geq a\}$, etc.; $(-\infty, \infty) = \mathbb{R}$. Intervals consist of *real* numbers: $\{x \in \mathbb{Z} \mid 2 \leq x \leq 5\}$ is not $[2, 5]$.

Exercise 0.27**from the book**

Let $A = \{x \in [10] \mid x \text{ is a power of } 2\}$, $B = \{2x \mid x \in [10]\}$ and $C = \{-99, -97, \dots, 97, 99\}$. (a) List A . (b) List B . (c) Write C in set-builder notation using some $[n]$.

(a) $[10] = \{0, 1, \dots, 9\}$; the powers of 2 in it are 1, 2, 4, 8. So $A = \{1, 2, 4, 8\}$.

(b) $B = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18\}$.

(c) The elements are $2k - 99$ for $k = 0, 1, \dots, 99$: $C = \{2k - 99 \mid k \in [100]\}$. $\square$

Exercise 0.30**recommended**

Express each illustrated set in the most appropriate notation. (The figures show, on a number line: (a) filled dot at -2 , hollow at 4 , segment between; (b) filled dots at -2 and 4 , segment between; (c) filled dots at $-2, -1, 0, 1, 2, 3, 4$ only; (d) hollow dot at 4 , shaded to the left without end; (e) filled dot at -2 , shaded to the right without end; (f) the whole line shaded.)

(a) $[-2, 4)$. (b) $[-2, 4]$. (c) $\{-2, -1, 0, 1, 2, 3, 4\} = \{n \in \mathbb{Z} \mid -2 \leq n \leq 4\}$ (not an interval: only integers). (d) $(-\infty, 4)$. (e) $[-2, \infty)$. (f) $(-\infty, \infty) = \mathbb{R}$. $\square$

C NUMBER BASES

Definition 0.31, Notation 0.34 · base- b expansion

Let $b > 1$. The **base- b expansion** of $n \in \mathbb{N}$ is the string $d_r d_{r-1} \dots d_0$ with (i) $n = d_r b^r + \dots + d_1 b + d_0$; (ii) $0 \leq d_i < b$; (iii) $d_r \neq 0$ if $n > 0$ (no leading zeros; the expansion of 0 is 0). We write $d_r \dots d_0(b)$ for the number with that expansion; no subscript means base 10. Digits beyond 9: $A = 10, B = 11, \dots$

The algorithm (page 17): repeated division

Divide n by b : the remainder is d_0 , the quotient is n_0 . Divide n_0 by b : remainder d_1 , quotient n_1 . Repeat until the quotient is 0. The expansion is $d_i d_{i-1} \dots d_0$ (last remainder first). Check by evaluating $\sum d_k b^k$.

Exercise 0.33

from the book

Find the binary, ternary, octal, decimal, hexadecimal and base-36 expansions of 21127.

Repeated division (for base 16: $21127 = 1320 \cdot 16 + 7$, $1320 = 82 \cdot 16 + 8$, $82 = 5 \cdot 16 + 2$, $5 = 0 \cdot 16 + 5$, read upwards: 5287). The others the same way:

$$21127 = 101001010000111_{(2)} = 1001222111_{(3)} = 51207_{(8)} = 21127_{(10)} = 5287_{(16)} = GAV_{(36)}.$$

Check of the last one: $G = 16, A = 10, V = 31$: $16 \cdot 36^2 + 10 \cdot 36 + 31 = 20736 + 360 + 31 = 21127$. $\square$

NOW YOU 0.47 · solutions in Appendix S

0.47 Find the base-17 expansion of 408 735 787 and the base-36 expansion of 1 442 151 747. (Both answers spell something.)

D DIVISION OF INTEGERS

Definition 0.36 · divides

Let $a, b \in \mathbb{Z}$. b **divides** a if $a = qb$ for some integer q . Also said: a is divisible by b , b is a divisor (factor) of a , a is a multiple of b . **No division is involved**: it is a statement about multiplication, so 0 divides 0 ($0 = q \cdot 0$).

How divisibility is used in proofs

To prove “ b divides a ”: write a as (an explicit integer) $\cdot b$ and say why the integer is an integer. **To use** “ b divides a ”: introduce a new letter, “ $a = qb$ for some $q \in \mathbb{Z}$ ”, and compute with it. (This is Proposition 0.39’s proof, which the book dissects in Example 1.1.1.)

Proposition 0.39

Let $a, b, c \in \mathbb{Z}$. If c divides b and b divides a , then c divides a . *Proof*: $b = qc$, $a = rb$ for some $q, r \in \mathbb{Z}$; then $a = r(qc) = (rq)c$ with $rq \in \mathbb{Z}$.

Exercise 0.38

recommended

Prove that 1 divides every integer, and that every integer divides 0.

Let $a \in \mathbb{Z}$. Then $a = a \cdot 1$ with $a \in \mathbb{Z}$, so 1 divides a by Definition 0.36. Let $b \in \mathbb{Z}$. Then $0 = 0 \cdot b$ with $0 \in \mathbb{Z}$, so b divides 0. $\square$

Exercise 0.40

essential

Let $a, b, d \in \mathbb{Z}$. Suppose that d divides a and d divides b . Given integers u and v , prove that d divides $au + bv$.

Since d divides a and d divides b , there are integers q, r with $a = qd$ and $b = rd$ (Definition 0.36). Then

$$au + bv = (qd)u + (rd)v = (qu + rv)d.$$

Since $qu + rv$ is an integer (closure of $\mathbb{Z}$), d divides $au + bv$ by Definition 0.36. $\square$

Definition 0.41 and Theorem 0.42 · even, odd, the division theorem

An integer n is **even** if it is divisible by 2; otherwise n is **odd**.

Division theorem. Let $a, b \in \mathbb{Z}$ with $b \neq 0$. There is exactly one way to write $a = qb + r$ with $q, r \in \mathbb{Z}$ and $0 \leq r < |b|$. q is the **quotient**, r the **remainder**. (“Exactly one” is the clause you use to prove that two expressions give the *same* remainder.)

Proposition 0.44

If a leaves remainder $r > 0$ when divided by b , then $-a$ leaves remainder $b - r$. *Proof:* $a = qb + r$ gives $-a = -(q+1)b + (b-r)$ with $0 < b-r < b$.

Exercise 0.45

recommended

Prove that if an integer a leaves a remainder of r when divided by an integer b , then a leaves a remainder of r when divided by $-b$.

Suppose $a = qb + r$ with $q \in \mathbb{Z}$ and $0 \leq r < |b|$. Then $a = (-q)(-b) + r$, where $-q \in \mathbb{Z}$ and $0 \leq r < |b| = |-b|$. By the uniqueness clause of Theorem 0.42, $-q$ is the quotient and r the remainder of a when divided by $-b$. $\square$

Watch out

- “ b divides a ” is not “ a/b is an integer”. Writing $\frac{a}{b} \in \mathbb{Z}$ in a proof about divisibility is marked wrong; produce the integer q .
- A remainder is never negative: $-7 = (-2) \cdot 4 + 1$, so -7 leaves remainder 1 on division by 4.
- Using the division theorem gives a *case split*: $n = 2k$ or $n = 2k + 1$; $n = 3k$, $3k + 1$ or $3k + 2$. Cover every case.

E IRRATIONAL NUMBERS

Definition 0.48, Assumption 0.49

An **irrational number** is a real number that is not rational. $\sqrt{2}$ is irrational (assumed here, proved in Proposition 4.3.12).

Proposition 0.50

There exist irrational numbers a and b such that ab is rational. *Proof:* $a = b = \sqrt{2}$ gives $ab = 2 = \frac{2}{1}$.

“It is possible that” means: *exhibit one example*. That is all Exercise 0.51 asks for.

Exercise 0.51**recommended**

Let r be a rational number and a an irrational number. Prove that it is possible that ra be rational, and possible that ra be irrational.

Take $r = 0$ and $a = \sqrt{2}$: then $ra = 0 = \frac{0}{1}$ is rational. Take $r = 1$ and $a = \sqrt{2}$: then $ra = \sqrt{2}$ is irrational (Assumption 0.49). Both possibilities occur. $\square$

F POLYNOMIALS AND COMPLEX NUMBERS

Not in the planning; included so that the 0.E questions that use it can be done.

Definitions 0.52–0.62

A **polynomial over S** (S one of $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$) is $a_0 + a_1x + \cdots + a_nx^n$ with all $a_k \in S$; its **degree** is the largest k with $a_k \neq 0$. Degrees $\leq 0, 1, 2, 3, 4, 5$: constant, linear, quadratic, cubic, quartic, quintic. A **root** is a number α with $p(\alpha) = 0$ (0.56). For $p(x) = ax^2 + bx + c$ the **discriminant** is $\Delta = b^2 - 4ac$ (0.59). The **imaginary unit** i satisfies $i^2 = -1$; for $a < 0$, $\sqrt{a} = \sqrt{|a|}i$; a **complex number** is $a + bi$ with $a, b \in \mathbb{R}$; $\mathbb{C}$ is the set of them (0.62). Compute with i as a variable, replacing i^2 by -1 ; $\frac{1}{a+bi} = \frac{a-bi}{a^2+b^2}$.

Theorem 0.60 · quadratic formula

Let $p(x) = ax^2 + bx + c$ with $a, b, c \in \mathbb{R}$, $a \neq 0$, and suppose $\Delta \geq 0$. Then the roots of $p(x)$ are precisely $\alpha = \frac{-b+\sqrt{\Delta}}{2a}$ and $\beta = \frac{-b-\sqrt{\Delta}}{2a}$.

Proof shape: complete the square, $p(x) = a(x + \frac{b}{2a})^2 - \frac{\Delta}{4a}$; check $p(\alpha) = p(\beta) = 0$; then show any root γ equals α or β (“precisely” means both directions).

Exercise 0.58**from the book**

Let $p(x) = ax + b$ be a linear polynomial over $\mathbb{Z}$. (a) Find a root. (b) Prove it is the only root. (c) When is the root an integer?

(a) Linear means degree 1, so $a \neq 0$. Put $\gamma = -\frac{b}{a}$; then $p(\gamma) = a \cdot (-\frac{b}{a}) + b = 0$.

(b) Let δ be any root: $a\delta + b = 0$. Then $a\delta = -b$ and, since $a \neq 0$, $\delta = -\frac{b}{a} = \gamma$.

(c) $-\frac{b}{a} \in \mathbb{Z}$ iff $b = qa$ for some $q \in \mathbb{Z}$ (take $q = -\frac{b}{a}$), i.e. iff a divides b (Definition 0.36). $\square$

Exercise 0.61

from the book

Let $p(x) = ax^2 + bx + c$ over $\mathbb{R}$ ($a \neq 0$) with discriminant Δ . Prove that $p(x)$ has exactly two real roots if $\Delta > 0$, exactly one if $\Delta = 0$, and none if $\Delta < 0$.

$\Delta > 0$. By Theorem 0.60 the roots are exactly α and β , and $\alpha - \beta = \frac{\sqrt{\Delta}}{a} \neq 0$, so they are two distinct real roots.

$\Delta = 0$. Then $\sqrt{\Delta} = 0$, so $\alpha = \beta = -\frac{b}{2a}$: by Theorem 0.60 this is a root and the only root.

$\Delta < 0$. Suppose $\gamma \in \mathbb{R}$ is a root. Completing the square, $0 = p(\gamma) = a(\gamma + \frac{b}{2a})^2 - \frac{\Delta}{4a}$, so $(\gamma + \frac{b}{2a})^2 = \frac{\Delta}{4a^2} < 0$. But the square of a real number is ≥ 0 : contradiction. So there is no real root. $\square$

NOW YOU 0.55, 0.64–0.68 · solutions in Appendix S

0.55 Find a quartic $s(w)$ over $\mathbb{R}$ that is not over $\mathbb{Q}$ but with $s(3) \in \mathbb{Q}$. **0.64** Express i^n in the form $a + bi$ for each $n \in \mathbb{Z}$. **0.65** Verify the conclusion of Theorem 0.60 for every discriminant. **0.66** Show $\alpha = a + bi$ is a root of a quadratic over $\mathbb{R}$ and find the other root. **0.67** State and prove the quadratic formula over $\mathbb{C}$. **0.68** Interpret subtraction and division geometrically on the number line.

G CHAPTER 0 EXERCISES (0.E)

Worked here: the essential and recommended ones. All others are in Appendix S. The True–False and Always–Sometimes–Never formats recur in every chapter: *true/always* needs a proof for all cases; *false/never* needs a counterexample (or a proof that it fails for all cases); *sometimes* needs one example and one counterexample.

Exercise 0.E.3

essential

Suppose m leaves remainder i and n leaves remainder j when divided by 3 ($0 \leq i, j < 3$). Prove that $m + n$ and $i + j$ leave the same remainder when divided by 3.

By Theorem 0.42, $m = 3q + i$ and $n = 3q' + j$ for some $q, q' \in \mathbb{Z}$. Divide $i + j$ by 3: $i + j = 3s + t$ with $s, t \in \mathbb{Z}$ and $0 \leq t < 3$, so the remainder of $i + j$ is t . Then

$$m + n = 3q + 3q' + i + j = 3(q + q' + s) + t, \quad 0 \leq t < 3,$$

and by the uniqueness clause of Theorem 0.42 the remainder of $m + n$ is t as well. $\square$

Exercise 0.E.4

essential

What are the possible remainders of n^2 when divided by 3, where $n \in \mathbb{Z}$?

By Theorem 0.42, $n = 3k, 3k + 1$ or $3k + 2$ for some $k \in \mathbb{Z}$. Then $n^2 = 3(3k^2)$, or $9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$, or $9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1$. So the remainder is 0 or 1 (and both occur: $n = 0, n = 1$); it is never 2. $\square$ (This is Proposition 1.1.24.)

Exercise 0.E.17 and 0.E.18

recommended

True or false? **0.E.17** Every integer is a natural number. **0.E.18** Every integer is a rational number.

0.E.17 False: $-1 \in \mathbb{Z}$ but $-1 \notin \mathbb{N}$ since $-1 < 0$ (Definition 0.6).

0.E.18 True: let $n \in \mathbb{Z}$; then $n = \frac{n}{1}$ with $n, 1 \in \mathbb{Z}$ and $1 \neq 0$, so $n \in \mathbb{Q}$ (Definition 0.15). $\square$

Exercise 0.E.22**essential**

True or false? The square root of every positive rational number is a rational number.

False. 2 is a positive rational number, and its square root $\sqrt{2}$ is irrational by Assumption 0.49. $\square$

Exercise 0.E.27 and 0.E.28**essential**

Let $a, b, c \in \mathbb{Z}$ with a dividing c and b dividing c . Always, sometimes or never: **0.E.27** ab divides c ; **0.E.28** ab divides c^2 .

0.E.27 Sometimes. $a = b = c = 1$: 1 divides 1. $a = b = c = 2$: 2 divides 2, but $ab = 4$ does not divide 2 (if $2 = 4q$ then $q = \frac{1}{2} \notin \mathbb{Z}$).

0.E.28 Always. Write $c = qa$ and $c = rb$ with $q, r \in \mathbb{Z}$. Then $c^2 = (qa)(rb) = (qr)(ab)$ with $qr \in \mathbb{Z}$, so ab divides c^2 . $\square$

Exercise 0.E.29 and 0.E.30**recommended**

0.E.29 Let $x, y \in \mathbb{Q}$ and $a, b, c, d \in \mathbb{Z}$ with $cy + d \neq 0$. Then $\frac{ax+b}{cy+d} \in \mathbb{Q}$: always, sometimes, never? **0.E.30** Let $\frac{a}{b}$ be a rational number. Then $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$: always, sometimes, never?

0.E.29 Always. Integers are rational (0.E.18), so $a, b, c, d \in \mathbb{Q}$; by Theorem 0.16, $ax + b \in \mathbb{Q}$ and $cy + d \in \mathbb{Q}$, and since $cy + d \neq 0$ the quotient is rational.

0.E.30 Sometimes. $\frac{1}{2}$: $a = 1, b = 2$ are integers. $\frac{\sqrt{2}}{\sqrt{2}} = 1$ is rational, but $a = \sqrt{2} \notin \mathbb{Z}$. $\square$

NOW YOU 0.E, the rest · solutions in Appendix S

○ 0.E.1 (base 64), 0.E.2, 0.E.6–0.E.11 (closure), 0.E.13–0.E.16 (algebraic numbers), 0.E.19–0.E.21, 0.E.23–0.E.26, 0.E.31–0.E.33.

Chapter 1

Logical structure

Contents

- A** Propositional logic (§1.1)
- B** Variables and quantifiers (§1.2)
- C** Logical equivalence (§1.3)
- D** Chapter 1 exercises (1.E)

Planning

★ 1.1: 21, 25, 32, 33, 38, 47, 56, 57 · 1.2: 6, 15, 18, 39 · 1.3: 5, 13, 14, 22, 27, 44ace · 1.E: 18
★ 1.1: 37, 43, 54, 55, 65, 66 · 1.2: 21, 22, 28, 37ab · 1.3: 32, 35 · 1.E: 11, 28, 29, 30
Bonus: 1.2.14. §1.1 is in **Quiz 1**.

The chapter's one idea: a proposition's *logical structure* (which operators and quantifiers it is built from) tells you what to write. Each operator has **introduction rules** (how to prove a goal of that shape) and **elimination rules** (how to use an assumption of that shape); each rule is a **Strategy**.

A PROPOSITIONAL LOGIC

Example 1.1.1 · assumptions and goals

At every point of a proof you have a list of **assumptions** (what is known or supposed) and **goals** (what remains to be shown). Every sentence you write changes the lists: “Suppose P ” adds P to the assumptions; unfolding a definition (“ $b = qc$ for some $q \in \mathbb{Z}$ ”) replaces an assumption by its definition and introduces a new variable; “it suffices to show ...” replaces the goal. Keep both lists in your head (or on scrap paper) at all times.

Definitions 1.1.2–1.1.3

A **propositional variable** (p, q, r, s) is a symbol representing a proposition. A **propositional formula** is a propositional variable, or is built from simpler formulae using **logical operators** ($\wedge, \vee, \Rightarrow, \neg, \Leftrightarrow$). Its truth value is determined by the truth values of its subformulae.

Conjunction $\wedge$ (“and”)

Definition 1.1.4 · rules for $\wedge$

($\wedge$ I) if p and q are true then $p \wedge q$ is true; ($\wedge$ E₁) if $p \wedge q$ is true then p is true; ($\wedge$ E₂) if $p \wedge q$ is true then q is true. Watch for hidden “and”s: “7 is a prime factor of 28” is “7 is prime” $\wedge$ “7 divides 28”.

Strategies 1.1.7 and 1.1.10

Proving $p \wedge q$: prove p , then prove q (two proofs tied together). **Assuming $p \wedge q$:** you may assume p and assume q separately.

Rules are written in natural-deduction style, premises above a line and conclusion below; stacking rules gives a **proof tree**, which is a plan for a proof (page 36).

NOW YOU 1.1.6, 1.1.13, 1.1.14 · solutions in Appendix S

1.1.6 Express “John is a mathematician who lives in Pittsburgh” as $p \wedge q$. **1.1.13** Given $(p \wedge q) \wedge r$, prove q using Strategy 1.1.10. **1.1.14** Proof tree for $(p \wedge q) \wedge (r \wedge s)$; apply it to “2 is an even prime number and 3 is an odd prime number”.

Disjunction $\vee$ (“or”)**Definition 1.1.15 · rules for $\vee$**

($\vee I_1$) if p is true then $p \vee q$ is true; ($\vee I_2$) if q is true then $p \vee q$ is true; ($\vee E$) if $p \vee q$ is true, and r can be derived from p and also from q , then r is true. (“Or” is inclusive.)

Strategies 1.1.16 and 1.1.19

Proving $p \vee q$: it suffices to prove just one of p or q . **Assuming $p \vee q$ (proof by cases):** to derive r , first derive r from the temporary assumption p , then derive r from the temporary assumption q . Write “Case 1 ... Case 2 ... In both cases, r .”

The division theorem produces disjunctions to split on: $n = 3k$ or $n = 3k + 1$ or $n = 3k + 2$ (Proposition 1.1.24: n^2 leaves remainder 0 or 1 on division by 3).

Exercise 1.1.21**essential**

Let p, q, r be propositions and suppose $q \vee r$ is true. Prove that $(p \vee q) \vee (p \vee r)$ is true, using Strategies 1.1.19 and 1.1.16.

We split into cases on the assumption $q \vee r$ (Strategy 1.1.19).

Case 1: q is true. Then $p \vee q$ is true by Strategy 1.1.16 (rule $\vee I_2$), and hence $(p \vee q) \vee (p \vee r)$ is true by Strategy 1.1.16 ($\vee I_1$).

Case 2: r is true. Then $p \vee r$ is true ($\vee I_2$), and hence $(p \vee q) \vee (p \vee r)$ is true ($\vee I_2$).

In both cases $(p \vee q) \vee (p \vee r)$ is true. □

Exercise 1.1.25**essential**

Let n be an integer. Prove that n^2 leaves a remainder of 0, 1 or 4 when divided by 5.

By the division theorem (Theorem 0.42), $n = 5k + r$ for some $k \in \mathbb{Z}$ and $r \in \{0, 1, 2, 3, 4\}$. Then

$$n^2 = 25k^2 + 10kr + r^2 = 5(5k^2 + 2kr) + r^2.$$

We split into cases on r . If $r = 0$: $n^2 = 5(5k^2)$, remainder 0. If $r = 1$: $n^2 = 5(5k^2 + 2k) + 1$, remainder 1. If $r = 2$: $n^2 = 5(5k^2 + 4k) + 4$, remainder 4. If $r = 3$: $r^2 = 9 = 5 + 4$, so $n^2 = 5(5k^2 + 6k + 1) + 4$, remainder 4. If $r = 4$: $r^2 = 16 = 15 + 1$, so $n^2 = 5(5k^2 + 8k + 3) + 1$, remainder 1.

In all cases the number in brackets is an integer and the remainder is 0, 1 or 4 (uniqueness in Theorem 0.42). □

NOW YOU 1.1.23, 1.1.26 · solutions in Appendix S

1.1.23 Let n be 8 or 9. Prove that n is even or a perfect square. **1.1.26** Let $a, b \in \mathbb{R}$ with $a^2 - 4b \neq 0$ and let α, β be the roots of $x^2 + ax + b$. Prove there is $c \in \mathbb{R}$ with $\alpha - \beta = c$ or $\alpha - \beta = ci$.

Implication $\Rightarrow$ (“if ... then”)

Definition 1.1.27 · rules for $\Rightarrow$

($\Rightarrow$ I) if q can be derived from the assumption that p is true, then $p \Rightarrow q$ is true; ($\Rightarrow$ E, *modus ponens*) if $p \Rightarrow q$ and p are true, then q is true.

Strategies 1.1.28 and 1.1.34

Proving $p \Rightarrow q$: assume p , derive q . First words: “Suppose p .” **Assuming $p \Rightarrow q$:** if you also have p , conclude q . Used to reduce a goal: if $p \Rightarrow q$ is known and p is easy, prove p instead of q (Example 1.1.36).

Proposition 1.1.30

Let $x, y \in \mathbb{R}$. If x and $x + y$ are rational, then y is rational. *Proof:* suppose $x = \frac{a}{b}$, $x + y = \frac{c}{d}$; then $y = (x + y) - x = \frac{bc - ad}{bd} \in \mathbb{Q}$.

Exercise 1.1.32

essential

Let r and s be propositions. Prove that $r \Rightarrow (s \Rightarrow s)$ is true, using the strategies for implications.

By Strategy 1.1.28 it suffices to assume r and derive $s \Rightarrow s$. So assume r . To prove $s \Rightarrow s$, again by Strategy 1.1.28 it suffices to assume s and derive s . So assume s . Then s is true, since it is assumed. Hence $s \Rightarrow s$ is true, and therefore $r \Rightarrow (s \Rightarrow s)$ is true. $\square$

Exercise 1.1.33

essential

Let a, b, c be propositions. Prove that $(a \vee b) \Rightarrow (c \Rightarrow ((c \wedge a) \vee b))$ is true.

Assume $a \vee b$ (Strategy 1.1.28). We must derive $c \Rightarrow ((c \wedge a) \vee b)$, so assume c as well. Now split into cases on $a \vee b$ (Strategy 1.1.19).

Case 1: a is true. Since c and a are both true, $c \wedge a$ is true (Strategy 1.1.7), and so $(c \wedge a) \vee b$ is true (Strategy 1.1.16).

Case 2: b is true. Then $(c \wedge a) \vee b$ is true (Strategy 1.1.16).

In both cases $(c \wedge a) \vee b$ holds. Hence $c \Rightarrow ((c \wedge a) \vee b)$, and hence the whole implication, is true. $\square$

Exercise 1.1.37

recommended

Let p, q, r be propositions. Prove that $(p \Rightarrow (q \Rightarrow r)) \Rightarrow (q \Rightarrow (p \Rightarrow r))$ is true.

Assume $p \Rightarrow (q \Rightarrow r)$. To prove $q \Rightarrow (p \Rightarrow r)$, assume q ; to prove $p \Rightarrow r$, assume p . By modus ponens (Strategy 1.1.34) applied to $p \Rightarrow (q \Rightarrow r)$ and p , we get $q \Rightarrow r$; by modus ponens again with q , we get r . Hence $p \Rightarrow r$, hence $q \Rightarrow (p \Rightarrow r)$, hence the whole formula. $\square$

Exercise 1.1.38

essential

Let p, q, r be propositions. Prove that $(p \Rightarrow p \wedge q) \Rightarrow ((p \vee q) \Rightarrow q)$ is true.

Assume $p \Rightarrow (p \wedge q)$. To prove $(p \vee q) \Rightarrow q$, assume $p \vee q$ and split into cases.

Case 1: p is true. By modus ponens, $p \wedge q$ is true, so q is true by Strategy 1.1.10 ($\wedge E_2$).

Case 2: q is true. Then q is true.

In both cases q holds, so $(p \vee q) \Rightarrow q$ is true, and hence the whole formula is true. $\square$

NOW YOU 1.1.31 · solutions in Appendix S

1.1.31 Let $p(x)$ be a polynomial over $\mathbb{C}$. Prove that if α is a root of $p(x)$ and $a \in \mathbb{C}$, then α is a root of $(x - a)p(x)$.

Converse and biconditional $\Leftrightarrow$ **Definitions 1.1.39 and 1.1.40**

The **converse** of $p \Rightarrow q$ is $q \Rightarrow p$. Ways of saying $p \Rightarrow q$: “if p then q ”, “ p only if q ”, “ p is sufficient for q ”. Ways of saying $q \Rightarrow p$: “ p if q ”, “ p is necessary for q ”.

$p \Leftrightarrow q$ means $(p \Rightarrow q) \wedge (q \Rightarrow p)$, read “ p if and only if q ”. So: **to prove $p \Leftrightarrow q$, prove both implications**, labelled $(\Rightarrow)$ and $(\Leftarrow)$; **to use $p \Leftrightarrow q$** , use either implication.

Examples 1.1.44 and 1.1.45 · solving an equation is a biconditional

Rearranging an equation only proves “if x is a solution then $x = \dots$ ” (it narrows the candidates). You must also check the converse by substituting back: for $x + \sqrt{x} = 0$ the rearrangement gives $x = 0$ or $x = 1$, but 1 is not a solution.

Example 1.1.46 · divisibility tests

To prove “ n is divisible by 8 iff its last three digits form a number divisible by 8”: write $n = n' + n''$ with n' the last three digits and $n'' = n - n' = 1000 \cdot (\dots)$, note 8 divides n'' , and use Exercise 0.40 in both directions ($n' = 1 \cdot n + (-1) \cdot n''$ and $n = 1 \cdot n' + 1 \cdot n''$).

Exercise 1.1.43**recommended**

Let p and q be propositions. Prove that $(p \Leftrightarrow q) \Leftrightarrow (q \Leftrightarrow p)$ is true.

By Definition 1.1.40 we prove both implications.

$(\Rightarrow)$ Assume $p \Leftrightarrow q$, i.e. $(p \Rightarrow q) \wedge (q \Rightarrow p)$. By Strategy 1.1.10 we have $q \Rightarrow p$ and $p \Rightarrow q$, so by Strategy 1.1.7, $(q \Rightarrow p) \wedge (p \Rightarrow q)$ is true, which is $q \Leftrightarrow p$.

$(\Leftarrow)$ Assume $q \Leftrightarrow p$, i.e. $(q \Rightarrow p) \wedge (p \Rightarrow q)$. The same two steps give $(p \Rightarrow q) \wedge (q \Rightarrow p)$, i.e. $p \Leftrightarrow q$. $\square$

Exercise 1.1.47**essential**

Prove that a natural number n is divisible by 3 if and only if the sum of its base-10 digits is divisible by 3.

Let $d_r \dots d_1 d_0$ be the base-10 expansion of n , so $n = \sum_{i=0}^r d_i 10^i$, and let $s = \sum_{i=0}^r d_i$ be the digit sum. Put $n'' = n - s = \sum_{i=0}^r d_i (10^i - 1)$.

Claim: 3 divides n'' . For each i , $10^i - 1 = (10 - 1)(10^{i-1} + 10^{i-2} + \dots + 1) = 9m_i$ with $m_i \in \mathbb{N}$ (for $i = 0$, $10^0 - 1 = 0 = 9 \cdot 0$), so $10^i - 1 = 3 \cdot (3m_i)$. Hence $n'' = 3 \sum_i d_i \cdot 3m_i$ is divisible by 3.

$(\Rightarrow)$ Suppose 3 divides n . Since 3 divides n'' , Exercise 0.40 gives that 3 divides $1 \cdot n + (-1) \cdot n'' = n - n'' = s$.

$(\Leftarrow)$ Suppose 3 divides s . Since 3 divides n'' , Exercise 0.40 gives that 3 divides $1 \cdot s + 1 \cdot n'' = n$. $\square$

Negation $\neg$ (“not”) and contradiction $\perp$ **Definitions 1.1.48 and 1.1.50**

A **contradiction** is a proposition known or assumed to be false; $\perp$ denotes an arbitrary one (e.g. $0 = 1$, $\sqrt{2} \in \mathbb{Q}$). Rules for $\neg$: ($\neg$ I) if a contradiction can be derived from the assumption that p is true, then $\neg p$ is true; ($\neg$ E) if $\neg p$ and p are both true, a contradiction may be derived.

Strategies 1.1.51 and 1.1.59 · proof by contradiction

Proving $\neg p$: assume p and derive a contradiction. First words: “Suppose p (towards a contradiction).”

Assuming $\neg p$: any derivation of p is a contradiction; this is how the contradiction is usually reached. Statements that are negations in disguise: “ x is irrational” ($= \neg(x \in \mathbb{Q})$), “ b does not divide a ”, “there is no ...”, “ X is empty”.

Propositions 1.1.53 and 1.1.58

1.1.53 If $r \in \mathbb{Q}$ and a is irrational, then $r + a$ is irrational. *Proof:* suppose $r + a \in \mathbb{Q}$; then $a = (r + a) - r \in \mathbb{Q}$ by Proposition 1.1.30, contradicting the assumption.

1.1.58 An integer a is odd iff $a = 2b + 1$ for some $b \in \mathbb{Z}$. *Proof:* $(\Rightarrow)$ by the division theorem $a = 2b$ or $a = 2b + 1$; the first contradicts “odd”. $(\Leftarrow)$ $a = 2b + 1$ leaves remainder 1, and even numbers leave remainder 0, so a is not even.

Exercise 1.1.54**recommended**

Let p and q be propositions. Prove that $(p \Rightarrow \neg q) \Rightarrow (q \Rightarrow \neg p)$ is true.

Assume $p \Rightarrow \neg q$. To prove $q \Rightarrow \neg p$, assume q . To prove $\neg p$, we use Strategy 1.1.51: assume p and derive a contradiction. By modus ponens from $p \Rightarrow \neg q$ and p we obtain $\neg q$. But q is also assumed, so by $(\neg E)$ we have a contradiction. Hence $\neg p$ is true, so $q \Rightarrow \neg p$ is true, so the whole formula is true. $\square$

Exercise 1.1.55**recommended**

Let p and q be propositions. Prove that $\neg p \Rightarrow (\neg q \Rightarrow \neg(p \vee q))$ is true.

Assume $\neg p$ and assume $\neg q$. To prove $\neg(p \vee q)$, assume $p \vee q$ and derive a contradiction, by cases. *Case 1:* p . This contradicts $\neg p$. *Case 2:* q . This contradicts $\neg q$. In both cases we have a contradiction, so $\neg(p \vee q)$ is true. Hence $\neg q \Rightarrow \neg(p \vee q)$, and hence the whole formula, is true. $\square$

Exercise 1.1.56**essential**

Let $x \in \mathbb{R}$. Prove by contradiction that if x is irrational then $-x$ and $\frac{1}{x}$ are irrational.

Suppose x is irrational. (Note $x \neq 0$, since $0 = \frac{0}{1}$ is rational.)

$-x$ is irrational. Suppose, towards a contradiction, that $-x$ is rational, say $-x = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b \neq 0$. Then $x = -\frac{a}{b}$ with $-a \in \mathbb{Z}$, so x is rational, contradicting the assumption. Hence $-x$ is irrational.

$\frac{1}{x}$ is irrational. Suppose $\frac{1}{x} = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b \neq 0$. Since $\frac{1}{x} \neq 0$ we have $a \neq 0$, and then $x = \frac{b}{a}$ with $b, a \in \mathbb{Z}$ and $a \neq 0$, so x is rational: contradiction. Hence $\frac{1}{x}$ is irrational. $\square$

Exercise 1.1.57**essential**

Prove by contradiction that there is no least positive real number: there is no positive real a such that $a \leq b$ for all positive real b .

Suppose, towards a contradiction, that a is a positive real number such that $a \leq b$ for every positive real number b . Then $b = \frac{a}{2}$ is a positive real number, so $a \leq \frac{a}{2}$. Multiplying by 2 gives $2a \leq a$, hence $a \leq 0$, contradicting $a > 0$. So no such a exists. $\square$

Two logical axioms

Axioms 1.1.60 and 1.1.67

Law of excluded middle (LEM): for every propositional formula p , the formula $p \vee (\neg p)$ is true.

Principle of explosion: from a contradiction, any consequence may be derived.

Strategy 1.1.61 · using LEM

To prove q , you may split into the cases “ p true” and “ p false” for any proposition p you like, and prove q in each case. (Proposition 1.1.63: if ab is even then a or b is even; split on “ a even”.)

Exercise 1.1.65**recommended**

Let p and q be propositions. Prove that $(\neg q \Rightarrow \neg p) \Rightarrow (p \Rightarrow q)$ is true.

Assume $\neg q \Rightarrow \neg p$, and assume p ; we must derive q . By the law of excluded middle, q or $\neg q$.

Case 1: q is true. Done.

Case 2: $\neg q$ is true. By modus ponens, $\neg p$ is true. Together with p this is a contradiction ($\neg E$), from which q follows by the principle of explosion.

In both cases q holds. Hence $p \Rightarrow q$, and hence the whole formula, is true. $\square$

Exercise 1.1.66**recommended**

Let a and b be irrational. By considering $\sqrt{2}^{\sqrt{2}}$, prove that it is possible that a^b be rational.

By the law of excluded middle, $\sqrt{2}^{\sqrt{2}}$ is rational or it is not.

Case 1: $\sqrt{2}^{\sqrt{2}}$ is rational. Take $a = b = \sqrt{2}$, both irrational (Assumption 0.49); then $a^b = \sqrt{2}^{\sqrt{2}}$ is rational.

Case 2: $\sqrt{2}^{\sqrt{2}}$ is irrational. Take $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$, both irrational. Then $a^b = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{\sqrt{2} \cdot \sqrt{2}} = \sqrt{2}^2 = 2$, which is rational.

In both cases we found irrational a, b with a^b rational. $\square$ (We never learn which case holds: a *non-constructive* proof, which is why LEM is called non-constructive.)

NOW YOU 1.1.64 · solutions in Appendix S

1.1.64 In the proof of Proposition 1.1.63, where is LEM used, where is contradiction used, and what is the contradiction? Prove the proposition again twice, once without LEM and once without contradiction.

Watch out

- Keep the direction straight: to prove $p \Rightarrow q$ assume p ; proving $q \Rightarrow p$ instead (the converse) proves a different proposition (1.E.12).
- When you use the law of excluded middle or proof by cases, the cases must be exhaustive and *every* case must reach the goal.
- In a proof by contradiction, say what the contradiction is: “this contradicts the assumption that x is irrational”.

B VARIABLES AND QUANTIFIERS

Definitions 1.2.1, 1.2.2, 1.2.4

In a statement, a variable x (understood to range over a set X) is **free** if it makes sense to substitute elements of X for it, and **bound** otherwise (“there exist integers a, b with $x = a^2 + b^2$ ”: x free, a, b bound). A **predicate** $p(x_1, \dots, x_n)$ is a symbol with a list of free variables, each with a **domain of discourse**; it becomes a proposition when values are substituted. A **logical formula** is built from predicates with logical operators and the **quantifiers** $\forall$ (“for all $x \in X$ ”), $\exists$ (“there exists $x \in X$ such that”), $\exists!$ (“there is exactly one”). Quantifiers go *in front*: “ $n = 2k$ for some $k \in \mathbb{Z}$ ” is written $\exists k \in \mathbb{Z}, n = 2k$.

Translating (Examples 1.2.5, 1.2.7)

English $\rightarrow$ symbols: introduce a variable for each noun phrase, unfold definitions (“ n is even” is $\exists k \in \mathbb{Z}, n = 2k$), and restrict with an implication inside $\forall$ (“every positive real has a square root”: $\forall a \in \mathbb{R}, (a > 0 \Rightarrow \exists b \in \mathbb{R}, b^2 = a)$). Symbols $\rightarrow$ English: read symbol by symbol, then remove bound variables by naming the concept they encode (“ $\exists b \in \mathbb{R}, a = b^2$ ” is “ a has a real square root”).

Exercise 1.2.6

essential

Find logical formulae representing: (a) there is an integer that is divisible by every integer; (b) there is no greatest odd integer; (c) between any two distinct rational numbers is a third distinct rational number; (d) if an integer has a rational square root, then that root is an integer.

- (a) $\exists n \in \mathbb{Z}, \forall m \in \mathbb{Z}, \exists q \in \mathbb{Z}, n = qm.$ { m divides n unfolded }
 (b) $\neg \exists n \in \mathbb{Z}, ([\exists k \in \mathbb{Z}, n = 2k + 1] \wedge \forall m \in \mathbb{Z}, ([\exists \ell \in \mathbb{Z}, m = 2\ell + 1] \Rightarrow m \leq n));$ equivalently $\forall n \in \mathbb{Z}, (n \text{ odd} \Rightarrow \exists m \in \mathbb{Z}, (m \text{ odd} \wedge m > n)).$
 (c) $\forall a \in \mathbb{Q}, \forall b \in \mathbb{Q}, (a < b \Rightarrow \exists c \in \mathbb{Q}, (a < c \wedge c < b)).$
 (d) $\forall n \in \mathbb{Z}, \forall x \in \mathbb{Q}, (x^2 = n \Rightarrow x \in \mathbb{Z}).$ □

NOW YOU 1.2.8 · solutions in Appendix S

1.2.8 Plain English, as few variables as possible: (a) $\exists q \in \mathbb{Z}, a = qb$; (b) $\exists a \in \mathbb{Z}, \exists b \in \mathbb{Z}, (b \neq 0 \wedge bx = a)$; (c) $\forall d \in \mathbb{N}, [(\exists q \in \mathbb{Z}, n = qd) \Rightarrow (d = 1 \vee d = n)];$ (d) $\forall a \in \mathbb{R}, [a > 0 \Rightarrow \exists b \in \mathbb{R}, (b > 0 \wedge a < b)].$

Universal quantifier $\forall$

Definition 1.2.9 · rules for $\forall$

($\forall$ I) if $p(x)$ can be derived from the assumption that x is an *arbitrary* element of X , then $\forall x \in X, p(x)$;
 ($\forall$ E) if $\forall x \in X, p(x)$ is true and $a \in X$, then $p(a)$ is true.

Strategies 1.2.10 and 1.2.19

Proving $\forall x \in X, p(x)$: “Let $x \in X$ ” (or “fix”, “take”), then prove $p(x)$ assuming nothing about x except $x \in X$. **Assuming** $\forall x \in X, p(x)$: you may use $p(a)$ for any particular $a \in X$ you need.

Common error (page 54)

“Let n be an arbitrary integer, say $n = 17$ ” proves $\exists n$, not $\forall n$. The *reader* chooses n , not you. A correct proof of $\forall n \in \mathbb{Z}, n^2 \geq 0$: let $n \in \mathbb{Z}$; either $n \geq 0$ or $n < 0$; in each case $n^2 \geq 0$.

Propositions 1.2.11 and 1.2.12

The square of every odd integer is odd ($n = 2k + 1 \Rightarrow n^2 = 2(2k^2 + 2k) + 1$). The last decimal digit of a square is 0, 1, 4, 5, 6 or 9 (write $n = 10m + d_0$; then $n^2 = 10m(10m + 2d_0) + d_0^2$ ends in the last digit of d_0^2).

Exercise 1.2.15**essential**

Prove that for every integer n , n is even if and only if n^2 is even.

Let $n \in \mathbb{Z}$. We prove both directions.

($\Rightarrow$) Suppose n is even, so $n = 2k$ for some $k \in \mathbb{Z}$. Then $n^2 = 4k^2 = 2(2k^2)$ with $2k^2 \in \mathbb{Z}$, so n^2 is even.

($\Leftarrow$) Suppose n^2 is even. By the division theorem, $n = 2k$ or $n = 2k + 1$ for some $k \in \mathbb{Z}$. If $n = 2k + 1$, then n^2 is odd by Proposition 1.2.11, contradicting that n^2 is even. So $n = 2k$, i.e. n is even. $\square$

Exercise 1.2.18**essential**

Prove that for all real numbers x and y , if x is irrational, then $x + y$ and $x - y$ are not both rational.

Let $x, y \in \mathbb{R}$ and suppose x is irrational. Suppose, towards a contradiction, that $x + y$ and $x - y$ are both rational. Then by Theorem 0.16 their sum $(x + y) + (x - y) = 2x$ is rational, and hence $x = \frac{2x}{2}$ is rational (Theorem 0.16 again, dividing by $2 \neq 0$). This contradicts the assumption that x is irrational. Hence $x + y$ and $x - y$ are not both rational. $\square$

Exercise 1.2.21 and 1.2.22**recommended**

1.2.21 Suppose $\forall x \in \mathbb{R}, P(x)$. Show that we can conclude $P(21)$. **1.2.22** Let $P(x), Q(x)$ be predicates with $x \in \mathbb{R}$. Prove that $(\forall x \in \mathbb{R}, P(x) \Rightarrow Q(x)) \Rightarrow (\forall x \in \mathbb{R}, P(x)) \Rightarrow (\forall x \in \mathbb{R}, Q(x))$.

1.2.21 Since $21 \in \mathbb{R}$ and $\forall x \in \mathbb{R}, P(x)$ holds, the rule $(\forall E)$ (Strategy 1.2.19) gives $P(21)$. $\square$

1.2.22 Assume $\forall x \in \mathbb{R}, (P(x) \Rightarrow Q(x))$ and assume $\forall x \in \mathbb{R}, P(x)$. To prove $\forall x \in \mathbb{R}, Q(x)$, let $x \in \mathbb{R}$ be arbitrary. By $(\forall E)$, $P(x) \Rightarrow Q(x)$ and $P(x)$ are both true, so $Q(x)$ is true by modus ponens. Since x was arbitrary, $\forall x \in \mathbb{R}, Q(x)$. $\square$

NOW YOU 1.2.14 (bonus), 1.2.16, 1.2.17 · solutions in Appendix S

1.2.14 Prove $\forall x \in \mathbb{R}, x = x$. **1.2.16** Prove that every integer is rational. **1.2.17** Prove that every linear polynomial over $\mathbb{Q}$ has a rational root.

Existential quantifier $\exists$ **Definition 1.2.23 · rules for $\exists$**

$(\exists I)$ if $a \in X$ and $p(a)$ is true, then $\exists x \in X, p(x)$; $(\exists E)$ if $\exists x \in X, p(x)$ is true, and q can be derived from the assumption that $p(a)$ holds for some fixed $a \in X$, then q is true.

Strategies 1.2.24 and 1.2.30

Proving $\exists x \in X, p(x)$: define a specific a (“Define $n = 9$ ”, “Take $z = \frac{x+y}{2}$ ”), check $a \in X$, and prove $p(a)$.

Assuming $\exists x \in X, p(x)$: “Take $a \in X$ such that $p(a)$ ” with a letter not yet in use. Divisibility assumptions are existential: from “3 divides n^3 ” take q with $n^3 = 3q$ (Proposition 1.2.31).

Exercise 1.2.28

recommended

Prove that there is an integer which is divisible by zero.

Define $n = 0$. Then $n \in \mathbb{Z}$, and $0 = 0 \cdot 0$ with $0 \in \mathbb{Z}$, so 0 divides n by Definition 0.36. Hence $\exists n \in \mathbb{Z}$ such that 0 divides n . $\square$

Exercise 1.2.29

from the book

Prove that for all $x, y \in \mathbb{Q}$, if $x < y$ then there is some $z \in \mathbb{Q}$ with $x < z < y$.

Let $x, y \in \mathbb{Q}$ and suppose $x < y$. Define $z = \frac{x+y}{2}$. Then $z \in \mathbb{Q}$ by Theorem 0.16 ($x+y \in \mathbb{Q}$ and $2 \neq 0$). Moreover $x < z$ because $2x < x+y$ (add x to $x < y$), and $z < y$ because $x+y < 2y$. So $x < z < y$, and z is the required element. $\square$

NOW YOU 1.2.27 · solutions in Appendix S

1.2.27 Prove that there is a real number that is irrational but whose square is rational.

Unique existence $\exists!$
Definition 1.2.32 and Proposition 1.2.38

$\exists! x \in X$, $p(x)$ is shorthand for

$$\underbrace{(\exists x \in X, p(x))}_{\text{existence}} \wedge \underbrace{(\forall a \in X, \forall b \in X, [(p(a) \wedge p(b)) \Rightarrow a = b])}_{\text{uniqueness}},$$

equivalently (1.2.38) $\exists x \in X, (p(x) \wedge \forall y \in X, (p(y) \Rightarrow y = x))$.

Strategy 1.2.35

Existence: exhibit a with $p(a)$. **Uniqueness:** “Let $a, b \in X$ with $p(a)$ and $p(b)$ ” ... “so $a = b$ ”. (Or: with the a from existence, show $\forall y, p(y) \Rightarrow y = a$.) Example 1.2.36: for $a > 0$ the positive square root is unique because $y^2 = z^2$ gives $(y-z)(y+z) = 0$ and $y+z > 0$.

Exercise 1.2.37 (a), (b)

recommended

Write as a formula with $\exists!$ and prove: (a) for each real a , the equation $x^2 + 2ax + a^2 = 0$ has exactly one real solution; (b) there is a unique real a for which $x^2 + a^2 = 0$ has a real solution.

(a) $\forall a \in \mathbb{R}, \exists! x \in \mathbb{R}, x^2 + 2ax + a^2 = 0$. Let $a \in \mathbb{R}$. *Existence:* $x = -a$ gives $a^2 - 2a^2 + a^2 = 0$. *Uniqueness:* let $x, y \in \mathbb{R}$ both be solutions. Since $x^2 + 2ax + a^2 = (x+a)^2$, we get $(x+a)^2 = 0$, so $x+a = 0$, so $x = -a$; likewise $y = -a$; hence $x = y$. $\square$

(b) $\exists! a \in \mathbb{R}, \exists x \in \mathbb{R}, x^2 + a^2 = 0$. *Existence:* $a = 0$ works, with $x = 0$. *Uniqueness:* let $a, b \in \mathbb{R}$ both have the property; take $x \in \mathbb{R}$ with $x^2 + a^2 = 0$. Since $x^2 \geq 0$ and $a^2 \geq 0$, the sum is 0 only if $a^2 = 0$, so $a = 0$. The same argument gives $b = 0$, so $a = b$. $\square$

NOW YOU 1.2.37 (c), 1.2.34 · solutions in Appendix S

1.2.37 (c) There is a unique natural number with exactly one positive divisor. **1.2.34 (Discussion)** Why does Definition 1.2.32 capture “exactly one”? Other possible definitions?

Quantifier alternation**Theorem 1.2.40**

$\exists y \in Y, \forall x \in X, p(x, y) \Rightarrow \forall x \in X, \exists y \in Y, p(x, y)$. *Proof:* fix $a \in X$; choose b with $\forall x, p(x, b)$; then $p(a, b)$.

The converse fails in general: in $\forall x \exists y$ the y may depend on x (“every door has a key”); in $\exists y \forall x$ one y works for all x (“a master key”).

Exercise 1.2.39**essential**

Let $p(x, y)$ be “ $x + y$ is even”. Prove that $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, p(x, y)$ is true, and that $\exists y \in \mathbb{Z}, \forall x \in \mathbb{Z}, p(x, y)$ is false.

First statement. Let $x \in \mathbb{Z}$. Define $y = x$. Then $y \in \mathbb{Z}$ and $x + y = 2x$ is even (it is $2 \cdot x$ with $x \in \mathbb{Z}$). So $\exists y \in \mathbb{Z}, p(x, y)$, and as x was arbitrary the statement holds.

Second statement. Suppose, towards a contradiction, that there is $y \in \mathbb{Z}$ with $x + y$ even for every $x \in \mathbb{Z}$. Take $x = 1 - y \in \mathbb{Z}$. Then $x + y = 1$, so 1 is even, i.e. $1 = 2k$ for some $k \in \mathbb{Z}$; but then $k = \frac{1}{2} \notin \mathbb{Z}$, a contradiction. Hence the statement is false. $\square$

C LOGICAL EQUIVALENCE**Definition 1.3.2 · logically equivalent**

Logical formulae p and q are **logically equivalent**, $p \equiv q$, if q can be derived from p and p can be derived from q . Equivalent formulae mean the same thing but suggest different proof strategies (Example 1.3.1: “every prime > 2 is odd” versus “there is no even prime > 2 ”).

Proving $p \equiv q$ from the definition (Examples 1.3.3, 1.3.4)

Two derivations: “First assume $p \dots$ hence q .” then “Now assume $q \dots$ hence p .” Inside each, use the strategies of §1.1 (cases for $\vee$, assume-and-derive for $\Rightarrow$, LEM and explosion where needed).

Exercise 1.3.5**essential**

Let p, q, r be propositional variables. Prove that $(p \vee q) \Rightarrow r$ is logically equivalent to $(p \Rightarrow r) \wedge (q \Rightarrow r)$.

First assume $(p \vee q) \Rightarrow r$. To prove $p \Rightarrow r$: assume p ; then $p \vee q$ by $(\vee I_1)$, so r by modus ponens. To prove $q \Rightarrow r$: assume q ; then $p \vee q$ by $(\vee I_2)$, so r by modus ponens. Hence $(p \Rightarrow r) \wedge (q \Rightarrow r)$ by $(\wedge I)$. Now assume $(p \Rightarrow r) \wedge (q \Rightarrow r)$, so $p \Rightarrow r$ and $q \Rightarrow r$ by $(\wedge E)$. To prove $(p \vee q) \Rightarrow r$, assume $p \vee q$ and split into cases: if p , then r by modus ponens; if q , then r by modus ponens. In both cases r , so $(p \vee q) \Rightarrow r$. Each formula can be derived from the other, so they are logically equivalent. $\square$

Truth tables

Theorem 1.3.6 and Definition 1.3.7

Two *propositional* formulae are logically equivalent iff they have the same truth value under every assignment of truth values to their variables. The **truth table** of a formula has one row per assignment (2^n rows for n variables) and one column per subformula. Basic tables: $\neg p$ is $\times$ iff p is $\checkmark$; $p \wedge q$ is $\checkmark$ only in row $\checkmark\checkmark$; $p \vee q$ is $\times$ only in row $\times\times$; $p \Rightarrow q$ is $\times$ only in row $\checkmark\times$.

Strategy 1.3.11

To prove propositional formulae are equivalent, show that their columns in a (combined) truth table are identical. Layout: variables first (p : four $\checkmark$ then four $\times$; q : two and two; r : alternating), then subformulae, then the main formulae.

Exercise 1.3.13**essential**

Use a truth table to prove that $p \Rightarrow q \equiv (\neg p) \vee q$.

p	q	$p \Rightarrow q$	$\neg p$	$(\neg p) \vee q$
$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
$\checkmark$	$\times$	$\times$	$\times$	$\times$
$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
$\times$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$

The columns for $p \Rightarrow q$ and $(\neg p) \vee q$ are identical, so by Strategy 1.3.11 the formulae are logically equivalent. $\square$

Exercise 1.3.14**essential**

Use a truth table to prove that $(p \vee q) \Rightarrow r \equiv (p \Rightarrow r) \wedge (q \Rightarrow r)$.

p	q	r	$p \vee q$	$(p \vee q) \Rightarrow r$	$p \Rightarrow r$	$q \Rightarrow r$	$(p \Rightarrow r) \wedge (q \Rightarrow r)$
$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
$\checkmark$	$\checkmark$	$\times$	$\checkmark$	$\times$	$\times$	$\times$	$\times$
$\checkmark$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
$\checkmark$	$\times$	$\times$	$\checkmark$	$\times$	$\times$	$\checkmark$	$\times$
$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
$\times$	$\checkmark$	$\times$	$\checkmark$	$\times$	$\checkmark$	$\times$	$\times$
$\times$	$\times$	$\checkmark$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
$\times$	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

The fifth and eighth columns agree in every row, so the formulae are logically equivalent. $\square$

NOW YOU 1.3.9 · solutions in Appendix S

1.3.9 Use the definitions of $\wedge$, $\vee$, $\Rightarrow$ to justify the basic truth tables.

Strategies from equivalences

Theorem 1.3.15 and Strategy 1.3.16 · double negation, indirect contradiction

$p \equiv \neg\neg p$ (two-row truth table). Hence **to prove p is true, you may assume p is false and derive a contradiction**. Compare Strategy 1.1.51 (to prove $\neg p$, assume p): that one is direct, this one is indirect. Example 1.3.17: $a, b, c \geq 0$ with $a^2 + b^2 = c^2$ implies $a + b \geq c$; assume $a + b < c$ and square.

Definition 1.3.18, Theorem 1.3.19, Strategy 1.3.20 · contraposition

The **contrapositive** of $p \Rightarrow q$ is $\neg q \Rightarrow \neg p$, and $p \Rightarrow q \equiv (\neg q) \Rightarrow (\neg p)$ (truth table). **Proof by contraposition:** to prove $p \Rightarrow q$, assume $\neg q$ and derive $\neg p$. Example 1.3.21: $mn > 64 \Rightarrow (m > 8 \vee n > 8)$; assume $m \leq 8$ and $n \leq 8$, get $mn \leq 64$. Announce it: “By contraposition, it suffices to ...”.

Exercise 1.3.22**essential**

Use the law of contraposition to prove that $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q))$, and use the technique this suggests to prove that an integer is even if and only if its square is even.

By Definition 1.1.40, $p \Leftrightarrow q$ is $(p \Rightarrow q) \wedge (q \Rightarrow p)$. By the law of contraposition (Theorem 1.3.19 applied to $q \Rightarrow p$), $q \Rightarrow p \equiv (\neg p) \Rightarrow (\neg q)$. Replacing the second conjunct by this equivalent formula gives $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q))$.

Technique: to prove “ p iff q ”, prove $p \Rightarrow q$ and then prove $\neg p \Rightarrow \neg q$.

Application. Let $n \in \mathbb{Z}$. ($p \Rightarrow q$) If n is even, $n = 2k$, then $n^2 = 2(2k^2)$ is even. ($\neg p \Rightarrow \neg q$) If n is odd, then n^2 is odd by Proposition 1.2.11. Hence n is even iff n^2 is even. $\square$

Exercise 1.3.23**from the book**

Prove that $p \vee q \equiv (\neg p) \Rightarrow q$. Use the strategy this suggests to prove that if two integers sum to an even number, then either both are even or both are odd.

By Exercise 1.3.13 (with $\neg p$ in place of p), $(\neg p) \Rightarrow q \equiv (\neg\neg p) \vee q \equiv p \vee q$ by the law of double negation.

Strategy: to prove $p \vee q$, assume $\neg p$ and derive q .

Application. Let $a, b \in \mathbb{Z}$ with $a + b$ even, say $a + b = 2m$. We prove “both even $\vee$ both odd” by assuming they are not both even and deriving that both are odd. So suppose a and b are not both even; then a is odd or b is odd (de Morgan, Theorem 1.3.24). The two cases are symmetric, so suppose a is odd: $a = 2k + 1$ (Proposition 1.1.58). Then $b = 2m - a = 2m - 2k - 1 = 2(m - k - 1) + 1$ is odd. So both are odd. $\square$

Negation: de Morgan’s laws and maximal negation**Theorems 1.3.24 and 1.3.29 · de Morgan’s laws**

Operators: (a) $\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$; (b) $\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$.

Quantifiers: (a) $\neg\forall x \in X, p(x) \equiv \exists x \in X, \neg p(x)$; (b) $\neg\exists x \in X, p(x) \equiv \forall x \in X, \neg p(x)$.

Also $\neg(p \Rightarrow q) \equiv p \wedge (\neg q)$ (Exercise 1.3.27) and $\neg\neg p \equiv p$. Quantifier laws cannot be checked with truth tables; the book derives each side from the other.

Strategy 1.3.30 · proof by counterexample

To prove $\forall x \in X, p(x)$ is false, exhibit one $a \in X$ with $p(a)$ false (that is $\exists x, \neg p(x)$). Example 1.3.31: not every integer is divisible by a prime; counterexample 1.

Exercise 1.3.27

essential

Prove that $\neg(p \Rightarrow q) \equiv p \wedge (\neg q)$ twice: with a truth table, and using Exercise 1.3.13 with de Morgan's laws and double negation.

Truth table.

p	q	$p \Rightarrow q$	$\neg(p \Rightarrow q)$	$\neg q$	$p \wedge (\neg q)$
✓	✓	✓	×	×	×
✓	×	×	✓	✓	✓
×	✓	✓	×	×	×
×	×	✓	×	✓	×

The fourth and sixth columns agree, so the formulae are equivalent.

Algebraically. $\neg(p \Rightarrow q) \equiv \neg((\neg p) \vee q)$ {Exercise 1.3.13} $\equiv (\neg\neg p) \wedge (\neg q)$ {de Morgan (b)} $\equiv p \wedge (\neg q)$ {double negation}. $\square$

Exercise 1.3.28

from the book

Prove that $\neg(p \wedge q) \equiv p \Rightarrow (\neg q)$. What strategy does this suggest for proving a conjunction false, and how does it compare with de Morgan's?

$p \Rightarrow (\neg q) \equiv (\neg p) \vee (\neg q)$ by Exercise 1.3.13, and $(\neg p) \vee (\neg q) \equiv \neg(p \wedge q)$ by de Morgan (a).

Strategy: to prove $p \wedge q$ is false, assume p and prove that q is false. De Morgan's law instead asks you to prove $\neg p$ or to prove $\neg q$, which requires deciding which one fails in advance; the new strategy lets you use p as an extra assumption while refuting q . $\square$

Exercise 1.3.32

recommended

Prove by counterexample that not every rational number can be expressed as $\frac{a}{b}$ with $a \in \mathbb{Z}$ even and $b \in \mathbb{Z}$ odd.

Counterexample: $\frac{1}{3}$. Suppose $\frac{1}{3} = \frac{a}{b}$ with a even and b odd. Then $b = 3a$, and $3a$ is even (it is $3 \cdot 2k = 2(3k)$), so b is even and odd at once, contradicting Definition 0.41. Hence $\frac{1}{3}$ has no such expression. $\square$

NOW YOU 1.3.25 · solutions in Appendix S

1.3.25 Prove de Morgan (b) three times: from the definition of $\equiv$, with a truth table, and from (a) plus double negation.

Definition 1.3.33 and Theorem 1.3.36 · maximally negated

A formula is **maximally negated** if every $\neg$ stands immediately before a predicate (or proposition) that contains no operators or quantifiers. Every formula is equivalent to a maximally negated one: push $\neg$ inwards with the table

$$\begin{aligned} \neg(p \wedge q) &\equiv (\neg p) \vee (\neg q) & \neg\forall x \in X, p(x) &\equiv \exists x \in X, \neg p(x) \\ \neg(p \vee q) &\equiv (\neg p) \wedge (\neg q) & \neg\exists x \in X, p(x) &\equiv \forall x \in X, \neg p(x) \\ \neg(p \Rightarrow q) &\equiv p \wedge (\neg q) & \neg\neg p &\equiv p \end{aligned}$$

one step at a time, ignoring the “stuff” inside (Example 1.3.37 negates the definition of convergence this way). Finally replace $\neg(x < y)$ by $x \geq y$, $\neg(a = b)$ by $a \neq b$.

Exercise 1.3.35**recommended**

Which of the following are maximally negated? (a) $\forall x \in X, (\neg p(x)) \Rightarrow \forall y \in X, \neg(r(x,y) \wedge s(x,y))$; (b) $\forall x \in X, (\neg p(x)) \Rightarrow \forall y \in X, (\neg r(x,y)) \vee (\neg s(x,y))$; (c) $\forall x \in \mathbb{R}, [x > 1 \Rightarrow (\exists y \in \mathbb{R}, [x < y \wedge \neg(x^2 \leq y)])]$; (d) $\neg \exists x \in \mathbb{R}, [x > 1 \wedge (\forall y \in \mathbb{R}, [x < y \Rightarrow x^2 \leq y])]$.

(a) No: $\neg(r \wedge s)$ has a negation in front of a conjunction. (b) Yes: every $\neg$ is directly before a predicate. (c) Yes: the only negation is $\neg(x^2 \leq y)$, in front of a predicate without operators (one may still rewrite it as $x^2 > y$). (d) No: the negation is in front of a quantified formula. $\square$

Exercise 1.3.39**from the book**

Maximally negate $\exists x \in \mathbb{R}, [x > 1 \wedge (\forall y \in \mathbb{R}, [x < y \Rightarrow x^2 \leq y])]$, then prove it true or prove it false.

Negation: $\forall x \in \mathbb{R}, [x \leq 1 \vee (\exists y \in \mathbb{R}, [x < y \wedge x^2 > y])]$ { $\neg \exists \rightarrow \forall \neg$; de Morgan (a); $\neg \forall \rightarrow \exists \neg$; $\neg(\Rightarrow) \rightarrow \wedge \neg$ }. The original statement is false; we prove the negation. Let $x \in \mathbb{R}$. If $x \leq 1$, the first disjunct holds. Otherwise $x > 1$; define $y = \frac{x+x^2}{2}$. Since $x > 1$ we have $x^2 > x$ (multiply $x > 1$ by $x > 0$), so y is strictly between x and x^2 : $x < y$ and $x^2 > y$. Hence the second disjunct holds. In both cases the negation holds. $\square$

NOW YOU 1.3.38 · solutions in Appendix S

1.3.38 Find a maximally negated formula equivalent to $\neg(p \Leftrightarrow q)$.

Tautologies**Definition 1.3.40, Strategy 1.3.41, Theorem 1.3.45**

A **tautology** is a formula that is true under every assignment of truth values to its variables and predicates. Any tautology may be assumed at any point of any proof (LEM says $p \vee \neg p$ is one). **Theorem 1.3.45:** q can be derived from p iff $p \Rightarrow q$ is a tautology; $p \equiv q$ iff $p \Leftrightarrow q$ is a tautology. (So “ $\equiv$ ” is a statement about formulae; “ $\Leftrightarrow$ ” is part of a formula.)

Proving that a formula is a tautology

Either give a derivation from no assumptions (assume the antecedents, derive the consequent), which proves it by Theorem 1.3.45(a); or, for a propositional formula, show that its truth-table column is all $\checkmark$ (Example 1.3.43 does both for $p \Rightarrow (q \Rightarrow p)$).

Exercise 1.3.44 (a), (c), (e)**essential**

Prove that each is a tautology and interpret it: (a) $[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (p \Rightarrow r)$; (c) $\exists y \in Y, \forall x \in X, p(x,y) \Rightarrow \forall x \in X, \exists y \in Y, p(x,y)$; (e) $(\neg \forall x \in X, p(x)) \Leftrightarrow (\exists x \in X, \neg p(x))$.

(a) Assume $(p \Rightarrow q) \wedge (q \Rightarrow r)$, so $p \Rightarrow q$ and $q \Rightarrow r$. Assume p . Modus ponens gives q , then r . Hence $p \Rightarrow r$; this derivation from no assumptions shows the formula is a tautology (Theorem 1.3.45(a)). *Meaning:* implications chain; in a proof you may pass from p to r through an intermediate q .
 (c) This is Theorem 1.2.40, whose proof is a derivation of the consequent from the antecedent; by Theorem 1.3.45(a) the implication is a tautology. *Meaning:* a “one size fits all” y gives a y for each x .
 (e) By Theorem 1.3.29(a), $\neg \forall x \in X, p(x) \equiv \exists x \in X, \neg p(x)$, so by Theorem 1.3.45(b) the biconditional is a tautology. *Meaning:* to refute a universal statement, give a counterexample (Strategy 1.3.30). $\square$

NOW YOU 1.3.44 (b), (d) · solutions in Appendix S

1.3.44 (b) $[p \Rightarrow (q \Rightarrow r)] \Rightarrow [(p \Rightarrow q) \Rightarrow (p \Rightarrow r)]$; **(d)** $[\neg(p \wedge q)] \Leftrightarrow [(\neg p) \vee (\neg q)]$.

D CHAPTER 1 EXERCISES (1.E)**Exercise 1.E.11**

recommended

Prove that $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q))$. How does this help to prove “ p if and only if q ”?

This is Exercise 1.3.22: replace $q \Rightarrow p$ in $(p \Rightarrow q) \wedge (q \Rightarrow p)$ by its contrapositive $(\neg p) \Rightarrow (\neg q)$ (Theorem 1.3.19). It helps when $\neg p$ is easier to work with than q : prove $p \Rightarrow q$, then assume $\neg p$ and derive $\neg q$ instead of proving $q \Rightarrow p$ (as in “ n even iff n^2 even”). $\square$

Exercise 1.E.18

essential

Let X be $\mathbb{Z}$ or $\mathbb{Q}$ and let p be $\forall x \in X, \exists y \in X, (x < y \wedge [\forall z \in X, \neg(x < z \wedge z < y)])$. Write $\neg p$ maximally negated. Prove that p is true for $X = \mathbb{Z}$ and false for $X = \mathbb{Q}$.

$\neg p \equiv \exists x \in X, \forall y \in X, (x \geq y \vee [\exists z \in X, (x < z \wedge z < y)])$. $\{\neg \forall \rightarrow \exists \neg, \neg \exists \rightarrow \forall \neg, \text{de Morgan (a) on the } \wedge, \neg \forall \neg \rightarrow \exists, \neg(x < y) \rightarrow x \geq y\}$

(p says: every x has a “next” element y with nothing strictly between.)

$X = \mathbb{Z}$: p is true. Let $x \in \mathbb{Z}$ and define $y = x + 1 \in \mathbb{Z}$. Then $x < y$. Let $z \in \mathbb{Z}$; there is no integer strictly between x and $x + 1$ (page 6), so $\neg(x < z \wedge z < y)$. Hence y has the required property.

$X = \mathbb{Q}$: p is false. We prove $\neg p$. Take $x = 0 \in \mathbb{Q}$. Let $y \in \mathbb{Q}$. If $y \leq 0$ then $x \geq y$. Otherwise $0 < y$; define $z = \frac{0+y}{2} = \frac{y}{2} \in \mathbb{Q}$, which satisfies $0 < z < y$ (Exercise 1.2.29), so the second disjunct holds. Hence $\neg p$ is true. $\square$

Exercise 1.E.28, 1.E.29, 1.E.30

recommended

True or false? Let ϕ be $\neg \forall x \geq 0, \exists y \in \mathbb{R}, y^2 = x$. **1.E.28** $\phi \equiv \forall x < 0, \exists y \notin \mathbb{R}, y^2 = x$. **1.E.29** $\phi \equiv \exists x \geq 0, \forall y \in \mathbb{R}, y^2 \neq x$. **1.E.30** $\phi \equiv \exists x < 0, \forall y \in \mathbb{R}, y^2 = x$.

Here “ $\forall x \geq 0$ ” means $\forall x \in \mathbb{R}, (x \geq 0 \Rightarrow \dots)$. Maximally negating: $\phi \equiv \exists x \in \mathbb{R}, (x \geq 0 \wedge \forall y \in \mathbb{R}, y^2 \neq x)$, i.e. $\exists x \geq 0, \forall y \in \mathbb{R}, y^2 \neq x$.

1.E.29 True: this is exactly the maximally negated form.

1.E.28 False: negation swaps the quantifiers and negates the predicate, but it never changes the domain of a bound variable ($x \geq 0$ stays, $y \in \mathbb{R}$ stays). The proposed formula is not obtained by any of the rules; and indeed ϕ is false (every $x \geq 0$ has the square root $y = \sqrt{x}$), while the proposed formula is true (for $x < 0$ take $y = 2i \notin \mathbb{R}$ if $x \neq -4$, and $y = i$ if $x = -4$). Different truth values, so not equivalent.

1.E.30 False: it negates the domain condition ($x < 0$) and keeps the predicate un-negated. As formulae they are not equivalent: treat “ $y^2 = x$ ” as an abstract predicate $q(x, y)$ and the domains as abstract sets; the correct negation is $\exists x \in A, \forall y, \neg q(x, y)$, which differs from $\exists x \in B, \forall y, q(x, y)$ (take A non-empty, B empty: then the first may be true while the second is false). $\square$

NOW YOU 1.E, the rest · solutions in Appendix S

1.E.1–1.E.10, 1.E.12–1.E.17, 1.E.19–1.E.27, 1.E.31–1.E.35.

** The three most common losses of marks in this chapter**

- Picking a value when the goal is $\forall$ (the Common error on page 54).
- Rearranging an equation and stopping: the converse must be checked (Examples 1.1.44–45).
- Negating by flipping everything: domains of quantifiers never change; $\neg(p \Rightarrow q)$ is $p \wedge \neg q$, not $\neg p \Rightarrow \neg q$.

Chapter 2

Sets

Contents

- A Sets (§2.1)
- B Set operations (§2.2 to Def. 2.2.46)

Planning

- ★ 2.1: 10, 27, 31, 35
- ★ 2.2: 5, 24, 26, 28, 35, 36, 37, 40
- ★ 2.1: 17, 18, 23, 24, 32, 36
- ★ 2.2: 4, 6, 9, 25, 41, 45
- Not in the uploaded pages: 2.2.46 onwards (2.2.51, 2.2.53) and 2.E. Tested in **Quiz 2**.

A SETS

Definitions 2.1.2 and 2.1.4 · sets, the universe, set-builder notation

A **set** is a collection of elements from a specified **universe of discourse** $\mathcal{U}$; $\mathcal{U}$ contains every mathematical object of interest but $\mathcal{U} \notin \mathcal{U}$. Unrestricted quantifiers range over $\mathcal{U}$: $\forall x, p(x)$ means $\forall x \in \mathcal{U}, p(x)$, and

$$\forall x \in X, p(x) \equiv \forall x, (x \in X \Rightarrow p(x)), \quad \exists x \in X, p(x) \equiv \exists x, (x \in X \wedge p(x)).$$

Set-builder notation: $a \in \{x \mid p(x)\} \Leftrightarrow a \in \mathcal{U} \wedge p(a)$; $\{x \in X \mid p(x)\} = \{x \mid x \in X \wedge p(x)\}$; $\{\text{expr}(x) \mid p(x)\} = \{y \mid \exists x, y = \text{expr}(x) \wedge p(x)\}$. A set written in set-builder notation need not be an element of $\mathcal{U}$ (e.g. $\{x \mid x = x\} = \mathcal{U}$); this is what dissolves Russell's paradox (Exercises 2.1.1, 2.1.5).

NOW YOU 2.1.1, 2.1.3, 2.1.5 · solutions in Appendix S

2.1.1 With $S = \{X \mid X \text{ is a set}\}$ and $R = \{X \in S \mid X \notin X\}$, determine the truth value of $R \in R$. **2.1.3** Why does “ $\forall x \in X$ ” expand with $\Rightarrow$ but “ $\exists x \in X$ ” with $\wedge$? Verify de Morgan's laws for the expanded forms. **2.1.5** Repeat 2.1.1 with the updated definitions.

Subsets

Definition 2.1.6

U is a **subset** of X , written $U \subseteq X$, if $\forall a \in U, a \in X$ (equivalently $\forall a, (a \in U \Rightarrow a \in X)$). $U \not\subseteq X$: not a subset. $U \subsetneq X$ (**proper subset**): $U \subseteq X$ and $U \neq X$. Example 2.1.11: $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$.

Strategy 2.1.7 · proving $U \subseteq X$

“Let $a \in U$.” Unfold what that means. ... “So $a \in X$.” (It is a $\forall$ -proof, Strategy 1.2.10.) Example 2.1.9: $[c, d] \subseteq (a, b)$ when $a < c < d < b$, because $c \leq x \leq d$ gives $a < x < b$.

Proposition 2.1.12 · transitivity

If $X \subseteq Y$ and $Y \subseteq Z$ then $X \subseteq Z$. *Proof:* let $a \in X$; then $a \in Y$ (as $X \subseteq Y$); then $a \in Z$ (as $Y \subseteq Z$).

Exercise 2.1.10**essential**

Let $a, b, c, d \in \mathbb{R}$ with $a < b$ and $c < d$. Prove that $[a, b] \subseteq (c, d]$ if and only if $a > c$ and $b \leq d$.

($\Leftarrow$) Suppose $a > c$ and $b \leq d$. Let $x \in [a, b]$, so $a \leq x < b$. Then $c < a \leq x$ gives $c < x$, and $x < b \leq d$ gives $x \leq d$. So $x \in (c, d]$. Hence $[a, b] \subseteq (c, d]$.

($\Rightarrow$) Suppose $[a, b] \subseteq (c, d]$. Since $a < b$, we have $a \in [a, b]$, so $a \in (c, d]$, which gives $c < a$ (and $a \leq d$). For $b \leq d$, suppose towards a contradiction that $b > d$. Define $x = \frac{d+b}{2}$. Then $d < x < b$, and $x > d \geq a$, so $x \in [a, b]$; but $x > d$ means $x \notin (c, d]$, contradicting $[a, b] \subseteq (c, d]$. Hence $b \leq d$. $\square$

Set equality**Axiom 2.1.14 · set extensionality**

$X = Y$ if and only if $\forall a, (a \in X \Leftrightarrow a \in Y)$, equivalently $X \subseteq Y$ and $Y \subseteq X$. Consequence: order and repetition in list notation do not matter.

Strategy 2.1.15 · proof by double containment

($\subseteq$) Let $a \in X \dots$ so $a \in Y$. ($\supseteq$) Let $a \in Y \dots$ so $a \in X$. Example 2.1.16: $\{x \in \mathbb{R} \mid x^2 \leq 1\} = [-1, 1]$, using $(1-a)(1+a) \geq 0$ for $\subseteq$ and $|a| \leq 1 \Rightarrow a^2 \leq 1$ for $\supseteq$.

Exercise 2.1.17**recommended**

Prove that $\{x \in \mathbb{R} \mid x^2 < x\} = (0, 1)$.

($\subseteq$) Let $a \in \{x \in \mathbb{R} \mid x^2 < x\}$, so $a \in \mathbb{R}$ and $a^2 < a$, i.e. $a(a-1) = a^2 - a < 0$. A product of two real numbers is negative only if one factor is positive and the other negative. If $a > 0$ and $a-1 < 0$ then $0 < a < 1$. If $a < 0$ and $a-1 > 0$ then $a < 0$ and $a > 1$, impossible. So $0 < a < 1$, i.e. $a \in (0, 1)$.

($\supseteq$) Let $a \in (0, 1)$, so $0 < a < 1$. Multiplying $a < 1$ by $a > 0$ gives $a^2 < a$; and $a \in \mathbb{R}$. So $a \in \{x \in \mathbb{R} \mid x^2 < x\}$. $\square$

Exercise 2.1.18**recommended**

Prove by double containment that $\{0, 1\} = \{1, 0\}$ and $\{0, 0\} = \{0\}$.

($\subseteq$) Let $a \in \{0, 1\}$. Then $a = 0$ or $a = 1$; in either case a is one of the listed elements $1, 0$, so $a \in \{1, 0\}$.

($\supseteq$) Let $a \in \{1, 0\}$; then $a = 1$ or $a = 0$, so $a \in \{0, 1\}$. Hence $\{0, 1\} = \{1, 0\}$.

($\subseteq$) Let $a \in \{0, 0\}$. Then $a = 0$ or $a = 0$, so $a = 0$, so $a \in \{0\}$. ($\supseteq$) Let $a \in \{0\}$; then $a = 0$, so $a \in \{0, 0\}$. Hence $\{0, 0\} = \{0\}$. $\square$

Empty and non-empty sets**Definition 2.1.19, Strategy 2.1.20, Theorem 2.1.25, Notation 2.1.26**

X is **non-empty** if it has at least one element ($\exists a, a \in X$); otherwise **empty** ($\neg \exists a, a \in X$). **To prove non-empty:** exhibit an element. **To prove empty:** assume there is $a \in X$ and derive a contradiction. Any two empty sets are equal (2.1.25), so there is one **empty set**, $\emptyset$. Examples 2.1.21–22: $\{x \in \mathbb{Q} \mid x^2 = 2\}$ and $[0]$ are empty.

Exercise 2.1.23**recommended**

Let $a, b \in \mathbb{R}$. Prove that $[a, b]$ is empty iff $a > b$, and that (a, b) is empty iff $a \geq b$.

$[a, b]$. ($\Leftarrow$) Suppose $a > b$. If $x \in [a, b]$ then $a \leq x \leq b$, so $a \leq b$, contradiction. So $[a, b]$ is empty. ($\Rightarrow$) By contraposition: suppose $a \leq b$. Then $a \leq a \leq b$, so $a \in [a, b]$ and $[a, b]$ is non-empty.

(a, b) . ($\Leftarrow$) Suppose $a \geq b$. If $x \in (a, b)$ then $a < x < b$, so $a < b$, contradiction. ($\Rightarrow$) By contraposition: suppose $a < b$. Then $\frac{a+b}{2} \in (a, b)$ (as in Exercise 1.2.29), so (a, b) is non-empty. $\square$

Exercise 2.1.24**recommended**

Let E be empty and $p(x)$ a predicate on E . Show that $\forall x \in E, p(x)$ is true and $\exists x \in E, p(x)$ is false. What does $\forall x \in E, x \neq x$ mean, and is it true?

$\forall x \in E, p(x)$ means $\forall x, (x \in E \Rightarrow p(x))$ (Definition 2.1.2). Let x be arbitrary and suppose $x \in E$. Since E is empty this is a contradiction, so $p(x)$ follows by the principle of explosion (Axiom 1.1.67). Hence the implication holds for every x , and the statement is true (“vacuously”).

$\exists x \in E, p(x)$ means $\exists x, (x \in E \wedge p(x))$; if it were true there would be some $x \in E$, contradicting emptiness. So it is false.

$\forall x \in E, x \neq x$ says “every element of the empty set is different from itself”. It is true, by the first part with $p(x)$ being $x \neq x$. $\square$

Exercise 2.1.27**essential**

Is it true that $\emptyset \in X$ for all sets X ? Is it true that $\emptyset \subseteq X$ for all sets X ?

$\emptyset \in X$ for all X is **false**: take $X = \emptyset$; it has no elements, so $\emptyset \notin \emptyset$.

$\emptyset \subseteq X$ for all X is **true**: by Definition 2.1.6 we need $\forall a \in \emptyset, a \in X$, which holds by Exercise 2.1.24 (vacuously). $\square$

NOW YOU 2.1.28 · solutions in Appendix S

2.1.28 Why are $\emptyset$, $\{\emptyset\}$, $\{\{\emptyset\}\}$ different? Truth values of $\emptyset \in \emptyset$, $\emptyset \subseteq \emptyset$, $\emptyset \in \{\emptyset\}$, $\emptyset \in \{\{\emptyset\}\}$, $\{\emptyset\} \subseteq \{\emptyset\}$, $\{\emptyset\} \subseteq \{\{\emptyset\}\}$; find more.

Power sets**Definition 2.1.29, Example 2.1.34**

$\mathcal{P}(X)$ is the set of all subsets of X : $U \in \mathcal{P}(X) \Leftrightarrow U \subseteq X$. $\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$. Confusions come from mixing $\in$ (element) with $\subseteq$ (subset): $U \subseteq X$ is the same as $U \in \mathcal{P}(X)$.

Exercise 2.1.31**essential**

Write out the elements of $\mathcal{P}(\{1, 2, 3\})$.

$\mathcal{P}(\{1, 2, 3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$: eight subsets, one for each choice of which of 1, 2, 3 to include. $\square$

Exercise 2.1.32**recommended**

Let X be a set. Show that $\emptyset \in \mathcal{P}(X)$ and $X \in \mathcal{P}(X)$.

By Exercise 2.1.27, $\emptyset \subseteq X$, so $\emptyset \in \mathcal{P}(X)$ by Definition 2.1.29. By Example 2.1.8, $X \subseteq X$, so $X \in \mathcal{P}(X)$. $\square$

Exercise 2.1.35**essential**

Let X and Y be sets. Prove that if $X \subseteq Y$, then $\mathcal{P}(X) \subseteq \mathcal{P}(Y)$.

Suppose $X \subseteq Y$. Let $U \in \mathcal{P}(X)$. Then $U \subseteq X$ (Definition 2.1.29). Since $U \subseteq X$ and $X \subseteq Y$, transitivity (Proposition 2.1.12) gives $U \subseteq Y$, so $U \in \mathcal{P}(Y)$. Hence $\mathcal{P}(X) \subseteq \mathcal{P}(Y)$ by Strategy 2.1.7. $\square$

Exercise 2.1.36**recommended**

True or false? (a) $\mathcal{P}(\emptyset) \in \{\emptyset\}$; (b) $\mathcal{P}(\emptyset) \in \mathcal{P}(\mathcal{P}(\emptyset))$; (c) $\mathcal{P}(\mathcal{P}(\emptyset)) \in \{\emptyset, \{\emptyset, \{\emptyset\}\}\}$. Repeat with $\in$ replaced by $\subseteq$.

First compute: $\mathcal{P}(\emptyset) = \{\emptyset\}$ and $\mathcal{P}(\mathcal{P}(\emptyset)) = \mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$.

(a) False: the only element of $\{\emptyset\}$ is $\emptyset$, and $\{\emptyset\} \neq \emptyset$ (it has an element).

(b) True: $\mathcal{P}(\emptyset) \subseteq \mathcal{P}(\emptyset)$ (Example 2.1.8), so $\mathcal{P}(\emptyset) \in \mathcal{P}(\mathcal{P}(\emptyset))$ by Definition 2.1.29.

(c) True: $\mathcal{P}(\mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\}$ is the second listed element.

With $\subseteq$: (a') $\{\emptyset\} \subseteq \{\emptyset\}$: true (Example 2.1.8). (b') $\{\emptyset\} \subseteq \{\emptyset, \{\emptyset\}\}$: true, its only element $\emptyset$ is listed.

(c') $\{\emptyset, \{\emptyset\}\} \subseteq \{\emptyset, \{\emptyset, \{\emptyset\}\}\}$: false, because $\{\emptyset\}$ is an element of the left set but not of the right one (the right set's elements are $\emptyset$ and $\{\emptyset, \{\emptyset\}\}$, and $\{\emptyset\}$ differs from both). $\square$

NOW YOU 2.1.33 · solutions in Appendix S

2.1.33 Write out $\mathcal{P}(\emptyset)$, $\mathcal{P}(\mathcal{P}(\emptyset))$ and $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$.

B SET OPERATIONS**Intersection****Definitions 2.2.1, 2.2.2, 2.2.10, Notation 2.2.13**

The **intersection** of a collection of sets is the set of objects that are elements of all of them. $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$, so $x \in A \cap B \Leftrightarrow x \in A \wedge x \in B$. A and B are **disjoint** if $A \cap B$ is empty. $A_0 \cap \cdots \cap A_{n-1} = \{x \mid x \in A_i \text{ for all } i \in [n]\}$.

Exercise 2.2.4**recommended**

Prove that $[0, \infty) \cap \mathbb{Z} = \mathbb{N}$.

($\subseteq$) Let $x \in [0, \infty) \cap \mathbb{Z}$. Then $x \in [0, \infty)$ and $x \in \mathbb{Z}$, so x is an integer with $x \geq 0$, i.e. a non-negative whole number, so $x \in \mathbb{N}$ (Definition 0.6). ($\supseteq$) Let $n \in \mathbb{N}$. Then n is a whole number, so $n \in \mathbb{Z}$, and $n \geq 0$, so $n \in [0, \infty)$. Hence $n \in [0, \infty) \cap \mathbb{Z}$. $\square$

Exercise 2.2.5

essential

Write down the elements of $\{0, 1, 4, 7\} \cap \{1, 2, 3, 4, 5\}$.

The elements common to both lists are 1 and 4: $\{0, 1, 4, 7\} \cap \{1, 2, 3, 4, 5\} = \{1, 4\}$. □

Exercise 2.2.6

recommended

Express $[-2, 5) \cap [4, 7)$ as a single interval.

x lies in both iff $-2 \leq x < 5$ and $4 \leq x < 7$, iff $4 \leq x < 5$. So $[-2, 5) \cap [4, 7) = [4, 5)$. □

Proposition 2.2.8

$X \subseteq Y$ if and only if $X \cap Y = X$. *Proof:* $(\Rightarrow)$ double containment, using $X \subseteq Y$ for $\supseteq$; $(\Leftarrow)$ if $a \in X = X \cap Y$ then $a \in Y$.

Exercise 2.2.9

recommended

Let X be a set. Prove that $X \cap \emptyset = \emptyset$ and $X \cap X = X$.

$X \cap \emptyset = \emptyset$. Suppose $X \cap \emptyset$ is non-empty, say $a \in X \cap \emptyset$. Then $a \in \emptyset$, contradicting that $\emptyset$ has no elements. So $X \cap \emptyset$ is empty, hence equal to $\emptyset$ (Theorem 2.1.25).

$X \cap X = X$. For any a : $a \in X \cap X \Leftrightarrow a \in X \wedge a \in X \Leftrightarrow a \in X$. So the sets are equal by extensionality (Axiom 2.1.14). □

NOW YOU 2.2.7, 2.2.12, 2.2.15 · solutions in Appendix S

2.2.7 For $A \subseteq \mathbb{R}$, express with $\cap$: (a) the positive elements of A ; (b) the integer elements of A . **2.2.12** Prove that (a, b) and (c, d) are disjoint iff $b \leq c$ or $d \leq a$. **2.2.15** Express the set of non-negative rational elements of A using $\cap$.

Indexed families**Definition 2.2.16, Notation 2.2.18**

An **indexed family** over a set I assigns a set A_i to each $i \in I$: $\{A_i \mid i \in I\}$. $\bigcap_{i \in I} A_i = \{x \mid x \in A_i \text{ for all } i \in I\}$, so $x \in \bigcap_{i \in I} A_i \Leftrightarrow \forall i \in I, x \in A_i$. For $I = \{n \in \mathbb{N} \mid n \geq 1\}$ one writes $\bigcap_{n \geq 1} A_n$ or $\bigcap_{n=1}^{\infty} A_n$.

Example 2.2.19 · the technique for intersections over $n \geq 1$

To prove $\bigcap_{n \geq 1} [0, 1 + \frac{1}{n}] = [0, 1]$: $(\supseteq)$ let $x \in [0, 1]$ and let $n \geq 1$ be arbitrary; check $x < 1 + \frac{1}{n}$. $(\subseteq)$ let x be in every $[0, 1 + \frac{1}{n}]$ and suppose $x > 1$; choose $N = \lfloor \frac{1}{x-1} \rfloor + 1$, which gives $N \geq 1$ and $N \geq \frac{1}{x-1}$, hence $x \geq 1 + \frac{1}{N}$, contradicting $x \in [0, 1 + \frac{1}{N}]$. **The invention is always the same: build an index N from x that breaks the inequality.**

NOW YOU 2.2.20 · solutions in Appendix S

2.2.20 For $n \in \mathbb{Z}$ let $A_n = \{x \in \mathbb{R} \mid x \neq n\}$. Describe $\bigcap_{n \in \mathbb{Z}} A_n$ in plain terms.

Union

Definitions 2.2.21, 2.2.22, Notations 2.2.29, 2.2.33

The **union** of a collection of sets is the set of objects that are elements of at least one of them. $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$, so $x \in A \cup B \Leftrightarrow x \in A \vee x \in B$. $A_0 \cup \cdots \cup A_{n-1} = \{x \mid x \in A_i \text{ for some } i \in [n]\}$. $\bigcup_{i \in I} A_i = \{x \mid x \in A_i \text{ for some } i \in I\}$, so $x \in \bigcup_{i \in I} A_i \Leftrightarrow \exists i \in I, x \in A_i$. Example 2.2.23: $E \cup O = \mathbb{Z}$ and $E \cap O = \emptyset$ for the even and odd integers.

Exercise 2.2.24**essential**

Write down the elements of $\{0, 1, 4, 7\} \cup \{1, 2, 3, 4, 5\}$.

Everything in either list, each element once: $\{0, 1, 2, 3, 4, 5, 7\}$. □

Exercise 2.2.25**recommended**

Express $[-2, 5) \cup [4, 7)$ as a single interval.

x lies in the union iff $-2 \leq x < 5$ or $4 \leq x < 7$; since the two ranges overlap on $[4, 5)$, this is iff $-2 \leq x < 7$. So $[-2, 5) \cup [4, 7) = [-2, 7)$. □

Exercise 2.2.26**essential**

Let X and Y be sets. Prove that $X \subseteq Y$ if and only if $X \cup Y = Y$.

($\Rightarrow$) Suppose $X \subseteq Y$; we prove $X \cup Y = Y$ by double containment. ($\subseteq$) Let $a \in X \cup Y$. Then $a \in X$ or $a \in Y$. If $a \in X$ then $a \in Y$ since $X \subseteq Y$; if $a \in Y$ then $a \in Y$. In both cases $a \in Y$. ($\supseteq$) Let $a \in Y$. Then $a \in X \cup Y$ by definition of union.

($\Leftarrow$) Suppose $X \cup Y = Y$. Let $a \in X$. Then $a \in X \cup Y$ by definition of union, so $a \in Y$ since $X \cup Y = Y$. Hence $X \subseteq Y$. □

Example 2.2.27 · distributivity of $\cap$ over $\cup$

$X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$, by double containment; the $\subseteq$ direction splits into cases on $x \in Y \cup Z$.

Exercise 2.2.28**essential**

Let X, Y, Z be sets. Prove that $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$.

($\subseteq$) Let $x \in X \cup (Y \cap Z)$. Then $x \in X$ or $x \in Y \cap Z$. If $x \in X$, then $x \in X \cup Y$ and $x \in X \cup Z$, so $x \in (X \cup Y) \cap (X \cup Z)$. If $x \in Y \cap Z$, then $x \in Y$ and $x \in Z$, so $x \in X \cup Y$ and $x \in X \cup Z$, and again $x \in (X \cup Y) \cap (X \cup Z)$. ($\supseteq$) Let $x \in (X \cup Y) \cap (X \cup Z)$, so $x \in X \cup Y$ and $x \in X \cup Z$. By the law of excluded middle, $x \in X$ or $x \notin X$. If $x \in X$, then $x \in X \cup (Y \cap Z)$. If $x \notin X$, then from $x \in X \cup Y$ we get $x \in Y$, and from $x \in X \cup Z$ we get $x \in Z$; so $x \in Y \cap Z$, and $x \in X \cup (Y \cap Z)$. □

NOW YOU 2.2.31, 2.2.32 · solutions in Appendix S

2.2.31 Write $X = \{x \in \mathbb{R} \mid x^4 - 5x^2 + 4 > 0\}$ as a union of intervals. **2.2.32** State and prove the distributivity laws for finitary unions and intersections.

Exercise 2.2.35**essential**

Express $\bigcup_{n \geq 1} [0, 1 - \frac{1}{n}]$ as an interval.

We claim $\bigcup_{n \geq 1} [0, 1 - \frac{1}{n}] = [0, 1)$. ($\subseteq$) Let x be in the union; take $n \geq 1$ with $x \in [0, 1 - \frac{1}{n}]$. Then $0 \leq x \leq 1 - \frac{1}{n} < 1$, so $x \in [0, 1)$. ($\supseteq$) Let $x \in [0, 1)$, so $0 \leq x < 1$ and $1 - x > 0$. Define $N = \lceil \frac{1}{1-x} \rceil$; since $\frac{1}{1-x} \geq 1$ we have $N \geq 1$, and $N \geq \frac{1}{1-x}$ gives $\frac{1}{N} \leq 1 - x$, i.e. $x \leq 1 - \frac{1}{N}$. So $x \in [0, 1 - \frac{1}{N}]$ and hence x is in the union. $\square$

Exercise 2.2.36**essential**

Prove that $\bigcap_{n \in \mathbb{N}} [n] = \emptyset$ and $\bigcup_{n \in \mathbb{N}} [n] = \mathbb{N}$.

Intersection. Suppose $x \in \bigcap_{n \in \mathbb{N}} [n]$. Then $x \in [n]$ for all $n \in \mathbb{N}$, in particular $x \in [0]$. But $[0] = \{k \in \mathbb{N} \mid k \leq 0\}$ is empty (Example 2.1.22): contradiction. So the intersection is empty.

Union. ($\subseteq$) If $x \in [n]$ for some n , then $x \in \mathbb{N}$ by definition of $[n]$. ($\supseteq$) Let $k \in \mathbb{N}$. Then $k < k + 1$ and $k + 1 \in \mathbb{N}$, so $k \in [k + 1]$, hence $k \in \bigcup_{n \in \mathbb{N}} [n]$. $\square$

Exercise 2.2.37**essential**

Find a family $\{X_n \mid n \in \mathbb{N}\}$ with (i) $\bigcup_{n \in \mathbb{N}} X_n = \mathbb{N}$; (ii) $\bigcap_{n \in \mathbb{N}} X_n = \emptyset$; (iii) $X_i \cap X_j \neq \emptyset$ for all $i, j \in \mathbb{N}$.

Take $X_n = \{k \in \mathbb{N} \mid k \neq n\}$ (“ $\mathbb{N}$ with n removed”). (i) Each $X_n \subseteq \mathbb{N}$; and any $k \in \mathbb{N}$ lies in X_{k+1} since $k \neq k + 1$. (ii) If k were in every X_n then in particular $k \in X_k$, i.e. $k \neq k$: contradiction. (iii) Given i, j , the number $i + j + 1$ differs from both i and j , so $i + j + 1 \in X_i \cap X_j$. $\square$

Relative complement**Definition 2.2.38**

$X \setminus A = \{x \in X \mid x \notin A\}$, the **relative complement** of A in X (“ X minus A ”); $x \in X \setminus A \Leftrightarrow x \in X \wedge x \notin A$. The book uses no absolute complement A^c . Example 2.2.39: $\mathbb{Z} \setminus E$ is the set of odd integers.

Exercise 2.2.40**essential**

Write down the elements of $\{0, 1, 4, 7\} \setminus \{1, 2, 3, 4, 5\}$.

Keep the elements of the first set that are not in the second: $\{0, 7\}$. $\square$

Exercise 2.2.41**recommended**

Express $[-2, 5) \setminus [4, 7)$ and $[4, 7) \setminus [-2, 5)$ as intervals.

$x \in [-2, 5) \setminus [4, 7)$ iff $-2 \leq x < 5$ and not $4 \leq x < 7$, iff $-2 \leq x < 4$: the set is $[-2, 4)$.

$x \in [4, 7) \setminus [-2, 5)$ iff $4 \leq x < 7$ and not $-2 \leq x < 5$, iff $5 \leq x < 7$: the set is $[5, 7)$. $\square$

NOW YOU 2.2.42, 2.2.43 · solutions in Appendix S

2.2.42 A = the reals that are not integers: (a) write A as a relative complement; (b) as a union of intervals.

2.2.43 Prove $X \setminus (X \setminus A) = X \cap A$, and deduce that $A \subseteq X$ iff $X \setminus (X \setminus A) = A$.

Sets versus logic

The dictionary (page 92)

With $p: a \in X$, $q: a \in Y$, $r(i): a \in X_i$:

$$\begin{array}{lll} a \notin X & \neg p & a \in \bigcap_{i \in I} X_i \quad \forall i \in I, r(i) \\ a \in X \cap Y & p \wedge q & a \in \bigcup_{i \in I} X_i \quad \exists i \in I, r(i) \\ a \in X \cup Y & p \vee q & a \in X \setminus Y \quad p \wedge (\neg q) \end{array}$$

Every set identity in this section is a logical law in disguise; the proof unfolds membership, applies the law, folds back.

Theorem 2.2.44 · de Morgan's laws for sets

(a) $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$; (b) $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$; (c) $X \setminus \bigcup_{i \in I} A_i = \bigcap_{i \in I} (X \setminus A_i)$; (d) $X \setminus \bigcap_{i \in I} A_i = \bigcup_{i \in I} (X \setminus A_i)$. The book proves (a) by double containment, using Theorem 1.3.24 to turn $x \notin A \cup B$ into $x \notin A$ and $x \notin B$.

Exercise 2.2.45**recommended**

Complete the proof of de Morgan's laws for sets: parts (b), (c), (d).

(b) ($\subseteq$) Let $x \in X \setminus (A \cap B)$: $x \in X$ and $x \notin A \cap B$, i.e. $\neg(x \in A \wedge x \in B)$. By de Morgan (Theorem 1.3.24(a)), $x \notin A$ or $x \notin B$. If $x \notin A$ then $x \in X \setminus A$; if $x \notin B$ then $x \in X \setminus B$. Either way $x \in (X \setminus A) \cup (X \setminus B)$. ($\supseteq$) Let $x \in (X \setminus A) \cup (X \setminus B)$. If $x \in X \setminus A$: $x \in X$ and $x \notin A$, so $x \notin A \cap B$ (an element of $A \cap B$ would be in A), hence $x \in X \setminus (A \cap B)$. The case $x \in X \setminus B$ is the same.

(c) For any x : $x \in X \setminus \bigcup_i A_i \Leftrightarrow x \in X \wedge \neg(\exists i \in I, x \in A_i) \Leftrightarrow x \in X \wedge \forall i \in I, x \notin A_i$ {Theorem 1.3.29(b)}
 $\Leftrightarrow \forall i \in I, (x \in X \wedge x \notin A_i)$ { $x \in X$ does not depend on i ; see note} $\Leftrightarrow \forall i \in I, x \in X \setminus A_i \Leftrightarrow x \in \bigcap_i (X \setminus A_i)$.
Note: the third step needs I non-empty for the direction $\Leftarrow$; for $I = \emptyset$ one reads the conventions of the book for empty intersections.

(d) For any x : $x \in X \setminus \bigcap_i A_i \Leftrightarrow x \in X \wedge \neg(\forall i \in I, x \in A_i) \Leftrightarrow x \in X \wedge \exists i \in I, x \notin A_i$ {Theorem 1.3.29(a)}
 $\Leftrightarrow \exists i \in I, (x \in X \wedge x \notin A_i) \Leftrightarrow \exists i \in I, x \in X \setminus A_i \Leftrightarrow x \in \bigcup_i (X \setminus A_i)$. $\square$

Watch out

- From $x \in A \cup B$ you get a *disjunction*: do cases. From $x \in A \cap B$ you get two facts. From $x \notin A \cup B$ you get $x \notin A$ and $x \notin B$ (de Morgan), not just one of them.
- $X \setminus A = X \setminus B$ does not give $A = B$ (aside on page 91): the complement operation does not cancel like subtraction.
- An “express as an interval” question still deserves the one-line justification “ x is in the set iff ...”.

Where the uploaded pages stop

Definition 2.2.46 (ordered lists) begins the part on Cartesian products, $\prod$ and A^k , which continues to the end of §2.2 and is followed by the chapter exercises 2.E (planning: ★ 2.2.51; 2.E.6, 33, 38; ★ 2.2.53; 2.E.4, 5, 16, 28, 32, 41). Those pages were not in the upload; send them and this workbook can be extended.

Chapter 3

Functions

Contents

- A** Functions, images and preimages (§3.1)
- B** Injections, surjections and bijections (§3.2)
- C** Book-level examples
- D** Stretch exercises
- E** Review set 3

Planning

Essential: 3.1: 11, 15abc, 36; 3.E: 13, 17, 19, 34, 41
Essential: 3.2: 5, 8, 12, 14, 20, 40, 47; 3.E: 24, 25, 48
Recommended: 3.1: 8, 16, 19, 33, 40, 41; 3.2: 6, 16, 19, 25, 39ab, 45, 46; 3.E: 2, 35, 37, 38, 47
Bonus: 3.1: 21
 Tested in **Quiz 2** (7 Oct).

📖 Chapters 3–5: not yet built from the book

The uploaded book files stop at page 93 (Definition 2.2.46). Chapters 3, 4 and 5 below are the earlier material, written without the book: the theory and the proof techniques are the standard ones for these sections, and the single book page I have (page 176: Lemma 5.2.29 to Theorem 5.2.35) is used in Chapter 5. But definition and theorem numbers, wording, and exercise numbers are **not verified** against the Utrecht edition, and the examples are not the book's exercises. The purple planning boxes still list the real exercise numbers from the planning. Upload book pages 94–180 and these chapters will be rebuilt exactly like Chapters 0–2.

A FUNCTIONS, IMAGES AND PREIMAGES

Function

A **function** $f : X \rightarrow Y$ consists of a set X (the **domain**), a set Y (the **codomain**) and a rule that assigns to *each* $a \in X$ *exactly one* element $f(a) \in Y$ (the **value** of f at a). We write $a \mapsto f(a)$ for the rule.

A proposed rule is **well-defined** if (i) it produces a value for every $a \in X$, (ii) the value is unique (does not depend on how a is written or on choices made), and (iii) the value lies in Y .

Two functions are **equal** if they have the same domain, the same codomain and $f(a) = g(a)$ for all $a \in X$. So $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto x^2$ and $g : \mathbb{R} \rightarrow [0, \infty), x \mapsto x^2$ are different functions.

Functions you will meet constantly

identity $\text{id}_X : X \rightarrow X, a \mapsto a$; constant functions; the inclusion $U \hookrightarrow X$ for $U \subseteq X$; projections $\pi_1 : X \times Y \rightarrow X, (a, b) \mapsto a$; the **characteristic function** $\chi_U : X \rightarrow \{0, 1\}$ of $U \subseteq X$ ($\chi_U(a) = 1$ if $a \in U$, else 0); the floor $\lfloor \cdot \rfloor : \mathbb{R} \rightarrow \mathbb{Z}$; piecewise rules (“ $f(n) = n/2$ if n even, $f(n) = 3n + 1$ if n odd”); sequences (a_n) as functions $\mathbb{N} \rightarrow \mathbb{R}$; binary operations such as $+$: $\mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$.

Example 1 — well-defined or not?

Self Tutor

Which of the following are well-defined functions? (a) $f : \mathbb{Q} \rightarrow \mathbb{Z}, f(\frac{a}{b}) = a$. (b) $g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = 1/x$. (c) $h : \mathbb{Z} \rightarrow \mathbb{N}, h(n) = n^2$. (d) $k : \mathbb{R} \rightarrow \mathbb{Z}, k(x) = \text{the integer } n \text{ with } n \leq x < n + 1$.

(a) Not well-defined: $\frac{1}{2} = \frac{2}{4}$ but the rule gives 1 and 2: the value depends on how the input is written (fails

(ii)).

(b) Not well-defined: no value at $x = 0$ (fails (i)). It is a function $\mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$.(c) Well-defined: n^2 is a unique natural number for every integer n (iii: $n^2 \geq 0$).(d) Well-defined: for each x such an n exists and is unique (if $n \leq x < n+1$ and $m \leq x < m+1$ then $|n-m| < 1$, so $n = m$). This is $\lfloor x \rfloor$. $\square$ **Graph, composition**

- The **graph** of $f : X \rightarrow Y$ is $\text{Gr}(f) = \{(a, f(a)) \mid a \in X\} \subseteq X \times Y$. A subset $G \subseteq X \times Y$ is the graph of a function iff $\forall a \in X, \exists! b \in Y, (a, b) \in G$ (“vertical line test”).
- For $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ the **composite** is $g \circ f : X \rightarrow Z$, $(g \circ f)(a) = g(f(a))$. Composition is associative, $h \circ (g \circ f) = (h \circ g) \circ f$, and $f \circ \text{id}_X = f = \text{id}_Y \circ f$; it is *not* commutative.

Example 2 — composition is not commutative

Self Tutor

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x + 1$ and $g(x) = x^2$. Compute $g \circ f$ and $f \circ g$ and show $g \circ f \neq f \circ g$.

$(g \circ f)(x) = g(x+1) = (x+1)^2$ and $(f \circ g)(x) = f(x^2) = x^2 + 1$. At $x = 1$: $(g \circ f)(1) = 4 \neq 2 = (f \circ g)(1)$. Same domain and codomain, but the values differ at one point, so the functions are different. $\square$

Image and preimageLet $f : X \rightarrow Y$, $U \subseteq X$ and $V \subseteq Y$.

- The **image** of U is $f[U] = \{f(a) \mid a \in U\} \subseteq Y$. Rule: $b \in f[U] \Leftrightarrow \exists a \in U, f(a) = b$.
- The **preimage** of V is $f^{-1}[V] = \{a \in X \mid f(a) \in V\} \subseteq X$. Rule: $a \in f^{-1}[V] \Leftrightarrow f(a) \in V$.
- $f[X]$ is called *the image of f* . Preimages exist for every function: $f^{-1}[V]$ does not require an inverse function.

Example 3 — computing images and preimages

Self Tutor

Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$. Find $f([-1, 2])$, $f^{-1}[[1, 4]]$ and $f^{-1}[\{-1\}]$, with justification for the first.

$f([-1, 2]) = [0, 4]$. ($\subseteq$) If $x \in [-1, 2]$ then $0 \leq x^2 \leq 4$. ($\supseteq$) If $y \in [0, 4]$ then $\sqrt{y} \in [0, 2] \subseteq [-1, 2]$ and $f(\sqrt{y}) = y$. {witness for the $\exists$ in the image rule}
 $f^{-1}[[1, 4]] = \{x \in \mathbb{R} \mid 1 \leq x^2 \leq 4\} = [-2, -1] \cup [1, 2]$. $f^{-1}[\{-1\}] = \{x \in \mathbb{R} \mid x^2 = -1\} = \emptyset$. $\square$

Images and preimages versus set operationsLet $f : X \rightarrow Y$, $g : Y \rightarrow Z$, $U, U' \subseteq X$ and $V, V' \subseteq Y$. Then:

- | | |
|--|--|
| (a) $U \subseteq U' \Rightarrow f[U] \subseteq f[U']$ | (b) $V \subseteq V' \Rightarrow f^{-1}[V] \subseteq f^{-1}[V']$ |
| (c) $f[U \cup U'] = f[U] \cup f[U']$ | (d) $f^{-1}[V \cup V'] = f^{-1}[V] \cup f^{-1}[V']$ |
| (e) $f[U \cap U'] \subseteq f[U] \cap f[U']$ (equality can fail) | (f) $f^{-1}[V \cap V'] = f^{-1}[V] \cap f^{-1}[V']$ |
| (g) $U \subseteq f^{-1}[f[U]]$ (equality can fail) | (h) $f[f^{-1}[V]] \subseteq V$ (equality can fail) |
| (i) $f^{-1}[Y \setminus V] = X \setminus f^{-1}[V]$ | (j) $(g \circ f)[U] = g[f[U]]$, $(g \circ f)^{-1}[W] = f^{-1}[g^{-1}[W]]$ |

Preimages behave perfectly; images lose information. Equality in (e) and (g) holds for all U, U' iff f is injective; equality in (h) for all V iff f is surjective (Section B).

Example 4 — an image inclusion and why it is not an equality

Self Tutor

Prove that $f[U \cap U'] \subseteq f[U] \cap f[U']$, and give an example where the inclusion is strict.

Let $b \in f[U \cap U']$. Fix $a \in U \cap U'$ with $f(a) = b$. {image rule} Since $a \in U$ and $f(a) = b$, $b \in f[U]$; since $a \in U'$ and $f(a) = b$, $b \in f[U']$. Hence $b \in f[U] \cap f[U']$. $\square$

Strict: $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, $U = \{-1\}$, $U' = \{1\}$. Then $f[U \cap U'] = f[\emptyset] = \emptyset$ but $f[U] \cap f[U'] = \{1\}$. The proof of “ $\supseteq$ ” breaks because the $a \in U$ and the $a' \in U'$ with $f(a) = f(a') = b$ need not be the same element.

Example 5 — preimage of a complement

Self Tutor

Prove that $f^{-1}[Y \setminus V] = X \setminus f^{-1}[V]$.

For any a : $a \in f^{-1}[Y \setminus V] \Leftrightarrow a \in X \wedge f(a) \in Y \setminus V \Leftrightarrow a \in X \wedge f(a) \in Y \wedge f(a) \notin V \Leftrightarrow a \in X \wedge \neg(f(a) \in V) \Leftrightarrow a \in X \wedge a \notin f^{-1}[V] \Leftrightarrow a \in X \setminus f^{-1}[V]$, using that $f(a) \in Y$ is automatic for $a \in X$. $\square$

Example 6 — $U \subseteq f^{-1}[f[U]]$

Self Tutor

Prove that $U \subseteq f^{-1}[f[U]]$ for every $U \subseteq X$, and find f , U with $U \neq f^{-1}[f[U]]$.

Let $a \in U$. Then $f(a) \in f[U]$ (witness a itself), so $a \in f^{-1}[f[U]]$ by the preimage rule. $\square$

Strict: $f(x) = x^2$ on $\mathbb{R}$, $U = \{1\}$: $f[U] = \{1\}$ and $f^{-1}[\{1\}] = \{-1, 1\} \neq U$.

Watch out

- $f[U]$ lives in the codomain, $f^{-1}[V]$ in the domain. Start a proof about $f[U]$ with “Let $b \in f[U]$; fix $a \in U$ with $f(a) = b$ ”. Start a proof about $f^{-1}[V]$ with “Let $a \in f^{-1}[V]$; then $f(a) \in V$ ”.
- Do not write $f^{-1}(b)$ for a preimage unless f is a bijection. The book’s $f^{-1}[V]$ is a set.
- A function is domain + codomain + rule. Changing the codomain changes the function (and can change surjectivity).

EXERCISES 3.1 and 3.E (first part)

Essential: 3.1: 11, 15abc, 36; 3.E: 13, 17, 19, 34, 41 **Recommended:** 3.1: 8, 16, 19, 33, 40, 41; 3.E: 2, 37, 38 **Bonus:** 3.1: 21

Skills: checking well-definedness; computing composites, images and preimages of concrete sets (intervals, finite sets); proving the inclusions and equalities in the table above from the two membership rules; building counterexamples with $x \mapsto x^2$ or small finite functions.

B INJECTIONS, SURJECTIONS AND BIJECTIONS**Injective, surjective, bijective**

Let $f: X \rightarrow Y$.

- f is **injective** (one-to-one) if $\forall a, b \in X, (f(a) = f(b) \Rightarrow a = b)$ (equivalently $a \neq b \Rightarrow f(a) \neq f(b)$).
Not injective: $\exists a, b \in X, a \neq b \wedge f(a) = f(b)$.
- f is **surjective** (onto) if $\forall b \in Y, \exists a \in X, f(a) = b$ (equivalently $f[X] = Y$).
Not surjective: $\exists b \in Y, \forall a \in X, f(a) \neq b$.
- f is **bijective** if it is both injective and surjective.

Templates

- **Injective:** “Let $a, b \in X$ and suppose $f(a) = f(b)$.” Compute until “ $a = b$ ”. (Do not start from $a = b$.)
- **Not injective:** two explicit different inputs with the same output.
- **Surjective:** “Let $b \in Y$.” Solve $f(a) = b$ for a on scratch paper; in the proof: “Put $a = \dots$. Then $a \in X$ {check!} and $f(a) = \dots = b$.”
- **Not surjective:** one explicit $b \in Y$ and a proof that $f(a) \neq b$ for all a (often a parity or sign argument).
- **Bijective:** both of the above, *or* exhibit an inverse (see below) and verify both compositions.

Example 7 — a bijection, directly

Self Tutor

Prove that $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 3x - 2$ is bijective.

Injective. Let $x, y \in \mathbb{R}$ with $f(x) = f(y)$. Then $3x - 2 = 3y - 2$, so $3x = 3y$, so $x = y$.

Surjective. Let $y \in \mathbb{R}$. Put $x = \frac{y+2}{3} \in \mathbb{R}$. Then $f(x) = 3 \cdot \frac{y+2}{3} - 2 = y$.

Hence f is bijective. □

Example 8 — neither injective nor surjective

Self Tutor

Let $g : \mathbb{Z} \rightarrow \mathbb{Z}$, $g(n) = n^2 + n$. Show that g is neither injective nor surjective.

Not injective: $g(0) = 0 = g(-1)$ and $0 \neq -1$.

Not surjective: we show $1 \notin g[\mathbb{Z}]$. Let $n \in \mathbb{Z}$. Then $g(n) = n(n+1)$ is even by the fact that $n(n+1)$ is even (one of $n, n+1$ is even), while 1 is odd, so $g(n) \neq 1$. □

Example 9 — a bijection $\mathbb{Z} \rightarrow \mathbb{N}$ with cases

Self Tutor

Define $h : \mathbb{Z} \rightarrow \mathbb{N}$ by $h(n) = 2n$ if $n \geq 0$ and $h(n) = -2n - 1$ if $n < 0$. Prove that h is a bijection.

First, h is well-defined: for $n \geq 0$, $2n \in \mathbb{N}$; for $n < 0$, $-2n - 1 \geq 2 - 1 = 1 \in \mathbb{N}$. Note $h(n)$ is even when $n \geq 0$ and odd when $n < 0$.

Injective. Let $m, n \in \mathbb{Z}$ with $h(m) = h(n)$. Since $h(m) = h(n)$ have the same parity, m, n are both ≥ 0 or both < 0 . If both ≥ 0 : $2m = 2n$, so $m = n$. If both < 0 : $-2m - 1 = -2n - 1$, so $m = n$.

Surjective. Let $k \in \mathbb{N}$. If k is even, $k = 2j$ with $j \in \mathbb{N}$, and $h(j) = 2j = k$. If k is odd, $k = 2j + 1$ with $j \in \mathbb{N}$; put $n = -(j+1) < 0$, then $h(n) = 2(j+1) - 1 = k$. □

Inverses

Let $f : X \rightarrow Y$. A function $g : Y \rightarrow X$ is

- a **left inverse** of f if $g \circ f = \text{id}_X$ (i.e. $g(f(a)) = a$ for all $a \in X$);
- a **right inverse** of f if $f \circ g = \text{id}_Y$ (i.e. $f(g(b)) = b$ for all $b \in Y$);
- an **inverse** of f if it is both. Then f has exactly one inverse, written f^{-1} , and $(f^{-1})^{-1} = f$.

Inverses and the three properties

Let $f : X \rightarrow Y$ with X inhabited.

- 1 f is injective $\Leftrightarrow f$ has a left inverse.
- 2 f is surjective $\Leftrightarrow f$ has a right inverse.
- 3 f is bijective $\Leftrightarrow f$ has an inverse. Then f^{-1} is also a bijection and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ for bijections f, g .

Composition

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$.

- 1 If f and g are injective (surjective, bijective), then so is $g \circ f$.
- 2 If $g \circ f$ is injective, then f is injective. If $g \circ f$ is surjective, then g is surjective.

Example 10 — injectivity of a composite

Self Tutor

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. Prove that if $g \circ f$ is injective then f is injective. Show by example that g need not be injective.

Suppose $g \circ f$ is injective. Let $a, b \in X$ with $f(a) = f(b)$. Applying g gives $g(f(a)) = g(f(b))$, i.e. $(g \circ f)(a) = (g \circ f)(b)$, so $a = b$ by injectivity of $g \circ f$. Hence f is injective. $\square$

Example: $X = \{0\}$, $Y = \{0, 1\}$, $Z = \{0\}$, $f(0) = 0$, $g(0) = g(1) = 0$. Then $g \circ f$ is injective (one-element domain) but g is not.

Example 11 — left inverse $\Rightarrow$ injective

Self Tutor

Let $f : X \rightarrow Y$ and suppose $g : Y \rightarrow X$ satisfies $g \circ f = \text{id}_X$. Prove that f is injective.

Let $a, b \in X$ with $f(a) = f(b)$. Then $a = \text{id}_X(a) = g(f(a)) = g(f(b)) = \text{id}_X(b) = b$. $\square$

Converse idea: if f is injective and $x_0 \in X$, define $g(b) =$ the unique a with $f(a) = b$ when $b \in f[X]$, and $g(b) = x_0$ otherwise; then $g \circ f = \text{id}_X$.

Example 12 — bijective via an explicit inverse

Self Tutor

Let $f : \mathbb{R} \setminus \{2\} \rightarrow \mathbb{R} \setminus \{1\}$, $f(x) = \frac{x}{x-2}$. Prove that f is a bijection and find f^{-1} .

Scratch: solve $y = \frac{x}{x-2}$: $y(x-2) = x \Rightarrow x(y-1) = 2y \Rightarrow x = \frac{2y}{y-1}$.

Define $g : \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R} \setminus \{2\}$ by $g(y) = \frac{2y}{y-1}$. It is well-defined: $y \neq 1$, and $g(y) = 2$ would mean $2y = 2y - 2$, impossible, so $g(y) \neq 2$.

For $x \neq 2$: $g(f(x)) = \frac{2 \cdot \frac{x}{x-2}}{\frac{x}{x-2} - 1} = \frac{2x}{x - (x-2)} = x$. For $y \neq 1$: $f(g(y)) = \frac{\frac{2y}{y-1}}{\frac{2y}{y-1} - 2} = \frac{2y}{2y - 2(y-1)} = y$.

So $g \circ f = \text{id}$ and $f \circ g = \text{id}$; hence f is a bijection with $f^{-1} = g$. $\square$

Example 13 — injective $\Rightarrow$ images preserve intersections

Self Tutor

Let $f : X \rightarrow Y$ be injective. Prove that $f[U \cap U'] = f[U] \cap f[U']$ for all $U, U' \subseteq X$.

“ $\subseteq$ ” is Example 3.4. ($\supseteq$) Let $b \in f[U] \cap f[U']$. Fix $a \in U$ with $f(a) = b$ and $a' \in U'$ with $f(a') = b$. Then $f(a) = f(a')$, so $a = a'$ by injectivity. Hence $a \in U \cap U'$ and $f(a) = b$, so $b \in f[U \cap U']$. $\square$

Watch out

- In an injectivity proof the hypothesis is $f(a) = f(b)$ and the goal is $a = b$. Writing “ $a = b \Rightarrow f(a) = f(b)$ ” proves nothing (every function does that).
- Surjectivity is about the *codomain*: $x \mapsto x^2$ is surjective $\mathbb{R} \rightarrow [0, \infty)$ but not $\mathbb{R} \rightarrow \mathbb{R}$. Always check that your witness a lies in the domain.
- f^{-1} as a function exists only for bijections; “ $f^{-1}(b)$ ” in a proof must be preceded by “since f is bijective”.

- For piecewise functions, injectivity needs the cross-cases too (an input from each piece); separate the pieces by a property of the outputs, as in Example 3.9.

EXERCISES 3.2 and 3.E (second part)

Essential: 3.2: 5, 8, 12, 14, 20, 40, 47; 3.E: 24, 25, 48 **Recommended:** 3.2: 6, 16, 19, 25, 39ab, 45, 46; 3.E: 35, 47

Skills: proving or refuting injectivity/surjectivity for concrete functions on $\mathbb{N}, \mathbb{Z}, \mathbb{R}$, products and power sets; finding inverses and verifying both compositions; proving the general facts about composites and left/right inverses; combining injectivity/surjectivity with images and preimages (equality cases of the Section A table).

C BOOK-LEVEL EXAMPLES

The hardest Chapter 3 exercises characterise injectivity or surjectivity by a property of *all* images, preimages or compositions, or ask for a bijection between two sets that look different. The invention is usually “apply the hypothesis to a cleverly chosen small set”.

Example 14 — surjectivity through preimages**Book level**

Let $f : X \rightarrow Y$. Prove that f is surjective if and only if $f[f^{-1}[V]] = V$ for every $V \subseteq Y$.

Invention: in the $\Leftarrow$ direction, apply the hypothesis to $V = \{b\}$.

($\Rightarrow$) Suppose f is surjective and let $V \subseteq Y$. The inclusion $f[f^{-1}[V]] \subseteq V$ holds for every f : if $b \in f[f^{-1}[V]]$, fix $a \in f^{-1}[V]$ with $f(a) = b$; then $b = f(a) \in V$. For $\supseteq$: let $b \in V$. By surjectivity fix $a \in X$ with $f(a) = b$. Then $f(a) \in V$, so $a \in f^{-1}[V]$, so $b = f(a) \in f[f^{-1}[V]]$.

($\Leftarrow$) Suppose $f[f^{-1}[V]] = V$ for all $V \subseteq Y$, and let $b \in Y$. Apply the hypothesis to $V = \{b\}$: then $b \in f[f^{-1}[\{b\}]]$, so there is $a \in f^{-1}[\{b\}]$ with $f(a) = b$. This a witnesses surjectivity at b . $\square$

Example 15 — injectivity through images**Book level**

Let $f : X \rightarrow Y$. Prove that f is injective if and only if $f^{-1}[f[U]] = U$ for every $U \subseteq X$.

Invention: apply the hypothesis to a one-element set.

($\Rightarrow$) Suppose f is injective and $U \subseteq X$. $U \subseteq f^{-1}[f[U]]$ holds always (Example 3.6). Conversely let $a \in f^{-1}[f[U]]$, so $f(a) \in f[U]$; fix $u \in U$ with $f(u) = f(a)$. By injectivity $a = u \in U$.

($\Leftarrow$) Suppose $f^{-1}[f[U]] = U$ for all U . Let $a, a' \in X$ with $f(a) = f(a')$. Then $f(a') \in f[\{a\}]$, so $a' \in f^{-1}[f[\{a\}]] = \{a\}$, hence $a' = a$. $\square$

Example 16 — a bijection $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ **Book level**

Define $g : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ by $g(m, n) = 2^m(2n + 1) - 1$. Prove that g is a bijection.

Invention: a lemma about the shape of positive integers, proved by strong induction.

Lemma. Every integer $k \geq 1$ can be written as $k = 2^m(2n + 1)$ with $m, n \in \mathbb{N}$, and m, n are unique. *Existence* is Example 4.16 (Chapter 4). *Uniqueness:* suppose $2^m(2n + 1) = 2^{m'}(2n' + 1)$ and, without loss of generality, $m \leq m'$. Dividing by 2^m gives $2n + 1 = 2^{m'-m}(2n' + 1)$. If $m' > m$, the right-hand side is even and the left-hand side is odd, impossible; so $m' = m$, and then $2n + 1 = 2n' + 1$ gives $n = n'$.

Injective. If $g(m, n) = g(m', n')$ then $2^m(2n + 1) = 2^{m'}(2n' + 1)$, so $(m, n) = (m', n')$ by uniqueness.

Surjective. Let $k \in \mathbb{N}$. Then $k + 1 \geq 1$, so $k + 1 = 2^m(2n + 1)$ for some $m, n \in \mathbb{N}$ by existence, i.e.

$$g(m, n) = k.$$

□

Example 17 — subsets are the same as $\{0, 1\}$ -valued functions

Book level

Let X be a set and write $\{0, 1\}^X$ for the set of all functions $X \rightarrow \{0, 1\}$. Prove that $\chi : \mathcal{P}(X) \rightarrow \{0, 1\}^X$, $U \mapsto \chi_U$, is a bijection.

Invention: write down the inverse, $g \mapsto g^{-1}[\{1\}]$, and check both composites.

Define $\psi : \{0, 1\}^X \rightarrow \mathcal{P}(X)$ by $\psi(g) = g^{-1}[\{1\}] = \{a \in X \mid g(a) = 1\}$.

$\psi \circ \chi = \text{id}$: for $U \subseteq X$, $\psi(\chi_U) = \{a \in X \mid \chi_U(a) = 1\} = \{a \in X \mid a \in U\} = U$.

$\chi \circ \psi = \text{id}$: let $g : X \rightarrow \{0, 1\}$ and $a \in X$. If $g(a) = 1$ then $a \in \psi(g)$, so $\chi_{\psi(g)}(a) = 1 = g(a)$. If $g(a) = 0$ then $a \notin \psi(g)$, so $\chi_{\psi(g)}(a) = 0 = g(a)$. Thus $\chi_{\psi(g)} = g$ (same domain, codomain and values).

So χ has an inverse and is a bijection. □

Consequence: a set with n elements has 2^n subsets, because there are 2^n functions from it to $\{0, 1\}$ (two choices per element).

Example 18 — injective means left-cancellable

Book level

Let $f : X \rightarrow Y$. Prove that f is injective if and only if for every set Z and all functions $g, h : Z \rightarrow X$: if $f \circ g = f \circ h$ then $g = h$.

Invention: in the $\Leftarrow$ direction, use a one-point set Z .

$(\Rightarrow)$ Suppose f is injective and $f \circ g = f \circ h$ with $g, h : Z \rightarrow X$. Let $z \in Z$. Then $f(g(z)) = f(h(z))$, so $g(z) = h(z)$. Since z was arbitrary and g, h share domain and codomain, $g = h$.

$(\Leftarrow)$ Suppose the cancellation property holds. Let $a, a' \in X$ with $f(a) = f(a')$. Let $Z = \{*\}$ and define $g, h : Z \rightarrow X$ by $g(*) = a$ and $h(*) = a'$. Then $(f \circ g)(*) = f(a) = f(a') = (f \circ h)(*)$, so $f \circ g = f \circ h$, so $g = h$ by hypothesis, so $a = g(*) = h(*) = a'$. □

Example 19 — one-sided inverses that meet in the middle

Book level

Let $f : X \rightarrow Y$. Suppose $g : Y \rightarrow X$ satisfies $g \circ f = \text{id}_X$ and $h : Y \rightarrow X$ satisfies $f \circ h = \text{id}_Y$. Prove that $g = h$, and deduce that f is a bijection with inverse g .

Invention: insert an identity and re-bracket.

$g = g \circ \text{id}_Y = g \circ (f \circ h) = (g \circ f) \circ h = \text{id}_X \circ h = h$, using associativity of composition.

Hence $g \circ f = \text{id}_X$ and $f \circ g = f \circ h = \text{id}_Y$, so g is a two-sided inverse of f ; thus f is a bijection and $f^{-1} = g$. The same computation shows that inverses are unique: if g and g' are both inverses of f , run the argument with $h = g'$. □

D STRETCH EXERCISES**STRETCH EXERCISES 3 · solutions in Appendix S**

S3.1 Let $f : X \rightarrow Y$ and $V \subseteq Y$. Prove that $f[f^{-1}[V]] = V \cap f[X]$.

S3.2 Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be bijections. Prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. [Hint: check both composites and use uniqueness of inverses]

S3.3 Let $f : X \rightarrow Y$. Prove that f is injective if and only if $f[X \setminus U] \subseteq Y \setminus f[U]$ for every $U \subseteq X$. [Hint: for $\Leftarrow$, take $U = \{a'\}$]

S3.4 Define $f : \mathbb{Z} \rightarrow \mathbb{Z}$ by $f(n) = n + 1$ if n is even and $f(n) = n - 1$ if n is odd. Prove that $f \circ f = \text{id}_{\mathbb{Z}}$ and deduce that f is a bijection.

S3.5 Let $f : X \rightarrow Y$ have a left inverse g and a right inverse h . Prove that $g = h$.

S3.6 Prove that $f : X \rightarrow Y$ is surjective if and only if for every set Z and all $g, h : Y \rightarrow Z$: $g \circ f = h \circ f$ implies $g = h$. [Hint: for $\Leftarrow$, suppose $b \notin f[X]$ and build $g, h : Y \rightarrow \{0, 1\}$ that differ only at b]

S3.7 Prove that for every set X there is no surjection $f : X \rightarrow \mathcal{P}(X)$. [Hint: consider $D = \{a \in X \mid a \notin f(a)\}$]

S3.8 Let $a < b$ be real numbers. Prove that $x \mapsto a + (b - a)x$ is a bijection $(0, 1) \rightarrow (a, b)$ by exhibiting its inverse.

E REVIEW SET 3

- Function = domain + codomain + rule; well-defined = every input, one output, in the codomain.
- $b \in f[U] \Leftrightarrow \exists a \in U, f(a) = b$; $a \in f^{-1}[V] \Leftrightarrow f(a) \in V$.
- Preimages commute with $\cap, \cup, \setminus$; images only with $\cup$ (and $\subseteq$ for $\cap$).
- Injective: $f(a) = f(b) \Rightarrow a = b$. Surjective: every b has an a . Bijective: both $\Leftrightarrow$ inverse exists.
- Left inverse $\Leftrightarrow$ injective; right inverse $\Leftrightarrow$ surjective.
- $g \circ f$ injective $\Rightarrow f$ injective; $g \circ f$ surjective $\Rightarrow g$ surjective.
- $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
- Counterexample factory: $x \mapsto x^2, n \mapsto 2n, n \mapsto n(n+1)$, tiny finite sets.

Chapter 4

Mathematical induction

Contents

- A** Weak induction (§4.2)
- B** Strong induction and well-ordering (§4.3)
- C** Book-level examples
- D** Stretch exercises
- E** Review set 4

Planning

Essential: 4.2: 5, 11; 4.E: 10, 11, 16, 17, 18, 20, 25
Essential: 4.3: 8, 9, 14, 15; 4.E: 13, 19, 26, 27, 29, 30, 31, 32
Recommended: 4.2: 8, 17, 20; 4.E: 12, 14, 15, 21, 24, 33; 4.3: 10, 18; 4.E: 28, 34, 35
 (§4.1, Peano's axioms, is not in the planning.)

A WEAK INDUCTION

Principle of (weak) induction

Let $p(n)$ be a statement about natural numbers n and let $b \in \mathbb{N}$. Suppose

- **(base case)** $p(b)$ is true, and
- **(induction step)** for all $n \geq b$: if $p(n)$ is true then $p(n+1)$ is true.

Then $p(n)$ is true for all $n \geq b$. (Most often $b = 0$ or $b = 1$.)

In the induction step the assumption “ $p(n)$ ” is the **induction hypothesis**. You prove $p(n+1)$ *from* it, for an arbitrary $n \geq b$; you do not get to assume $p(n+1)$.

The induction template (use these words)

- 1 “For $n \in \mathbb{N}$ (with $n \geq b$) let $p(n)$ be the statement: ...”. $p(n)$ is a *statement*, not a number.
- 2 “**Base case** $n = b$: ...” Verify $p(b)$ by direct computation.
- 3 “**Induction step.** Let $n \geq b$ and suppose $p(n)$ holds, i.e. ... (IH). We prove $p(n+1)$, i.e. ...”
Start from the $n+1$ expression, isolate the n expression inside it, apply the IH, finish with algebra.
- 4 “By induction, $p(n)$ holds for all $n \geq b$.” $\square$

Notation that appears in the exercises

$\sum_{k=1}^n a_k = a_1 + \cdots + a_n$ and $\prod_{k=1}^n a_k = a_1 \cdots a_n$; an empty sum is 0, an empty product is 1. Recursive definitions: $0! = 1$, $(n+1)! = (n+1) \cdot n!$; a sequence given by a_0 and a rule for a_{n+1} in terms of a_n . The key identity for the step is $\sum_{k=1}^{n+1} a_k = (\sum_{k=1}^n a_k) + a_{n+1}$.

Example 1 — a sum formula

Self Tutor

Prove that $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ for all $n \in \mathbb{N}$.

For $n \in \mathbb{N}$ let $p(n)$ be the statement $\sum_{k=1}^n k = \frac{n(n+1)}{2}$.
 Base case $n = 0$: the empty sum is 0 and $\frac{0 \cdot 1}{2} = 0$. So $p(0)$ holds.

Induction step. Let $n \in \mathbb{N}$ and suppose $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ (IH). Then

$$\sum_{k=1}^{n+1} k = \left(\sum_{k=1}^n k \right) + (n+1) \stackrel{\text{IH}}{=} \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2} = \frac{(n+1)((n+1)+1)}{2},$$

which is $p(n+1)$.

By induction, $p(n)$ holds for all $n \in \mathbb{N}$. □

Example 2 — geometric sum

Self Tutor

Let $r \in \mathbb{R}$ with $r \neq 1$. Prove that $\sum_{k=0}^n r^k = \frac{r^{n+1} - 1}{r - 1}$ for all $n \in \mathbb{N}$.

Base case $n = 0$: LHS = $r^0 = 1$, RHS = $\frac{r-1}{r-1} = 1$.

Induction step: let $n \in \mathbb{N}$ and assume the formula for n . Then $\sum_{k=0}^{n+1} r^k = \frac{r^{n+1} - 1}{r - 1} + r^{n+1} = \frac{r^{n+1} - 1 + r^{n+2} - r^{n+1}}{r - 1} = \frac{r^{n+2} - 1}{r - 1}$, the formula for $n + 1$.

By induction the formula holds for all $n \in \mathbb{N}$. □

Example 3 — divisibility

Self Tutor

Prove that $3 \mid 4^n - 1$ for all $n \in \mathbb{N}$.

Base case $n = 0$: $4^0 - 1 = 0 = 3 \cdot 0$.

Induction step: let $n \in \mathbb{N}$ and suppose $3 \mid 4^n - 1$; fix $k \in \mathbb{Z}$ with $4^n - 1 = 3k$. Then

$$4^{n+1} - 1 = 4 \cdot 4^n - 1 = 4(4^n - 1) + 3 = 4 \cdot 3k + 3 = 3(4k + 1),$$

so $3 \mid 4^{n+1} - 1$. By induction, $3 \mid 4^n - 1$ for all $n \in \mathbb{N}$. □

Trick: write the $n + 1$ quantity as (something) · (the n quantity) + (something divisible).

Example 4 — an inequality with a larger base case

Self Tutor

Prove that $2^n > n^2$ for all $n \geq 5$.

Base case $n = 5$: $2^5 = 32 > 25 = 5^2$.

Induction step: let $n \geq 5$ and suppose $2^n > n^2$. Then $2^{n+1} = 2 \cdot 2^n > 2n^2$, so it suffices to show $2n^2 \geq (n+1)^2$, i.e. $n^2 - 2n - 1 \geq 0$, i.e. $(n-1)^2 \geq 2$, which holds since $n-1 \geq 4$. Hence $2^{n+1} > (n+1)^2$.

By induction, $2^n > n^2$ for all $n \geq 5$. □

Pattern for inequalities: bound the $n + 1$ side by the IH, then prove a separate “gap” inequality for $n \geq b$.

Example 5 — closed form of a recursive sequence

Self Tutor

Let $a_0 = 1$ and $a_{n+1} = 2a_n + 1$ for $n \in \mathbb{N}$. Prove that $a_n = 2^{n+1} - 1$ for all $n \in \mathbb{N}$.

Base case: $a_0 = 1 = 2^1 - 1$. Induction step: let $n \in \mathbb{N}$ with $a_n = 2^{n+1} - 1$. Then $a_{n+1} = 2a_n + 1 = 2(2^{n+1} - 1) + 1 = 2^{n+2} - 1$. By induction, $a_n = 2^{n+1} - 1$ for all n . □

Binomial coefficients and the binomial theorem

For $n, k \in \mathbb{N}$ with $k \leq n$, $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the number of k -element subsets of an n -element set ($\binom{n}{k} = 0$ for $k > n$).

Pascal's rule: $\binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1}$. **Binomial theorem:** for $x, y \in \mathbb{R}$ and $n \in \mathbb{N}$,

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Both are proved by induction on n (the theorem's step uses Pascal's rule). Consequences: $\sum_{k=0}^n \binom{n}{k} = 2^n$ ($x = y = 1$), $\sum_k (-1)^k \binom{n}{k} = 0$ for $n \geq 1$.

Example 6 — Bernoulli's inequality

Self Tutor

Let $x \in \mathbb{R}$ with $x \geq -1$. Prove that $(1+x)^n \geq 1+nx$ for all $n \in \mathbb{N}$.

Base case $n = 0$: $(1+x)^0 = 1 \geq 1+0$. Induction step: let $n \in \mathbb{N}$ and suppose $(1+x)^n \geq 1+nx$. Since $1+x \geq 0$ we may multiply the IH by $1+x$ without flipping the inequality:

$$(1+x)^{n+1} = (1+x)(1+x)^n \geq (1+x)(1+nx) = 1 + (n+1)x + nx^2 \geq 1 + (n+1)x,$$

as $nx^2 \geq 0$. By induction the inequality holds for all $n \in \mathbb{N}$. □

Watch out

- No base case, or the wrong base case ($n = 0$ when the claim is for $n \geq 1$): the proof is invalid. The classic “all horses have the same colour” fallacy has a correct-looking step that fails exactly from $n = 1$ to $n = 2$.
- Never start the step with the statement $p(n+1)$ and manipulate it to something true; start from one side of $p(n+1)$ and transform it, or show both sides equal a common expression.
- State $p(n)$ explicitly and say where you use the IH. “Assume true for n ” without saying what is true is not acceptable.
- If the step uses $p(n-1)$ as well as $p(n)$, you are doing strong induction: see Section B for the extra base cases.

EXERCISES 4.2 and 4.E (weak induction)

Essential: 4.2: 5, 11; 4.E: 10, 11, 16, 17, 18, 20, 25 **Recommended:** 4.2: 8, 17, 20; 4.E: 12, 14, 15, 21, 24, 33

Skills: sum/product formulas, divisibility, inequalities with a base case $b > 0$, recursively defined sequences, statements about finite sets (e.g. number of subsets), binomial coefficients and the binomial theorem. Every one of them is the template above with a different algebraic middle.

B STRONG INDUCTION AND WELL-ORDERING**Principle of strong induction**

Let $p(n)$ be a statement about natural numbers and $b \in \mathbb{N}$. Suppose that for every $n \geq b$:

if $p(k)$ holds for all k with $b \leq k < n$, then $p(n)$ holds.

Then $p(n)$ holds for all $n \geq b$.

In practice: prove the first few values $p(b), \dots, p(b+m)$ directly (**base cases**), then for $n > b+m$ prove $p(n)$ assuming $p(k)$ for all $b \leq k < n$ (or just for the specific smaller k you need, such as $n-1$ and $n-2$).

Weak and strong induction are equivalent principles, but strong induction is the right tool whenever the “reason” for $p(n)$ involves a smaller value that is not $n-1$: factoring $n = ab$, the recurrence $f_n = f_{n-1} + f_{n-2}$, subtracting 4 from n , dividing n by b .

Strong induction template

- 1 “Let $p(n)$ be the statement ...”. Decide which smaller values the step will need, and hence how many base cases.
- 2 “**Base cases.**” Check each one directly.
- 3 “**Induction step.** Let $n \geq \dots$ and suppose $p(k)$ holds for all $b \leq k < n$ (IH). ... Therefore $p(n)$.” Say explicitly which $k < n$ you apply the IH to, and check that $k \geq b$.
- 4 “By strong induction, $p(n)$ holds for all $n \geq b$.”

Example 7 — prime factorisation exists

Self Tutor

Prove that every natural number $n \geq 2$ is a product of (one or more) primes.

Let $p(n)$: “ n is a product of primes”. Let $n \geq 2$ and suppose $p(k)$ for all $2 \leq k < n$ (IH).

Case 1: n is prime. Then n is a product of one prime.

Case 2: n is not prime. Then $n = ab$ with $1 < a, b < n$ (a positive divisor other than 1 and n exists). By the IH, a and b are products of primes, so $n = ab$ is too.

In both cases $p(n)$ holds, so by strong induction every $n \geq 2$ is a product of primes. □

No separate base case was needed: for $n = 2$ the IH is vacuous and Case 1 applies.

Example 8 — Fibonacci bound, two base cases

Self Tutor

Let $f_0 = 0$, $f_1 = 1$ and $f_{n+2} = f_{n+1} + f_n$. Prove that $f_n \leq 2^{n-1}$ for all $n \geq 1$.

Base cases: $f_1 = 1 \leq 2^0$ and $f_2 = 1 \leq 2^1$.

Step: let $n \geq 3$ and suppose $f_k \leq 2^{k-1}$ for all $1 \leq k < n$. Since $n-1 \geq 1$ and $n-2 \geq 1$ the IH applies to both:

$$f_n = f_{n-1} + f_{n-2} \leq 2^{n-2} + 2^{n-3} \leq 2^{n-2} + 2^{n-2} = 2^{n-1}.$$

By strong induction, $f_n \leq 2^{n-1}$ for all $n \geq 1$. □

Two base cases, because the step uses $n-2$ and is therefore only valid from $n = 3$.

Example 9 — stamps: every $n \geq 12$ is $4a + 5b$

Self Tutor

Prove that every $n \in \mathbb{N}$ with $n \geq 12$ can be written as $n = 4a + 5b$ with $a, b \in \mathbb{N}$.

Base cases: $12 = 4 \cdot 3$, $13 = 4 \cdot 2 + 5$, $14 = 4 + 5 \cdot 2$, $15 = 5 \cdot 3$.

Step: let $n \geq 16$ and suppose the claim for all $12 \leq k < n$. Then $12 \leq n-4 < n$, so $n-4 = 4a + 5b$ for some $a, b \in \mathbb{N}$, and $n = 4(a+1) + 5b$.

By strong induction the claim holds for all $n \geq 12$. □

Example 10 — the division theorem, existence part

Self Tutor

Let $b \geq 1$. Prove that for every $a \in \mathbb{N}$ there exist $q, r \in \mathbb{N}$ with $a = qb + r$ and $0 \leq r < b$.

By strong induction on a . Let $a \in \mathbb{N}$ and suppose the claim for all $a' < a$.

If $a < b$: take $q = 0, r = a$. If $a \geq b$: then $0 \leq a - b < a$, so by the IH $a - b = q'b + r$ with $0 \leq r < b$; hence $a = (q' + 1)b + r$. $\square$

Uniqueness: if $qb + r = q'b + r'$ with $0 \leq r, r' < b$ then $b \mid r - r'$ and $|r - r'| < b$, so $r = r'$, then $q = q'$.

Extend to negative a via $-a$.

Example 11 — base- b expansions exist

Self Tutor

Let $b \geq 2$. Prove that every $n \geq 1$ has a base- b expansion.

Strong induction on n . Let $n \geq 1$ and assume all $1 \leq k < n$ have expansions. Write $n = qb + d$ with $0 \leq d < b$ (division theorem). If $q = 0$ then $n = d$ is a one-digit expansion. If $q \geq 1$ then $1 \leq q < n$ (as $b \geq 2$), so $q = (d_r \dots d_0)_b$ by the IH, and then $n = qb + d = (d_r \dots d_0 d)_b$. $\square$

Well-ordering principle

Every inhabited subset of $\mathbb{N}$ has a least element. This is equivalent to induction, and gives the **minimal counterexample** method: to prove $\forall n \in \mathbb{N}, p(n)$, suppose not; then $\{n \in \mathbb{N} \mid \neg p(n)\}$ is inhabited, so it has a least element m ; derive a contradiction by producing a smaller counterexample (or by checking m directly if m is small).

Example 12 — well-ordering

Self Tutor

Prove that there is no integer n with $0 < n < 1$.

Suppose, for a contradiction, that $S = \{n \in \mathbb{Z} \mid 0 < n < 1\}$ is inhabited. Then $S \subseteq \mathbb{N}$, so by well-ordering S has a least element m . From $0 < m < 1$ we get, multiplying by $m > 0$, $0 < m^2 < m < 1$. So $m^2 \in S$ and $m^2 < m$, contradicting the minimality of m . $\square$

Watch out

- Count the base cases: if the step uses $n - d$, you need base cases $b, b + 1, \dots, b + d - 1$ and the step starts at $n = b + d$. Always check that the smaller value you feed into the IH is still $\geq b$.
- In the step, the IH is “ $p(k)$ for all $b \leq k < n$ ”, and you prove $p(n)$ (not $p(n + 1)$). Mixing the two conventions produces off-by-one errors.
- “ n is not prime, so $n = ab$ ”: say why $1 < a, b < n$. Without that the IH does not apply.

EXERCISES 4.3 and 4.E (strong induction)

Essential: 4.3: 8, 9, 14, 15; 4.E: 13, 19, 26, 27, 29, 30, 31, 32 **Recommended:** 4.3: 10, 18; 4.E: 28, 34, 35

Skills: recognising when weak induction is not enough; proofs with several base cases (Fibonacci-type recurrences, $n - d$ steps); existence of factorisations/expansions; proofs via the well-ordering principle and minimal counterexamples.

C BOOK-LEVEL EXAMPLES

The hardest Chapter 4 exercises need algebra that must be *aimed* at the induction hypothesis, a recurrence that reaches back two steps, or a statement about pictures or sets rather than formulas. The invention is usually the choice of $p(n)$ or the rearrangement that exposes the hypothesis.

Example 13 — sum of squares

Book level

Prove that $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbb{N}$.

Technique: factor out what the target contains, here $(n+1)$, before expanding.

Base case $n = 0$: both sides are 0. Induction step: let $n \in \mathbb{N}$ and assume the formula for n . Then

$$\begin{aligned} \sum_{k=1}^{n+1} k^2 &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{(n+1)[n(2n+1) + 6(n+1)]}{6} \\ &= \frac{(n+1)(2n^2 + 7n + 6)}{6} = \frac{(n+1)(n+2)(2n+3)}{6}, \end{aligned}$$

which is the formula for $n+1$, since $2(n+1) + 1 = 2n+3$. By induction the formula holds for all $n \in \mathbb{N}$.

□

Example 14 — Cassini's identity: rearranging to expose the hypothesis

Book level

With $f_0 = 0$, $f_1 = 1$ and $f_{n+2} = f_{n+1} + f_n$, prove that $f_{n+1}f_{n-1} - f_n^2 = (-1)^n$ for all $n \geq 1$.

Base case $n = 1$: $f_2f_0 - f_1^2 = 0 - 1 = -1 = (-1)^1$.

Induction step: let $n \geq 1$ and assume $f_{n+1}f_{n-1} - f_n^2 = (-1)^n$. Then

$$\begin{aligned} f_{n+2}f_n - f_{n+1}^2 &= (f_{n+1} + f_n)f_n - f_{n+1}^2 \quad \{\text{recurrence}\} \\ &= f_n^2 - f_{n+1}(f_{n+1} - f_n) = f_n^2 - f_{n+1}f_{n-1} \quad \{f_{n+1} - f_n = f_{n-1} \text{ since } n \geq 1\} \\ &= -(f_{n+1}f_{n-1} - f_n^2) = -(-1)^n = (-1)^{n+1}. \quad \{\text{IH}\} \end{aligned}$$

By induction the identity holds for all $n \geq 1$.

□

Example 15 — Binet's formula by strong induction

Book level

Let $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. Prove that $f_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$ for all $n \in \mathbb{N}$.

Invention: φ and ψ are the roots of $x^2 = x + 1$, so their powers satisfy the Fibonacci recurrence.

First, $\varphi^2 = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2} = \varphi + 1$, and similarly $\psi^2 = \psi + 1$. Multiplying by φ^n : $\varphi^{n+2} = \varphi^{n+1} + \varphi^n$; likewise for ψ .

Base cases. $n = 0$: $(1-1)/\sqrt{5} = 0 = f_0$. $n = 1$: $(\varphi - \psi)/\sqrt{5} = \sqrt{5}/\sqrt{5} = 1 = f_1$.

Step. Let $n \geq 2$ and assume the formula for all $k < n$, in particular for $n-1$ and $n-2$ (both ≥ 0). Then

$$f_n = f_{n-1} + f_{n-2} = \frac{(\varphi^{n-1} + \varphi^{n-2}) - (\psi^{n-1} + \psi^{n-2})}{\sqrt{5}} = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

By strong induction the formula holds for all $n \in \mathbb{N}$.

□

Example 16 — a power of two times an odd number

Book level

Prove that every integer $k \geq 1$ can be written as $k = 2^m(2n+1)$ with $m, n \in \mathbb{N}$.

Invention: strong induction, because the smaller case is $k/2$, not $k-1$.

Let $k \geq 1$ and assume the claim for all j with $1 \leq j < k$. If k is odd, then $k = 2n+1$ for some $n \in \mathbb{N}$, and $k = 2^0(2n+1)$. If k is even, then $k = 2j$ with $1 \leq j < k$; by the hypothesis $j = 2^m(2n+1)$, so $k = 2^{m+1}(2n+1)$. By strong induction the claim holds for all $k \geq 1$. $\square$ (Uniqueness of m, n : Example 3.16.)

Example 17 — a statement about pictures

Book level

Say n lines in the plane are in *general position* if no two are parallel and no three pass through one point. Prove that n lines in general position divide the plane into $\frac{n^2+n+2}{2}$ regions.

Invention: find the recurrence $R(n+1) = R(n) + (n+1)$ and prove that in the step.

Let $p(n)$ be: “any n lines in general position cut the plane into $\frac{n^2+n+2}{2}$ regions”. Base case $n=0$: no lines, one region, and $\frac{0+0+2}{2} = 1$.

Step: let $n \in \mathbb{N}$, assume $p(n)$, and take $n+1$ lines in general position. Remove one of them, ℓ . The remaining n lines are in general position, so by $p(n)$ they give $\frac{n^2+n+2}{2}$ regions. Now put ℓ back: it meets each of the other n lines in exactly one point (no two parallel), and these n points are distinct (no three concurrent), so they cut ℓ into $n+1$ pieces. Each piece splits one existing region into two, adding exactly $n+1$ regions. Hence the count is

$$\frac{n^2+n+2}{2} + (n+1) = \frac{n^2+3n+4}{2} = \frac{(n+1)^2 + (n+1) + 2}{2},$$

which is $p(n+1)$. By induction $p(n)$ holds for all $n \in \mathbb{N}$. $\square$

Example 18 — well-ordering implies induction

Book level

Assume the well-ordering principle. Let $p(n)$ be a statement with $p(0)$ true and $p(n) \Rightarrow p(n+1)$ for all $n \in \mathbb{N}$. Prove that $p(n)$ holds for all $n \in \mathbb{N}$.

Invention: look at the least counterexample.

Suppose, for a contradiction, that $S = \{n \in \mathbb{N} \mid \neg p(n)\}$ is inhabited. By well-ordering S has a least element m . Since $p(0)$ holds, $m \neq 0$, so $m-1 \in \mathbb{N}$. As $m-1 < m$ and m is least in S , $m-1 \notin S$, i.e. $p(m-1)$ holds. By the induction step $p(m)$ holds, contradicting $m \in S$. Hence $S = \emptyset$. $\square$

Example 19 — infinite descent

Book level

Use the well-ordering principle to prove that there are no integers $a, b \geq 1$ with $a^2 = 2b^2$.

Invention: among all solutions pick one with the smallest b , then build a smaller one.

Suppose solutions exist, and let $S = \{b \geq 1 \mid \exists a \geq 1, a^2 = 2b^2\}$, an inhabited subset of $\mathbb{N}$. Let b be its least element and fix $a \geq 1$ with $a^2 = 2b^2$. Then a^2 is even, so a is even by the Lemma in Chapter 0; write $a = 2c$ with $c \geq 1$. Then $4c^2 = 2b^2$, so $b^2 = 2c^2$. Thus (b, c) is again a solution, so $c \in S$. But $c^2 = b^2/2 < b^2$ gives $c < b$, contradicting the minimality of b . $\square$

This is the “infinite descent” of the book’s title, and a second proof that $\sqrt{2}$ is irrational.

D STRETCH EXERCISES**STRETCH EXERCISES 4 · solutions in Appendix S**

S4.1 Prove that $\sum_{k=1}^n k \cdot k! = (n+1)! - 1$ for all $n \in \mathbb{N}$.

S4.2 Prove that $6 \mid n^3 - n$ for all $n \in \mathbb{N}$. [Hint: $(n+1)^3 - (n+1) = (n^3 - n) + 3n(n+1)$]

S4.3 Define $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ for $0 \leq k \leq n$ and $\binom{n}{k} = 0$ for $k > n$. Prove Pascal's rule $\binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1}$ from this formula, and then prove by induction on n that $\binom{n}{k} \in \mathbb{N}$ for all k .

S4.4 Prove that for every $n \geq 1$, a $2^n \times 2^n$ board with one square removed can be tiled by L-shaped pieces of three squares. [Hint: split into four quadrants and place one piece at the centre]

S4.5 Prove that for all $n \in \mathbb{N}$ there is no integer $d \geq 2$ dividing both f_n and f_{n+1} . [Hint: if $d \mid f_{n+1}$ and $d \mid f_{n+2}$ then $d \mid f_n$]

S4.6 Prove that $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} \geq \sqrt{n}$ for all $n \geq 1$. [Hint: the gap inequality is $\sqrt{n} + \frac{1}{\sqrt{n+1}} \geq \sqrt{n+1}$]

S4.7 Prove that every $n \geq 1$ is a sum of distinct powers of 2. [Hint: strong induction; subtract the largest power of 2 not exceeding n]

S4.8 Prove that $\sum_{k=1}^n f_k = f_{n+2} - 1$ for all $n \geq 1$.

S4.9 Prove that every $n \geq 8$ can be written as $3a + 5b$ with $a, b \in \mathbb{N}$. [Hint: how many base cases does a step of size 3 need?]

E REVIEW SET 4

- Weak: $p(b)$ and $\forall n \geq b, p(n) \Rightarrow p(n+1)$.
- Strong: $\forall n \geq b, (\forall k, b \leq k < n \Rightarrow p(k)) \Rightarrow p(n)$; base cases as many as the step needs.
- Template words: “Let $p(n)$ be”, “Base case”, “Induction step”, “(IH)”, “By induction”.
- Sums: $\sum_1^{n+1} = \sum_1^n + a_{n+1}$. Divisibility: $x_{n+1} = c \cdot x_n +$ (multiple). Inequalities: IH then a gap inequality.
- Binomial theorem $(x+y)^n = \sum \binom{n}{k} x^k y^{n-k}$; Pascal's rule.
- Well-ordering: inhabited $S \subseteq \mathbb{N}$ has a least element; minimal counterexample.

Chapter 5

Relations

Contents

- A** Relations and their properties (§5.1)
- B** Equivalence relations and partitions (§5.2)
- C** Book-level examples
- D** Stretch exercises
- E** Review set 5

Planning

Essential: 5.1: 6, 13, 22, 23, 28, 39, 40, 41; 5.2: 24, 27, 33; 5.E: 18, 22, 23
Recommended: 5.1: 10, 11, 18, 24, 29, 33, 34; 5.2: 25, 28, 30, 34, 36
 Tested in **Quiz 3** (21 Oct) together with modular arithmetic (§7.3).

A RELATIONS AND THEIR PROPERTIES

Relation

A **relation** R from a set X to a set Y is a statement $a R b$ about pairs (a, b) with $a \in X, b \in Y$; equivalently, it is its **graph** $\text{Gr}(R) = \{(a, b) \in X \times Y \mid a R b\} \subseteq X \times Y$. A relation *on* X is a relation from X to X . Every subset of $X \times Y$ is the graph of exactly one relation, so a set with n elements has 2^{n^2} relations on it. Examples on $\mathbb{Z}$: $=, \leq, <, |, "a \equiv b \pmod{n}"$ (i.e. $n \mid a - b$), “ a and b have the same parity”. On $\mathcal{P}(X)$: $\subseteq$. Between X and Y : “ $f(a) = b$ ” for a function $f : X \rightarrow Y$ (its graph is $\text{Gr}(f)$).

Properties of a relation R on X

R is ...	definition	to disprove, exhibit ...
reflexive	$\forall a \in X, a R a$	one a with $\neg(a R a)$
symmetric	$\forall a, b \in X, (a R b \Rightarrow b R a)$	a, b with $a R b$ and $\neg(b R a)$
transitive	$\forall a, b, c \in X, (a R b \wedge b R c \Rightarrow a R c)$	a, b, c with $a R b, b R c, \neg(a R c)$
antisymmetric	$\forall a, b \in X, (a R b \wedge b R a \Rightarrow a = b)$	$a \neq b$ with $a R b$ and $b R a$

relation	reflexive	symmetric	transitive	antisymmetric
$=$ on any set	✓	✓	✓	✓
$\leq$ on $\mathbb{R}$; $\subseteq$ on $\mathcal{P}(X)$; $ $ on $\mathbb{N}$	✓	✗	✓	✓
$<$ on $\mathbb{R}$	✗	✗	✓	✓ (vacuously)
$\equiv \pmod{n}$ on $\mathbb{Z}$	✓	✓	✓	✗ ($n \geq 2$)
$\neq$ on a set with ≥ 2 elements	✗	✓	✗	✗

Proving properties

Each property is a $\forall$ -statement, usually with an implication inside; write exactly that.

- Reflexive: “Let $a \in X$. Then ..., so $a R a$.”
- Symmetric: “Let $a, b \in X$ and suppose $a R b$ Hence $b R a$.”
- Transitive: “Let $a, b, c \in X$ and suppose $a R b$ and $b R c$ Hence $a R c$.”

- Antisymmetric: “Let $a, b \in X$ and suppose $a R b$ and $b R a$ Hence $a = b$.”
Unfold “ $a R b$ ” into its definition on the first line (“so $3 \mid a - b$; fix k with $a - b = 3k$ ”).

Example 1 — congruence modulo 3

Self Tutor

On $\mathbb{Z}$ define $a \sim b$ to mean $3 \mid a - b$. Prove that $\sim$ is reflexive, symmetric and transitive, and show that it is not antisymmetric.

Reflexive. Let $a \in \mathbb{Z}$. Then $a - a = 0 = 3 \cdot 0$, so $3 \mid a - a$, i.e. $a \sim a$.

Symmetric. Let $a, b \in \mathbb{Z}$ with $a \sim b$; fix $k \in \mathbb{Z}$ with $a - b = 3k$. Then $b - a = 3(-k)$, so $3 \mid b - a$, i.e. $b \sim a$.

Transitive. Let $a, b, c \in \mathbb{Z}$ with $a \sim b$ and $b \sim c$; fix $k, l \in \mathbb{Z}$ with $a - b = 3k$ and $b - c = 3l$. Then $a - c = (a - b) + (b - c) = 3(k + l)$, so $a \sim c$.

Not antisymmetric. $0 \sim 3$ and $3 \sim 0$ (both differences are ± 3), but $0 \neq 3$. □

Example 2 — divisibility is antisymmetric on $\mathbb{N}$ but not on $\mathbb{Z}$

Self Tutor

Prove that $\mid$ is antisymmetric on $\mathbb{N}$. Show that it is not antisymmetric on $\mathbb{Z}$.

Let $a, b \in \mathbb{N}$ with $a \mid b$ and $b \mid a$; fix $k, l \in \mathbb{N}$ with $b = ka$ and $a = lb$ (the quotients are ≥ 0 since $a, b \geq 0$). Then $a = lka$. If $a = 0$ then $b = k \cdot 0 = 0 = a$. If $a \neq 0$ then $lk = 1$, and since $k, l \in \mathbb{N}$ this forces $k = l = 1$, so $a = b$.

On $\mathbb{Z}$: $2 \mid -2$ and $-2 \mid 2$ but $2 \neq -2$. □

Example 3 — a relation with some properties and not others

Self Tutor

On $\mathbb{Z}$ define $a R b$ to mean that $a + b$ is odd. Decide whether R is reflexive, symmetric, transitive.

Not reflexive: $0 + 0 = 0$ is even, so $\neg(0 R 0)$.

Symmetric: let $a, b \in \mathbb{Z}$ with $a R b$. Then $b + a = a + b$ is odd, so $b R a$.

Not transitive: $1 R 2$ and $2 R 3$ (sums 3 and 5), but $1 + 3 = 4$ is even, so $\neg(1 R 3)$. □

Relations that are functions

A relation R from X to Y is (the graph of) a function $X \rightarrow Y$ iff $\forall a \in X, \exists! b \in Y, a R b$: every input is related to exactly one output. Then $f(a)$ is that b .

Example 4 — is it a function?

Self Tutor

Which of the relations $G_1 = \{(x, y) \in \mathbb{R}^2 \mid y = x^2\}$ and $G_2 = \{(x, y) \in \mathbb{R}^2 \mid x = y^2\}$ is the graph of a function $\mathbb{R} \rightarrow \mathbb{R}$?

G_1 : for each $x \in \mathbb{R}$ there is exactly one y with $y = x^2$. So G_1 is the graph of $x \mapsto x^2$.

G_2 : for $x = 1$ both $(1, 1)$ and $(1, -1)$ lie in G_2 (not unique), and for $x = -1$ there is no y with $y^2 = -1$ (not existent). So G_2 is not the graph of a function. □

Watch out

- Reflexivity is about *all* elements. “Symmetric and transitive imply reflexive” is false: the empty relation on $\{1\}$ is symmetric and transitive but not reflexive (the argument “ $a R b$, so $b R a$, so $a R a$ ” needs some b to exist).

- Antisymmetric is not the negation of symmetric: $=$ is both; $\neq$ is neither.
- To refute a property one concrete counterexample is enough, but you must check it: state the pairs and why they are (not) related.

EXERCISES 5.1

Essential: 6, 13, 22, 23, 28, 39, 40, 41 **Recommended:** 10, 11, 18, 24, 29, 33, 34

Skills: writing a relation as a set of pairs and back; deciding and proving/refuting reflexive, symmetric, transitive, antisymmetric for concrete relations on $\mathbb{Z}$, $\mathbb{R}$, $\mathcal{P}(X)$, finite sets; relations as functions (the $\forall\exists!$ test); counting relations.

B EQUIVALENCE RELATIONS AND PARTITIONS**Equivalence relation, class, quotient**

An **equivalence relation** on X is a relation $\sim$ that is reflexive, symmetric and transitive. For $a \in X$, the **equivalence class** of a is

$$[a]_{\sim} = \{b \in X \mid a \sim b\} \subseteq X,$$

and the **quotient** of X by $\sim$ is the set of all classes, $X/\sim = \{[a]_{\sim} \mid a \in X\} \subseteq \mathcal{P}(X)$. Membership rule: $b \in [a] \Leftrightarrow a \sim b$.

Lemma: classes are equal or disjoint

Let $\sim$ be an equivalence relation on X and $a, b \in X$. Then

$$a \sim b \Leftrightarrow [a] = [b] \Leftrightarrow [a] \cap [b] \text{ is inhabited.}$$

Moreover $a \in [a]$ for every a , so every class is inhabited and every element lies in some class.

Example 5 — proof of the Lemma**Self Tutor**

Prove the Lemma.

$(a \sim b \Rightarrow [a] = [b])$ Suppose $a \sim b$. Let $c \in [a]$, i.e. $a \sim c$. By symmetry $b \sim a$, and by transitivity $(b \sim a, a \sim c) \Rightarrow b \sim c$, i.e. $c \in [b]$. So $[a] \subseteq [b]$; by symmetry of the situation $(b \sim a)$ also $[b] \subseteq [a]$.

$([a] = [b] \Rightarrow [a] \cap [b] \text{ inhabited})$ $a \in [a]$ by reflexivity, and $[a] = [b]$, so $a \in [a] \cap [b]$.

$([a] \cap [b] \text{ inhabited} \Rightarrow a \sim b)$ Fix $c \in [a] \cap [b]$: $a \sim c$ and $b \sim c$. By symmetry $c \sim b$; by transitivity $a \sim b$.

□

Partition

A **partition** of X is a set $\mathcal{U}$ of subsets of X such that (i) every $U \in \mathcal{U}$ is inhabited; (ii) distinct members are disjoint: $U, V \in \mathcal{U}, U \neq V \Rightarrow U \cap V = \emptyset$; (iii) they cover X : $\bigcup_{U \in \mathcal{U}} U = X$ (every $a \in X$ lies in some $U \in \mathcal{U}$).

Equivalently: every $a \in X$ lies in *exactly one* member of $\mathcal{U}$.

Fundamental theorem: equivalence relations = partitions

1 If $\sim$ is an equivalence relation on X , then $X/\sim$ is a partition of X (by the Lemma).

2 If $\mathcal{U}$ is a partition of X , then $a \sim_{\mathcal{U}} b \Leftrightarrow \exists U \in \mathcal{U}, (a \in U \wedge b \in U)$ defines an equivalence relation

on X whose classes are exactly the members of $\mathcal{U}$.

3 These constructions are inverse to each other: equivalence relations on X correspond bijectively to partitions of X .

Consequence for computations: to describe $X/\sim$, find one representative per class and prove (a) every element is equivalent to one of the representatives, (b) distinct representatives are not equivalent.

Example 6 — $\mathbb{Z}/n\mathbb{Z}$ has exactly n elements

Self Tutor

Let $n \geq 1$ and let $a \equiv b \pmod{n}$ mean $n \mid a - b$ (an equivalence relation, as in Example 5.1). Write $[a]_n$ for the class of a and $\mathbb{Z}/n\mathbb{Z}$ for the quotient. Prove that $\mathbb{Z}/n\mathbb{Z} = \{[0]_n, [1]_n, \dots, [n-1]_n\}$ and that these n classes are distinct.

(a) Let $a \in \mathbb{Z}$. By the division theorem $a = qn + r$ with $0 \leq r < n$. Then $a - r = qn$, so $a \equiv r$, so $[a]_n = [r]_n$ by the Lemma, with $r \in \{0, \dots, n-1\}$.

(b) Let $0 \leq r < s \leq n-1$. Then $0 < s - r < n$, so $n \nmid s - r$ (the only multiple of n in $(0, n)$ would be $\geq n$). Hence $r \not\equiv s$ and $[r]_n \neq [s]_n$ by the Lemma.

So $\mathbb{Z}/n\mathbb{Z}$ consists of exactly the n distinct classes $[0]_n, \dots, [n-1]_n$; the class of a is the class of its remainder. $\square$

Example 7 — describing the classes

Self Tutor

On $\mathbb{R}$ define $x \sim y$ iff $x - y \in \mathbb{Z}$. Show that $\sim$ is an equivalence relation, describe $[x]$, and show that every class contains exactly one element of $[0, 1)$.

Reflexive: $x - x = 0 \in \mathbb{Z}$. *Symmetric:* if $x - y \in \mathbb{Z}$ then $y - x = -(x - y) \in \mathbb{Z}$. *Transitive:* if $x - y, y - z \in \mathbb{Z}$ then $x - z = (x - y) + (y - z) \in \mathbb{Z}$.

$[x] = \{y \in \mathbb{R} \mid x - y \in \mathbb{Z}\} = \{x + k \mid k \in \mathbb{Z}\}$.

Exactly one representative in $[0, 1)$: existence: $r = x - \lfloor x \rfloor \in [0, 1)$ and $x - r = \lfloor x \rfloor \in \mathbb{Z}$, so $r \in [x]$.

Uniqueness: if $r, r' \in [x] \cap [0, 1)$ then $r - r' \in \mathbb{Z}$ (transitivity/symmetry) and $|r - r'| < 1$, so $r = r'$. $\square$

Example 8 — fractions: transitivity needs care

Self Tutor

On $X = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ define $(a, b) \sim (c, d)$ iff $ad = bc$. Prove that $\sim$ is transitive. (Its classes are the rational numbers: $[(a, b)]$ “is” $\frac{a}{b}$.)

Let $(a, b), (c, d), (e, f) \in X$ with $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$, i.e. $ad = bc$ and $cf = de$. We need $af = be$. Multiply the first equation by f and use the second:

$$adf = bcf = b(cf) = bde, \quad \text{so} \quad d(af - be) = 0.$$

Since $d \neq 0$ (second coordinates are non-zero), $af - be = 0$, i.e. $af = be$. Hence $(a, b) \sim (e, f)$. $\square$

Example 9 — from partition to relation and back

Self Tutor

Let $X = \{1, 2, 3, 4, 5\}$ and $\mathcal{U} = \{\{1, 3\}, \{2\}, \{4, 5\}\}$. Write the corresponding equivalence relation as a set of pairs. How many equivalence relations are there on $\{1, 2, 3\}$?

$a \sim b$ iff a, b lie in the same member: $\text{Gr}(\sim) = \{(1, 1), (3, 3), (1, 3), (3, 1), (2, 2), (4, 4), (5, 5), (4, 5), (5, 4)\}$ (nine pairs: $2^2 + 1^2 + 2^2$).

Equivalence relations on $\{1, 2, 3\}$ correspond to partitions: $\{\{1, 2, 3\}\}; \{\{1\}, \{2\}, \{3\}\}$; and three of the form $\{\{a, b\}, \{c\}\}$. Total: 5. $\square$

The quotient function

For an equivalence relation $\sim$ on X , the **quotient function** is $q_\sim : X \rightarrow X/\sim$, $q_\sim(a) = [a]_\sim$. It is surjective (every class is $[a]$ for some a), and $q_\sim(a) = q_\sim(b) \Leftrightarrow a \sim b$ (Lemma).

For $\equiv \pmod{n}$: $q_n : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ sends a to $[r]_n$ where r is the remainder of a on division by n (Example 5.6); so $q_n(a) = q_n(b)$ iff a and b have the same remainder.

Theorem 5.2.35: every surjection is a quotient

Let $p : X \rightarrow Z$ be a surjection. Define $a \sim b \Leftrightarrow p(a) = p(b)$. Then $\sim$ is an equivalence relation on X , and $f : X/\sim \rightarrow Z$, $f([a]) = p(a)$, is a well-defined bijection with $f \circ q_\sim = p$. The relation $\sim$ is the only equivalence relation on X for which such a bijection exists.

Example 10 — the induced bijection (proof of the core of Theorem 5.2.35)**Self Tutor**

Let $p : X \rightarrow Z$ be surjective and $a \sim b \Leftrightarrow p(a) = p(b)$. Prove that $\sim$ is an equivalence relation and that $f([a]) = p(a)$ defines a bijection $X/\sim \rightarrow Z$.

Equivalence relation. Reflexive: $p(a) = p(a)$. Symmetric: $p(a) = p(b) \Rightarrow p(b) = p(a)$. Transitive: $p(a) = p(b)$ and $p(b) = p(c)$ give $p(a) = p(c)$.

Well-defined. If $[a] = [b]$ then $a \sim b$ (Lemma), i.e. $p(a) = p(b)$: the proposed value does not depend on the representative.

Injective. Suppose $f([a]) = f([b])$, i.e. $p(a) = p(b)$. Then $a \sim b$, so $[a] = [b]$ by the Lemma.

Surjective. Let $z \in Z$. Since p is surjective, fix $a \in X$ with $p(a) = z$; then $f([a]) = z$. □

Defining a function on a quotient: well-definedness

To define $F : X/\sim \rightarrow Y$ by a formula $F([a]) = h(a)$ you must prove: $\forall a, b \in X, (a \sim b \Rightarrow h(a) = h(b))$. Otherwise the “function” assigns two values to one class. This check is mandatory in every exercise that defines something on $\mathbb{Z}/n\mathbb{Z}$, $X/\sim$, fractions, etc.

Example 11 — well-defined and not well-defined on $\mathbb{Z}/n\mathbb{Z}$ **Self Tutor**

(a) Prove that $[a]_n + [b]_n := [a + b]_n$ is a well-defined operation on $\mathbb{Z}/n\mathbb{Z}$. (b) Show that $F : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}$, $F([a]_n) = a$ is not well-defined (for $n \geq 1$).

(a) Suppose $[a]_n = [a']_n$ and $[b]_n = [b']_n$, i.e. $n \mid a - a'$ and $n \mid b - b'$; fix k, l with $a - a' = nk$, $b - b' = nl$. Then $(a + b) - (a' + b') = n(k + l)$, so $n \mid (a + b) - (a' + b')$, i.e. $[a + b]_n = [a' + b']_n$. The result does not depend on the representatives. □

(b) $[0]_n = [n]_n$ (since $n \mid n - 0$), but the rule gives $F([0]_n) = 0$ and $F([n]_n) = n \neq 0$. So F is not well-defined. □

Watch out

- $[a] = [b]$ does not mean $a = b$; it means $a \sim b$. An equivalence class is a *set*; $X/\sim$ is a set of sets.
- When you define anything on a quotient, write the well-definedness check before anything else. “Choose a representative” must be followed by “and the result is independent of the choice because ...”.
- To prove a relation is an equivalence relation you need all three properties, each with its own “Let ...” line. Reflexivity is the one people forget.
- Counting classes: existence of a representative for every element *and* distinctness of the representatives, both proved.

EXERCISES 5.2 and 5.E**Essential:** 5.2: 24, 27, 33; 5.E: 18, 22, 23 **Recommended:** 5.2: 25, 28, 30, 34, 36

Skills: proving a relation is an equivalence relation; computing and describing equivalence classes and quotients ($\mathbb{Z}/n\mathbb{Z}$, $\mathbb{R}/\mathbb{Z}$ -type examples, fractions, functions grouped by a value); converting between partitions and equivalence relations; the quotient function and Theorem 5.2.35 (surjections $\leftrightarrow$ quotients, Exercise 5.2.33 on remainders); proving well-definedness of functions and operations on quotients; using Lemma 5.2.29-type bijections class by class (Exercise 5.2.30).

C BOOK-LEVEL EXAMPLES

The hardest Chapter 5 exercises ask you to count, to combine relations, to recognise a quotient as the kernel of a function, or to build a function out of a quotient and prove it is well-defined. The exercise from the book page you have (5.2.30) is worked in full below.

Example 12 — counting relations**Book level**

Let X be a set with n elements. How many relations on X are there? How many of them are reflexive? How many are symmetric? Justify your answers.

Invention: a relation is a subset of $X \times X$, so count subsets with constraints.

$X \times X$ has n^2 elements, so it has 2^{n^2} subsets; every subset is the graph of exactly one relation, so there are 2^{n^2} relations.

Reflexive: the n pairs (a, a) must be in the graph; the other $n^2 - n$ pairs are free. So 2^{n^2-n} .

Symmetric: the graph is determined by which of the n pairs (a, a) it contains and which of the $\binom{n}{2} = \frac{n(n-1)}{2}$ unordered pairs $\{a, b\}$ with $a \neq b$ it contains (it contains (a, b) iff it contains (b, a)). So $2^{n + n(n-1)/2} = 2^{n(n+1)/2}$. $\square$

Example 13 — combining equivalence relations**Book level**

Let $\sim$ and $\approx$ be equivalence relations on X . Define $a R b$ to mean $a \sim b$ and $a \approx b$. Prove that R is an equivalence relation. Show by example that “ $a \sim b$ or $a \approx b$ ” need not be one.

Reflexive: $a \sim a$ and $a \approx a$, so $a R a$. *Symmetric:* if $a \sim b$ and $a \approx b$, then $b \sim a$ and $b \approx a$. *Transitive:* if $a R b$ and $b R c$, then $a \sim b$, $b \sim c$ give $a \sim c$, and $a \approx b$, $b \approx c$ give $a \approx c$; so $a R c$. $\square$

The “or” relation can fail transitivity. On $X = \{1, 2, 3\}$ let $\sim$ have classes $\{1, 2\}, \{3\}$ and $\approx$ have classes $\{1\}, \{2, 3\}$. Then $1 \sim 2$ and $2 \approx 3$, so 1 and 2 are related, and so are 2 and 3; but $1 \not\sim 3$ and $1 \not\approx 3$, so 1 and 3 are not.

Example 14 — a quotient of the plane**Book level**

On $\mathbb{R}^2$ define $(x, y) \sim (x', y')$ iff $x^2 + y^2 = x'^2 + y'^2$. Prove that $\sim$ is an equivalence relation, describe its classes, and show that $\mathbb{R}^2/\sim$ is in bijection with $[0, \infty)$.

Invention: recognise $\sim$ as “same value under $p(x, y) = x^2 + y^2$ ” and use Theorem 5.2.35.

Let $p: \mathbb{R}^2 \rightarrow [0, \infty)$, $p(x, y) = x^2 + y^2$. Then $(x, y) \sim (x', y')$ iff $p(x, y) = p(x', y')$, which is an equivalence relation by the first part of Example 5.10.

The class of (x, y) is $\{(x', y') \mid x'^2 + y'^2 = r^2\}$ with $r = \sqrt{x^2 + y^2}$: the circle of radius r about the origin, or $\{(0, 0)\}$ when $r = 0$.

p is surjective: for $t \geq 0$, $p(\sqrt{t}, 0) = t$. By Theorem 5.2.35, $[(x, y)] \mapsto x^2 + y^2$ is a well-defined bijection $\mathbb{R}^2/\sim \rightarrow [0, \infty)$. $\square$

Example 15 — pulling back an equivalence relation

Book level

Let $f : X \rightarrow Y$ be a function and $\sim$ an equivalence relation on Y . Define $a \approx b$ iff $f(a) \sim f(b)$. Prove that $\approx$ is an equivalence relation on X .

Reflexive: $f(a) \sim f(a)$ by reflexivity of $\sim$. *Symmetric:* if $f(a) \sim f(b)$ then $f(b) \sim f(a)$. *Transitive:* if $f(a) \sim f(b)$ and $f(b) \sim f(c)$ then $f(a) \sim f(c)$. $\square$

With $\sim$ equal to $=$ on Y this is the kernel relation of Theorem 5.2.35; the previous example is the case $f = p$.

Example 16 — Exercise 5.2.30: gluing bijections class by class

Book level

Let $\sim$ and $\approx$ be equivalence relations on X and Y . Suppose $p : X/\sim \rightarrow Y/\approx$ is a bijection, and that for each class $E \in X/\sim$ a bijection $h_E : E \rightarrow p(E)$ is given. Prove that there is a bijection $h : X \rightarrow Y$.

Invention: define h piecewise, $h(a) = h_{[a]}(a)$, and use that distinct classes are disjoint.

Define $h : X \rightarrow Y$ by $h(a) = h_{[a]}(a)$. This makes sense: $a \in [a]$, $h_{[a]}$ is a function $[a] \rightarrow p([a])$, and $p([a]) \subseteq Y$.

Injective. Let $a, b \in X$ with $h(a) = h(b)$, i.e. $h_{[a]}(a) = h_{[b]}(b)$; call this element y . Then $y \in p([a])$ and $y \in p([b])$, so these two classes of $\approx$ intersect, hence are equal (Lemma). Since p is injective, $[a] = [b]$. Now $h_{[a]}(a) = h_{[a]}(b)$ with $h_{[a]}$ injective, so $a = b$.

Surjective. Let $y \in Y$ and put $F = [y]_{\approx} \in Y/\approx$. Since p is surjective, fix $E \in X/\sim$ with $p(E) = F$. Since $h_E : E \rightarrow F$ is surjective and $y \in F$, fix $a \in E$ with $h_E(a) = y$. As $a \in E$ and E is a class, $E = [a]$; so $h(a) = h_{[a]}(a) = h_E(a) = y$. $\square$

Example 17 — well-defined or not, with a twist

Book level

Let $n \geq 1$. (a) Prove that $[a]_n \cdot [b]_n := [ab]_n$ is a well-defined operation on $\mathbb{Z}/n\mathbb{Z}$. (b) Deduce that $S([a]_n) = [a^2]_n$ is a well-defined function $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$. (c) Show that $T([a]_3) = [2^a]_3$ is *not* well-defined on $\mathbb{Z}/3\mathbb{Z}$.

Invention: write the difference of the two candidate outputs as a multiple of n .

(a) Suppose $[a] = [a']$ and $[b] = [b']$, so $a - a' = nk$ and $b - b' = nl$ for some $k, l \in \mathbb{Z}$. Then $ab - a'b' = a(b - b') + b'(a - a') = n(al + b'k)$, so $[ab] = [a'b']$.

(b) Take $b = a$ and $b' = a'$ in (a): if $[a] = [a']$ then $[a^2] = [a'^2]$.

(c) $[0]_3 = [3]_3$, but $2^0 = 1$ and $2^3 = 8$, and $8 - 1 = 7$ is not a multiple of 3, so $[2^0]_3 \neq [2^3]_3$. The rule assigns two different values to one class. $\square$

Example 18 — functions out of a quotient

Book level

Let $\sim$ be an equivalence relation on X with quotient function $q : X \rightarrow X/\sim$, and let Y be a set. Let C be the set of functions $h : X \rightarrow Y$ such that $a \sim b \Rightarrow h(a) = h(b)$ for all $a, b \in X$. Prove that

$$\Phi : \{\text{functions } X/\sim \rightarrow Y\} \rightarrow C, \quad \Phi(g) = g \circ q,$$

is a bijection.

Invention: injectivity is cancellation by the surjection q ; surjectivity is the well-definedness check.

First, $\Phi(g) \in C$: if $a \sim b$ then $[a] = [b]$, so $g(q(a)) = g([a]) = g([b]) = g(q(b))$.

Injective. Suppose $g \circ q = g' \circ q$. Let $E \in X/\sim$; since q is surjective fix a with $q(a) = E$. Then $g(E) = g(q(a)) = g'(q(a)) = g'(E)$. So $g = g'$.

Surjective. Let $h \in C$. Define $g : X/\sim \rightarrow Y$ by $g([a]) = h(a)$; this is well-defined because $[a] = [b]$ implies $a \sim b$, which implies $h(a) = h(b)$. Then $(g \circ q)(a) = g([a]) = h(a)$ for all a , so $\Phi(g) = h$. $\square$

D STRETCH EXERCISES

STRETCH EXERCISES 5 · solutions in Appendix S

- S5.1** On $\mathbb{Z}$ define $a R b$ iff $a \mid b$. Decide, with proof or counterexample, which of reflexive, symmetric, transitive, antisymmetric hold.
- S5.2** Let R be a symmetric and transitive relation on X such that for every $a \in X$ there is some $b \in X$ with $a R b$. Prove that R is reflexive.
- S5.3** On $\mathbb{R}$ define $x \sim y$ iff $x - y \in \mathbb{Q}$. Prove that $\sim$ is an equivalence relation, show that $[0] = \mathbb{Q}$ and $[\sqrt{2}] = \{\sqrt{2} + r \mid r \in \mathbb{Q}\}$, and prove that $[\sqrt{2}] \neq [\sqrt{3}]$. [Hint: $(\sqrt{3} - \sqrt{2})(\sqrt{3} + \sqrt{2}) = 1$]
- S5.4** How many equivalence relations are there on a set with 4 elements? [Hint: count partitions by shape]
- S5.5** On $\mathcal{P}(\mathbb{N})$ define $U \sim V$ iff $U \Delta V$ is finite. Prove that $\sim$ is an equivalence relation. [Hint: $U \Delta W \subseteq (U \Delta V) \cup (V \Delta W)$]
- S5.6** Let $\sim$ be an equivalence relation on X and $f : X \rightarrow Y$ a function with $a \sim b \Rightarrow f(a) = f(b)$. Let $g : X/\sim \rightarrow Y$ be the induced function $g([a]) = f(a)$. Prove that $g[X/\sim] = f[X]$ and that g is injective if and only if $f(a) = f(b) \Rightarrow a \sim b$ for all a, b .
- S5.7** Let $m, n \geq 1$. Prove that $[a]_n \mapsto [a]_m$ is a well-defined function $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$ if and only if $m \mid n$.
- S5.8** Prove that a relation R on X is an equivalence relation if and only if R is reflexive and satisfies: for all $a, b, c \in X$, if $a R b$ and $a R c$ then $b R c$.

E REVIEW SET 5

- Relation on X = subset of $X \times X$; 2^{n^2} of them.
- Reflexive $\forall a a R a$; symmetric $a R b \Rightarrow b R a$; transitive $a R b \wedge b R c \Rightarrow a R c$; antisymmetric $a R b \wedge b R a \Rightarrow a = b$.
- Function $\Leftrightarrow \forall a \exists! b, a R b$.
- Equivalence relation = R+S+T; $[a] = \{b \mid a \sim b\}$; $X/\sim = \{[a]\}$.
- $a \sim b \Leftrightarrow [a] = [b] \Leftrightarrow [a] \cap [b] \neq \emptyset$.
- Partition: inhabited, pairwise disjoint, cover. Equivalence relations $\leftrightarrow$ partitions.
- $\mathbb{Z}/n\mathbb{Z} = \{[0], \dots, [n-1]\}$, class = remainder.
- $q_\sim : X \rightarrow X/\sim$ surjective; every surjection p is $q_\sim$ up to the bijection $[a] \mapsto p(a)$, with $a \sim b \Leftrightarrow p(a) = p(b)$.
- Defining $F([a]) = h(a)$ needs $a \sim b \Rightarrow h(a) = h(b)$.

Appendix S

Solutions to the NOW YOU exercises

Contents

- A Chapter 0
- B Chapter 1
- C Chapter 2

Planning

Every book exercise in the covered pages that is not worked in the main text is solved here, in the order of the book. Compare line by line: did you introduce every variable, unfold every definition, and name the strategy?

A CHAPTER 0

0.2 (a) $2 + 2 = 4$. (b) $2 + 2 = 5$. (c) “There are infinitely many pairs of primes that differ by 2” (the twin prime conjecture; nobody knows its truth value). Any proposition whose truth value is unknown is fine, e.g. $\pi^\pi \in \mathbb{Q}$ (Exercise 0.21(f)).

0.47 Repeated division by 17: $408735787 = 24043281 \cdot 17 + 10$, $24043281 = 1414310 \cdot 17 + 11$, $1414310 = 83194 \cdot 17 + 12$, $83194 = 4893 \cdot 17 + 13$, $4893 = 287 \cdot 17 + 14$, $287 = 16 \cdot 17 + 15$, $16 = 0 \cdot 17 + 16$. Reading the remainders upwards, with $A = 10, \dots, G = 16$: $408735787 = GFEDCBA_{(17)}$. Likewise $1442151747 = 40059770 \cdot 36 + 27$, $40059770 = 1112771 \cdot 36 + 14$, $1112771 = 30910 \cdot 36 + 11$, $30910 = 858 \cdot 36 + 22$, $858 = 23 \cdot 36 + 30$, $23 = 0 \cdot 36 + 23$, and $27 = R$, $14 = E$, $11 = B$, $22 = M$, $30 = U$, $23 = N$: $1442151747 = NUMBER_{(36)}$.

0.55 $s(w) = \sqrt{2}w^4 - 81\sqrt{2}$ has coefficients $\sqrt{2}, -81\sqrt{2} \notin \mathbb{Q}$ (so it is not a polynomial over $\mathbb{Q}$), is quartic, and $s(3) = \sqrt{2}(81 - 81) = 0 \in \mathbb{Q}$.

0.64 Since $i^2 = -1$, $i^3 = -i$ and $i^4 = 1$. For $n \in \mathbb{Z}$ write $n = 4q + r$ with $r \in \{0, 1, 2, 3\}$ (Theorem 0.42); then $i^n = (i^4)^q i^r = i^r$. So $i^n = 1$ if $r = 0$; $i^n = 0 + 1 \cdot i$ if $r = 1$; $i^n = -1$ if $r = 2$; $i^n = 0 + (-1)i$ if $r = 3$. (For negative n this uses $i^{-1} = -i$, which is consistent with the formula.)

0.65 Let $p(x) = ax^2 + bx + c$ with $a \neq 0$ and $\Delta < 0$. By Definition 0.62, $\sqrt{\Delta} = \sqrt{|\Delta|}i$, so $(\sqrt{\Delta})^2 = |\Delta|i^2 = -|\Delta| = \Delta$. The computation in the proof of Theorem 0.60 used only $(\sqrt{\Delta})^2 = \Delta$ and the completed square $p(x) = a(x + \frac{b}{2a})^2 - \frac{\Delta}{4a}$, so $p(\alpha) = p(\beta) = 0$ holds verbatim. For “no other roots”: if $p(\gamma) = 0$ then $(\gamma + \frac{b}{2a})^2 = \frac{\Delta}{4a^2} = (\frac{\sqrt{\Delta}}{2a})^2$, so $(\gamma + \frac{b}{2a} - \frac{\sqrt{\Delta}}{2a})(\gamma + \frac{b}{2a} + \frac{\sqrt{\Delta}}{2a}) = 0$, and a product of complex numbers is 0 only if a factor is 0; hence $\gamma = \alpha$ or $\gamma = \beta$. The cases $\Delta \geq 0$ are Theorem 0.60.

0.66 Let $\alpha = a + bi$ and put $p(x) = x^2 - 2ax + (a^2 + b^2)$, a quadratic over $\mathbb{R}$. Then $p(\alpha) = (a + bi)^2 - 2a(a + bi) + a^2 + b^2 = a^2 + 2abi - b^2 - 2a^2 - 2abi + a^2 + b^2 = 0$. The other root is $a - bi$: $p(a - bi) = a^2 - 2abi - b^2 - 2a^2 + 2abi + a^2 + b^2 = 0$. (Indeed $p(x) = (x - (a + bi))(x - (a - bi))$.)

0.67 Quadratic formula over $\mathbb{C}$. Let $p(x) = ax^2 + bx + c$ with $a, b, c \in \mathbb{C}$, $a \neq 0$, and let $\delta \in \mathbb{C}$ be any number with $\delta^2 = b^2 - 4ac$ (every complex number has a square root; we take this for granted). Then the roots of $p(x)$ are precisely $\frac{-b+\delta}{2a}$ and $\frac{-b-\delta}{2a}$. *Proof:* completing the square, $p(x) = a(x + \frac{b}{2a})^2 - \frac{\delta^2}{4a}$; substituting the two values gives $a \cdot \frac{\delta^2}{4a^2} - \frac{\delta^2}{4a} = 0$; and if $p(\gamma) = 0$ then $(\gamma + \frac{b}{2a})^2 = (\frac{\delta}{2a})^2$, so $\gamma + \frac{b}{2a} = \pm \frac{\delta}{2a}$ by factorising the difference of squares.

0.68 Subtraction $a - b$: translate a by b units to the left (it is the addition of $-b$): put the 0 of the lower line under a and read $a - b$ above the point $-b$ of the lower line. Division a/b ($b \neq 0$): it is the inverse of

multiplication. Align the zeros, then stretch (and reflect if $b < 0$) the lower line until its point b lies beneath the point a of the upper line; then a/b is the point of the upper line above the point 1 of the lower line (because the scaling sends $1 \mapsto a/b$ and $b \mapsto a$).

0.E.1 With $A, \dots, Z \mapsto 0, \dots, 25$; $a, \dots, z \mapsto 26, \dots, 51$; $0, \dots, 9 \mapsto 52, \dots, 61$; $- \mapsto 62$; $_ \mapsto 63$, the digits of dQw4w9WgXcQ are

$$29, 16, 48, 56, 48, 61, 22, 32, 23, 28, 16, \quad \text{and} \quad \sum_k d_k 64^k = 33736714463647725328.$$

Repeated division of 7159047702620056984 by 64 gives the remainders 24, 28, 2, 29, 45, 18, 3, 44, 26, 13, 6; reading them last-first, 6, 13, 26, 44, 3, 18, 45, 29, 2, 28, 24 are the characters G, N, a, C, _, S, M, v, q, 2, Y, so the base-64 expansion is the string GNaC_SMvq2Y.

0.E.2 $(a + b\sqrt{2})(c + d\sqrt{2}) = (ac + 2bd) + (ad + bc)\sqrt{2}$. If $ad + bc = 0$ the product is $ac + 2bd \in \mathbb{Z}$. If $ad + bc \neq 0$ and the product were an integer m , then $\sqrt{2} = \frac{m - (ac + 2bd)}{ad + bc} \in \mathbb{Q}$, contradicting Assumption 0.49. So the product is an integer exactly when $ad + bc = 0$.

0.E.6–0.E.11 $a \odot b = a + b$: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ all closed (Assumptions 0.7, 0.9, Theorem 0.16, Assumption 0.5). $a - b$: $\mathbb{N}$ not closed ($1 - 2 = -1 \notin \mathbb{N}$); $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ closed. $(a - b)(a + b)$: $\mathbb{N}$ not closed ($0 \odot 1 = -1$); $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ closed (products and sums). $(a - 1)(b - 1) + 2(a + b) = ab + a + b + 1$: all four closed (sums and products only). $\frac{a}{b^2 + 1}$: $\mathbb{N}$ and $\mathbb{Z}$ not closed ($1 \odot 1 = \frac{1}{2}$); $\mathbb{Q}$ and $\mathbb{R}$ closed, as $b^2 + 1 \neq 0$. $\frac{a}{\sqrt{b^2 + 1}}$: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ not closed, since $1 \odot 1 = \frac{1}{\sqrt{2}}$ is irrational (if $\frac{1}{\sqrt{2}} = r \in \mathbb{Q}$ then $r \neq 0$ and $\sqrt{2} = \frac{1}{r} \in \mathbb{Q}$); $\mathbb{R}$ closed, as $\sqrt{b^2 + 1}$ is a non-zero real.

0.E.13 Let $x \in \mathbb{Q}$. The polynomial $p(t) = t - x$ is non-zero, has coefficients $1, -x \in \mathbb{Q}$, and $p(x) = 0$. So x is algebraic.

0.E.14 $p(t) = t^2 - 2$ is non-zero over $\mathbb{Q}$ and $p(\sqrt{2}) = 2 - 2 = 0$.

0.E.15 Let $x = \sqrt{2} + \sqrt{3}$. Then $x^2 = 5 + 2\sqrt{6}$, so $(x^2 - 5)^2 = 24$, i.e. $x^4 - 10x^2 + 1 = 0$. So x is a root of the non-zero polynomial $t^4 - 10t^2 + 1$ over $\mathbb{Q}$.

0.E.16 By Exercise 0.66, $x + yi$ is a root of $t^2 - 2xt + (x^2 + y^2)$, whose coefficients $1, -2x, x^2 + y^2$ are rational when $x, y \in \mathbb{Q}$ (Theorem 0.16), and which is non-zero. So $x + yi$ is algebraic.

0.E.19 True: $0 = 0 \cdot n$ for every $n \in \mathbb{Z}$ (Exercise 0.38).

0.E.20 True: $n^2 = n \cdot n$ with $n \in \mathbb{Z}$, so n divides n^2 .

0.E.21 True: if $x \in \mathbb{Q}$ then $x^2 = x \cdot x \in \mathbb{Q}$ by Theorem 0.16.

0.E.23 False: $a = 0, b = 5$: $0 = 0 \cdot 5 + 0$, remainder 0, which is not less than $a = 0$. (Also any negative a : $-3 = (-1) \cdot 5 + 2$ and $2 > -3$.)

0.E.24 False: x^2 has discriminant 0 and its only root is 0 (if $\gamma^2 = 0$ then $\gamma = 0$), so it does not have two distinct complex roots.

0.E.25 Always. Since $n < b_1 < b_2$, condition (ii) allows n itself as a single digit in both bases, and $n = n \cdot b^0$ with $n \neq 0$, so the base- b_1 expansion and the base- b_2 expansion of n are both the one-digit string n .

0.E.26 Never. Suppose the two expansions are the same string $d_r \dots d_0$. Since $n > b_1$, the base- b_1 expansion has at least two digits, so $r \geq 1$ and $d_r \neq 0$. Then $n = \sum_{i \leq r} d_i b_1^i$ and also $n = \sum_{i \leq r} d_i b_2^i$. But $b_1 < b_2$ gives $d_r b_1^r < d_r b_2^r$ and $d_i b_1^i \leq d_i b_2^i$ for the other terms, so $\sum d_i b_1^i < \sum d_i b_2^i$, i.e. $n < n$: contradiction.

0.E.31 Sometimes: $x = 1$ has $x^2 = 1 \in \mathbb{Q}$ and $x \in \mathbb{Q}$; $x = \sqrt{2}$ has $x^2 = 2 \in \mathbb{Q}$ but $x \notin \mathbb{Q}$.

0.E.32 Always. From $x^2 + 1 \in \mathbb{Q}$ we get $x^2 \in \mathbb{Q}$, hence $x^4 = (x^2)^2 \in \mathbb{Q}$ (Theorem 0.16); from $x^5 + 1 \in \mathbb{Q}$ we get $x^5 \in \mathbb{Q}$. If $x = 0$ then $x \in \mathbb{Q}$. If $x \neq 0$ then $x^4 \neq 0$ and $x = \frac{x^5}{x^4} \in \mathbb{Q}$ by Theorem 0.16.

0.E.33 Always. Since $u + vi$ with $v \neq 0$ is not real, $p(x)$ has a non-real root, so $\Delta < 0$ (by Exercise 0.61 and Theorem 0.60, $\Delta \geq 0$ gives only real roots). By Exercise 0.65 the roots are $\frac{-b \pm \sqrt{|\Delta|}i}{2a}$, which are of the form $u' \pm v'i$ with $u' = \frac{-b}{2a}$, $v' = \frac{\sqrt{|\Delta|}}{2a}$. One of them is $u + vi$, so $u = u'$ and $v = \pm v'$, and the other root is $u - vi$.

B CHAPTER 1

1.1.6 p : “John is a mathematician”; q : “John lives in Pittsburgh”; the statement is $p \wedge q$.

1.1.13 Assume $(p \wedge q) \wedge r$. By Strategy 1.1.10 we may assume $p \wedge q$ (and r). By Strategy 1.1.10 again, from $p \wedge q$ we may assume q . Hence q is true.

1.1.14 Proof tree: from p and q conclude $p \wedge q$ ($\wedge I$); from r and s conclude $r \wedge s$ ($\wedge I$); from $p \wedge q$ and $r \wedge s$ conclude $(p \wedge q) \wedge (r \wedge s)$ ($\wedge I$). With p : “2 is even”, q : “2 is prime”, r : “3 is odd”, s : “3 is prime”, a proof has four parts (prove 2 even, prove 2 prime, prove 3 odd, prove 3 prime), combined two at a time.

1.1.23 Since $n = 8$ or $n = 9$, we split into cases. Case $n = 8$: $8 = 2 \cdot 4$, so n is even. Case $n = 9$: $9 = 3^2$, so n is a perfect square. In both cases, n is even or a perfect square.

1.1.26 The discriminant is $\Delta = a^2 - 4b \neq 0$. If $\Delta > 0$, Theorem 0.60 gives $\alpha = \frac{-a + \sqrt{\Delta}}{2}$, $\beta = \frac{-a - \sqrt{\Delta}}{2}$ (in some order), so $\alpha - \beta = \pm \sqrt{\Delta}$; take $c = \pm \sqrt{\Delta} \in \mathbb{R}$. If $\Delta < 0$, Exercise 0.65 gives the roots $\frac{-a \pm \sqrt{|\Delta|}i}{2}$, so $\alpha - \beta = \pm \sqrt{|\Delta|}i = ci$ with $c = \pm \sqrt{|\Delta|} \in \mathbb{R}$. Either way such a c exists.

1.1.31 Suppose α is a root of $p(x)$ and $a \in \mathbb{C}$. Let $q(x) = (x - a)p(x)$. Then $q(\alpha) = (\alpha - a)p(\alpha) = (\alpha - a) \cdot 0 = 0$, so α is a root of $q(x)$.

1.1.64 In the proof of Proposition 1.1.63, the law of excluded middle is used to split into the cases “ a is even” and “ a is odd” (odd means not even). Proof by contradiction is used inside the second case: assuming b odd we derived that ab is odd, which contradicts the hypothesis that ab is even; the contradiction is “ ab is even and ab is not even”. *Version without LEM*: suppose ab is even, and suppose towards a contradiction that it is not the case that a or b is even; then (de Morgan) a is odd and b is odd, so $a = 2k + 1$, $b = 2\ell + 1$ and $ab = 2(2k\ell + k + \ell) + 1$ is odd, contradicting ab even. *Version with LEM and no contradiction*: by LEM, a is even or odd, and b is even or odd. If a is even or b is even we are done. In the remaining case both are odd, and then $ab = 2(2k\ell + k + \ell) + 1$ leaves remainder 1 on division by 2, so by the uniqueness clause of the division theorem ab is not of the form $2m$; this case is therefore excluded by the hypothesis that ab is even. (Any proof about “odd” numbers must use negation somewhere, because odd is defined as not even.)

1.2.8 (a) “ b divides a .” (b) “ x is rational.” (c) “The only natural numbers that divide n are 1 and n ” (true for $n = 1$ and for primes). (d) “There is no largest positive real number” (every positive real is exceeded by a positive real).

1.2.14 Let $x \in \mathbb{R}$. Every object is equal to itself (the basic assumption mentioned in Definition 0.3), so $x = x$. Since x was arbitrary, $\forall x \in \mathbb{R}, x = x$.

1.2.16 Let $n \in \mathbb{Z}$. Then $n = \frac{n}{1}$ with $n, 1 \in \mathbb{Z}$ and $1 \neq 0$, so $n \in \mathbb{Q}$ by Definition 0.15.

1.2.17 Let $p(x) = ax + b$ be a linear polynomial over $\mathbb{Q}$, so $a, b \in \mathbb{Q}$ and $a \neq 0$. Define $\gamma = -\frac{b}{a}$; then $\gamma \in \mathbb{Q}$ by Theorem 0.16, and $p(\gamma) = -b + b = 0$. So $p(x)$ has a rational root.

1.2.27 Define $x = \sqrt{2}$. Then x is irrational (Assumption 0.49) and $x^2 = 2 = \frac{2}{1}$ is rational.

1.2.34 The existence part gives at least one x with $p(x)$; the uniqueness part says any two elements satisfying p are equal, so there cannot be two different ones; together: exactly one. Alternative definitions: Proposition 1.2.38’s form $\exists x \in X, (p(x) \wedge \forall y \in X, (p(y) \Rightarrow y = x))$; or “ $\exists x, p(x)$ and $\neg \exists a, \exists b, (a \neq b \wedge p(a) \wedge p(b))$ ”.

1.2.37 (c) $\exists! n \in \mathbb{N}$, (n has exactly one positive divisor). *Existence*: $n = 1$: its positive divisors are the positive d with $1 = qd$, and then $d = 1$ (if $d \geq 2$ then $qd \neq 1$ for $q \in \mathbb{Z}$); so exactly one. *Uniqueness*: let $n \in \mathbb{N}$ have exactly one positive divisor. By Exercise 0.38, 1 divides n , so that divisor is 1. If $n \geq 2$ then n also divides n and $n \neq 1$: two positive divisors, contradiction. If $n = 0$ then every positive integer divides 0: more than one, contradiction. Hence $n = 1$. So any two such natural numbers are both 1.

1.3.9 $\neg p$: by $(\neg I)/(\neg E)$, $\neg p$ is true exactly when p is false. $p \wedge q$: by $(\wedge I)$ it is true when both are, and by $(\wedge E_1)$, $(\wedge E_2)$ it cannot be true when either is false. $p \vee q$: by $(\vee I)$ it is true when at least one is; if both are false, $p \vee q$ cannot be true, since by $(\vee E)$ we could then derive $\perp$ (both p and q give contradictions). $p \Rightarrow q$: by $(\Rightarrow E)$ it is false when p is true and q false; when q is true, q can be derived from anything, so $(\Rightarrow I)$ makes it true; when p is false, assuming p gives a contradiction and then q by explosion, so $(\Rightarrow I)$ makes it true.

1.3.25 (i) *Derivations*. Assume $\neg(p \vee q)$. To prove $\neg p$: assume p ; then $p \vee q$ ($\vee I_1$), contradicting $\neg(p \vee q)$; so $\neg p$. Likewise $\neg q$. Hence $(\neg p) \wedge (\neg q)$. Conversely assume $(\neg p) \wedge (\neg q)$. To prove $\neg(p \vee q)$, assume $p \vee q$ and split: p contradicts $\neg p$; q contradicts $\neg q$. So $\neg(p \vee q)$. (ii) *Truth table*: $\neg(p \vee q)$ is $\checkmark$ only in row $\times \times$, and so is $(\neg p) \wedge (\neg q)$. (iii) By part (a) applied to $\neg p, \neg q$: $\neg((\neg p) \wedge (\neg q)) \equiv (\neg \neg p) \vee (\neg \neg q) \equiv p \vee q$ (double negation). Negating both sides, $(\neg p) \wedge (\neg q) \equiv \neg \neg((\neg p) \wedge (\neg q)) \equiv \neg(p \vee q)$.

1.3.38 $\neg(p \Leftrightarrow q) \equiv \neg[(p \Rightarrow q) \wedge (q \Rightarrow p)] \equiv \neg(p \Rightarrow q) \vee \neg(q \Rightarrow p) \equiv (p \wedge \neg q) \vee (q \wedge \neg p)$, using Definition 1.1.40, de Morgan (a) and Exercise 1.3.27.

1.3.44 (b), (d) (b) Assume $p \Rightarrow (q \Rightarrow r)$; assume $p \Rightarrow q$; assume p . Modus ponens gives q (from $p \Rightarrow q$) and $q \Rightarrow r$ (from the first assumption), then r . So the formula is derivable from no assumptions, hence a tautology. *Meaning*: an implication can be distributed over an implication. (d) By Theorem 1.3.24(a), $\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$, so by Theorem 1.3.45(b) the biconditional is a tautology. *Meaning*: to show a conjunction fails, show one conjunct fails.

1.E.1 p : n even, q : n prime, r : $n = 2$. (a) “If n is an even prime, then $n = 2$ ”: true for all n (an even prime is divisible by 2, so 2 is one of its only positive divisors 1 and n , and $2 \neq 1$). (b) “If n is a prime other than 2, then n is odd”: true for all n (same reason). (c) “ n is odd, or n is not prime, or $n \neq 2$ ”: true for every n except $n = 2$, for which all three disjuncts fail; so true for some, false for some. (d) “ n is an even prime different from 2”: false for all n .

1.E.2 (a) p : guinea pigs are quiet; q : they are hungry; r : they are loud: $p \wedge (q \Rightarrow r)$. (b) p : 2 is even; q : 2 is prime: $p \wedge q$. (c) p : $\sqrt{2}$ is irrational; q : $\sqrt{2}$ is an integer: $p \wedge (p \Rightarrow \neg q)$ (hence $\neg q$).

1.E.3 “ A is necessary for B ” means $B \Rightarrow A$; “ A is sufficient for B ” means $A \Rightarrow B$. (a) $p \Rightarrow \neg q$: true. (b) $\neg q \Rightarrow p$: false. (c) $\neg q \Rightarrow p$: false. (d) $p \Rightarrow \neg q$: true.

1.E.4 To prove $(p \wedge q) \Rightarrow \neg r$: assume $p \wedge q$, so assume p and q ; then prove $\neg r$ by assuming r and deriving a contradiction.

1.E.5 Assume $p \vee q$; assume r ; prove s by cases: derive s from p (and r), then derive s from q (and r).

1.E.6 Prove both implications. $(\Rightarrow)$ Assume $p \Rightarrow q$; assume $\neg p$; prove $\neg q$ by assuming q and deriving a contradiction. $(\Leftarrow)$ Assume $\neg p \Rightarrow \neg q$; assume p ; prove q . *Warning*: this formula is not a tautology (with p true and q false, $p \Rightarrow q$ is false but $\neg p \Rightarrow \neg q$ is true), so the second step cannot succeed in general: $\neg p \Rightarrow \neg q$ is the *inverse*, equivalent to the converse $q \Rightarrow p$, not to $p \Rightarrow q$.

1.E.7 Prove one of the disjuncts: either prove p and prove $\neg q$ (assume q , derive a contradiction), or prove q and prove $\neg p$. (Not a tautology: false when p, q are both true.)

1.E.8 “Between any two distinct rational numbers there is an irrational number.” *Proof*. Let $a, b \in \mathbb{Q}$ and suppose $a < b$. Define $c = a + \frac{b-a}{\sqrt{2}}$. First, $\frac{b-a}{\sqrt{2}} = \frac{b-a}{2} \cdot \sqrt{2}$ is irrational: if it were rational, then $\sqrt{2} = \frac{b-a}{\sqrt{2}} \cdot \frac{2}{b-a}$ would be rational (Theorem 0.16, $b-a \neq 0$), contradicting Assumption 0.49. Hence c is irrational by Proposition 1.1.53. Second, since $\sqrt{2} > 1$ we have $0 < \frac{1}{\sqrt{2}} < 1$, so $0 < \frac{b-a}{\sqrt{2}} < b-a$ and therefore $a < c < b$. So c is the required number.

1.E.9 $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, y > x$. *Proof.* Let $x \in \mathbb{Z}$. Define $y = x + 1$. Then $y \in \mathbb{Z}$ (closure) and $y > x$.

1.E.10 $\exists a \in X, \exists b \in X, (a \neq b \wedge p(a) \wedge p(b) \wedge \forall x \in X, (p(x) \Rightarrow (x = a \vee x = b)))$. *Structure:* exhibit two distinct elements satisfying p , then show every element satisfying p is one of them. *Proof for $x^2 = 1$.* Take $a = 1, b = -1$: distinct, and $1^2 = (-1)^2 = 1$. Let $x \in \mathbb{R}$ with $x^2 = 1$. Then $(x - 1)(x + 1) = 0$, so $x - 1 = 0$ or $x + 1 = 0$, i.e. $x = 1$ or $x = -1$.

1.E.12 The truth tables of $p \Rightarrow q$ and $q \Rightarrow p$ differ in the rows $(\checkmark, \times)$ and $(\times, \checkmark)$: in the first, $p \Rightarrow q$ is $\times$ and $q \Rightarrow p$ is $\checkmark$. Example: p : “ $1 = 2$ ”, q : “ $1 = 1$ ”; $p \Rightarrow q$ is true (false hypothesis) and $q \Rightarrow p$ is false.

1.E.13 The column is $\checkmark$ exactly when p, q differ: $(p \wedge \neg q) \vee (q \wedge \neg p)$ (equivalently $\neg(p \Leftrightarrow q)$, Exercise 1.3.38).

1.E.14 The column equals the column of q : the formula q (or $(p \vee \neg p) \wedge q$).

1.E.15 The column is $\checkmark$ exactly in the rows where p and q have the same value, regardless of r : $p \Leftrightarrow q$.

1.E.16 The column is $\checkmark$ exactly when r is $\checkmark$ and $(p, q) \neq (\checkmark, \times)$: $r \wedge (p \Rightarrow q)$.

1.E.17 First note $p \uparrow q \equiv \neg(p \wedge q)$: if $\neg(p \wedge q)$ then by de Morgan $\neg p$ or $\neg q$, and rule (i) or (ii) gives $p \uparrow q$; conversely if $p \uparrow q$, assume $p \wedge q$, so p and q , and rule (iii) yields $\perp$; hence $\neg(p \wedge q)$. (a) $p \uparrow p \equiv \neg(p \wedge p) \equiv \neg p$; so $(p \uparrow p) \uparrow (p \uparrow p) \equiv \neg(p \uparrow p) \equiv \neg\neg p \equiv p$. (b) $(p \uparrow p) \uparrow (q \uparrow q) \equiv \neg(\neg p \wedge \neg q) \equiv p \vee q$ (de Morgan); $(p \uparrow q) \uparrow (p \uparrow q) \equiv \neg(p \uparrow q) \equiv \neg\neg(p \wedge q) \equiv p \wedge q$. (c) $p \Rightarrow q \equiv \neg p \vee q \equiv \neg(p \wedge \neg q) \equiv p \uparrow (\neg q) \equiv p \uparrow (q \uparrow q)$.

1.E.19 $\neg\exists!x \in X, p(x) \equiv \neg[(\exists x, p(x)) \wedge (\forall a, \forall b, (p(a) \wedge p(b)) \Rightarrow a = b)] \equiv (\forall x \in X, \neg p(x)) \vee (\exists a \in X, \exists b \in X, (p(a) \wedge p(b) \wedge a \neq b))$. *Strategy:* either show that no x satisfies p , or exhibit two distinct elements that do. *Application.* Let $a \in \mathbb{R}$ with $a \neq -1$ and $q(x) = x^4 - 2ax^2 + a^2 - 1 = (x^2 - a)^2 - 1 = (x^2 - a - 1)(x^2 - a + 1)$. If $a > -1$ then $a + 1 > 0$ and $x = \sqrt{a+1}, x = -\sqrt{a+1}$ are two distinct real roots (distinct because $a + 1 \neq 0$). If $a < -1$ then $a + 1 < 0$ and $a - 1 < 0$, so neither $x^2 = a + 1$ nor $x^2 = a - 1$ has a real solution: no real root at all. In both cases there is not a unique real root.

1.E.20 Define $\forall!x \in X, p(x)$ to mean $\neg\exists!x \in X, \neg p(x)$. Then $\neg\forall!x, p(x) \equiv \neg\neg\exists!x, \neg p(x) \equiv \exists!x, \neg p(x)$, and $\neg\exists!x, p(x) \equiv \neg\exists!x, \neg\neg p(x) \equiv \forall!x, \neg p(x)$, which are de Morgan’s laws with $\forall!, \exists!$. (It says: “it is not the case that exactly one x fails p ”.)

1.E.21 True: Theorem 1.3.19.

1.E.22 False: Exercise 1.E.12 ($p \Rightarrow q \neq q \Rightarrow p$).

1.E.23 True: by de Morgan and double negation, $p \wedge q \equiv \neg(\neg p \vee \neg q)$; replacing every conjunction this way (innermost first) gives an equivalent formula with only disjunctions and negations.

1.E.24 False: $p \wedge q$ is a counterexample. A formula built from variables with disjunctions only is true as soon as one of its variables is true; if it contains p it is true when p is true and q false, if it contains only q it is true when q is true and p false, and if it contains neither it does not depend on p, q at all; in every case it differs from $p \wedge q$ somewhere.

1.E.25 False: for p false and r true, $p \wedge (q \vee r)$ is false while $(p \wedge q) \vee r$ is true.

1.E.26 False: for p false, q true, r false, $p \vee (q \vee r)$ is true while $(p \vee q) \wedge (p \vee r)$ is $\checkmark \wedge \times = \times$.

1.E.27 False: the correct negation is $\exists x \in X, \exists y \in Y, \neg p(x, y)$. Counterexample: $X = Y = \{0\}$ with $p(0, 0)$ true; then $\neg\forall x\forall y p$ is false but $\exists x\forall y p$ is true.

1.E.31 Sometimes: if q is true, $p \Rightarrow q$ is true; if q is false (and p true), it is false.

1.E.32 Always: assume p ; together with $\neg p$ (i.e. p false) this is a contradiction, so q follows by the principle of explosion; hence $p \Rightarrow q$ (“vacuously true”).

1.E.33 Always: assume $\forall x \forall y p(x, y)$; let $y \in Y$ and $x \in X$; then $p(x, y)$ by $(\forall E)$ twice; so $\forall y \forall x p(x, y)$; and symmetrically.

1.E.34 Sometimes. Theorem 1.2.40 gives $\exists \forall \Rightarrow \forall \exists$ always. The converse can hold (e.g. $p(x, y)$: “ $y = y$ ”, both true) and can fail (Exercise 1.2.39: $p(x, y)$: “ $x + y$ even” on $\mathbb{Z}$).

1.E.35 Never. By de Morgan’s laws for quantifiers, $\exists x \exists y \neg p(x, y) \equiv \neg \forall x \forall y p(x, y)$, so the second formula is the negation of the first; a formula and its negation always have opposite truth values, so they are never equivalent.

C CHAPTER 2

2.1.1 If $R \in R$, then by definition of R we have $R \in S$ and $R \notin R$: contradiction. If $R \notin R$, then R is a set, so $R \in S$, and $R \notin R$, so R satisfies the defining property of R , giving $R \in R$: contradiction. Neither truth value is possible: a paradox.

2.1.3 “For all $x \in X$ ” must say nothing about objects outside X , and an implication with false hypothesis is true, so $x \notin X$ never produces a counterexample. “There exists $x \in X$ ” must produce a witness that is in X and satisfies p ; with $\Rightarrow$ instead, any object outside X would be a witness. De Morgan: $\neg \forall x, (x \in X \Rightarrow p(x)) \equiv \exists x, \neg(x \in X \Rightarrow p(x)) \equiv \exists x, (x \in X \wedge \neg p(x))$, which is $\exists x \in X, \neg p(x)$; and $\neg \exists x, (x \in X \wedge p(x)) \equiv \forall x, \neg(x \in X \wedge p(x)) \equiv \forall x, (x \in X \Rightarrow \neg p(x))$ (Exercise 1.3.28), which is $\forall x \in X, \neg p(x)$.

2.1.5 With Definition 2.1.4, $R \in R \Leftrightarrow R \in \mathcal{U} \wedge R \in S \wedge R \notin R$. If $R \in \mathcal{U}$ we would get the paradox of 2.1.1 again, so $R \notin \mathcal{U}$. Then $R \in R$ is simply false (the right-hand side has the false conjunct $R \in \mathcal{U}$), and $R \notin R$ no longer implies $R \in R$. No contradiction remains: R , like $\mathcal{U}$ itself, is a collection that is not an element of the universe.

2.1.13 Two sets should be equal when they have exactly the same elements, regardless of how they are described: this is Axiom 2.1.14.

2.1.28 $\emptyset$ has no elements; $\{\emptyset\}$ has one element, namely $\emptyset$; $\{\{\emptyset\}\}$ has one element, namely $\{\emptyset\}$, which is not $\emptyset$. (a) $\emptyset \in \emptyset$: false. (b) $\emptyset \subseteq \emptyset$: true. (c) $\emptyset \in \{\emptyset\}$: true. (d) $\emptyset \in \{\{\emptyset\}\}$: false (its only element is $\{\emptyset\} \neq \emptyset$). (e) $\{\emptyset\} \subseteq \{\emptyset\}$: true. (f) $\{\emptyset\} \subseteq \{\{\emptyset\}\}$: false ($\emptyset$ is in the left set but not in the right). More: $\{\emptyset\} \in \{\{\emptyset\}\}$ true; $\{\emptyset\} \in \{\emptyset\}$ false; $\emptyset \subseteq \{\{\emptyset\}\}$ true; $\{\{\emptyset\}\} \subseteq \{\emptyset\}$ false.

2.1.33 $\mathcal{P}(\emptyset) = \{\emptyset\}$; $\mathcal{P}(\mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\}$; $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset))) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$.

2.2.7 (a) $A \cap (0, \infty)$. (b) $A \cap \mathbb{Z}$.

2.2.12 $(\Leftarrow)$ Suppose $b \leq c$, and let $x \in (a, b) \cap (c, d)$. Then $x < b \leq c < x$, contradiction; so the intervals are disjoint. The case $d \leq a$ is symmetric. $(\Rightarrow)$ By contraposition: suppose $b > c$ and $d > a$. Then $\max(a, c) < \min(b, d)$ (each of a, c is less than each of b, d), so $x = \frac{\max(a, c) + \min(b, d)}{2}$ satisfies $a < x < b$ and $c < x < d$, and the intervals are not disjoint.

2.2.15 $A \cap [0, \infty) \cap \mathbb{Q}$.

2.2.20 $x \in \bigcap_{n \in \mathbb{Z}} A_n$ iff $x \in \mathbb{R}$ and $x \neq n$ for every integer n , i.e. iff x is a real number that is not an integer. So the intersection is the set of non-integer real numbers ($\mathbb{R} \setminus \mathbb{Z}$ in the notation of 2.2.38).

2.2.31 $x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$ is positive iff both factors are positive or both are negative, i.e. iff $x^2 > 4$ (then also $x^2 > 1$) or $x^2 < 1$ (then also $x^2 < 4$). Hence $X = (-\infty, -2) \cup (-1, 1) \cup (2, \infty)$.

2.2.32 For sets $X, A_0, \dots, A_{n-1}$: $X \cap (A_0 \cup \dots \cup A_{n-1}) = (X \cap A_0) \cup \dots \cup (X \cap A_{n-1})$ and $X \cup (A_0 \cap \dots \cap A_{n-1}) = (X \cup A_0) \cap \dots \cap (X \cup A_{n-1})$. *Proof of the first:* $a \in X \cap (A_0 \cup \dots \cup A_{n-1}) \Leftrightarrow a \in X \wedge \exists i \in [n], a \in A_i \Leftrightarrow \exists i \in [n], (a \in X \wedge a \in A_i) \Leftrightarrow \exists i \in [n], a \in X \cap A_i \Leftrightarrow a \in (X \cap A_0) \cup \dots \cup (X \cap A_{n-1})$. *Second:* $a \in$

$X \cup (A_0 \cap \cdots \cap A_{n-1}) \Leftrightarrow a \in X \vee \forall i \in [n], a \in A_i \Leftrightarrow \forall i \in [n], (a \in X \vee a \in A_i) \Leftrightarrow a \in (X \cup A_0) \cap \cdots \cap (X \cup A_{n-1})$; the middle step is the law $p \vee \forall i q(i) \equiv \forall i (p \vee q(i))$ when p does not involve i (prove it by cases on p).

2.2.42 (a) $A = \mathbb{R} \setminus \mathbb{Z}$. (b) $A = \bigcup_{n \in \mathbb{Z}} (n, n+1)$: a non-integer real x lies in $(\lfloor x \rfloor, \lfloor x \rfloor + 1)$, and every element of some $(n, n+1)$ is a real number strictly between consecutive integers, hence not an integer.

2.2.43 For any a : $a \in X \setminus (X \setminus A) \Leftrightarrow a \in X \wedge \neg(a \in X \wedge a \notin A) \Leftrightarrow a \in X \wedge (a \notin X \vee a \in A)$ {de Morgan}
 $\Leftrightarrow (a \in X \wedge a \notin X) \vee (a \in X \wedge a \in A)$ {distributivity, Example 1.3.3} $\Leftrightarrow a \in X \wedge a \in A$ {the first disjunct is a contradiction} $\Leftrightarrow a \in X \cap A$. *Deduction:* by Proposition 2.2.8 (with A in the role of X and X in the role of Y), $A \subseteq X$ iff $A \cap X = A$, i.e. iff $X \cap A = A$, i.e. iff $X \setminus (X \setminus A) = A$.

Solutions to the stretch exercises of Chapters 3–5

Contents

- A Chapter 3
- B Chapter 4
- C Chapter 5

Planning

Compare your own attempt line by line: did you introduce every variable, unfold every definition, and say which strategy you were using?

A CHAPTER 3

S3.1 $(\subseteq)$ Let $b \in f[f^{-1}[V]]$; fix $a \in f^{-1}[V]$ with $f(a) = b$. Then $b = f(a) \in V$ and $b \in f[X]$. $(\supseteq)$ Let $b \in V \cap f[X]$; fix $a \in X$ with $f(a) = b$. Since $f(a) \in V$, $a \in f^{-1}[V]$, so $b \in f[f^{-1}[V]]$.

S3.2 $(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \text{id}_Y \circ f = \text{id}_X$ and $(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = \text{id}_Z$. So $f^{-1} \circ g^{-1}$ is an inverse of $g \circ f$, and inverses are unique.

S3.3 $(\Rightarrow)$ Let f be injective, $U \subseteq X$, and $b \in f[X \setminus U]$: $b = f(a)$ with $a \notin U$. If $b \in f[U]$ then $b = f(u)$ with $u \in U$, and injectivity gives $a = u \in U$, a contradiction. So $b \in Y \setminus f[U]$. $(\Leftarrow)$ Suppose the inclusion holds for every U . Let $f(a) = f(a')$ and suppose $a \neq a'$. Take $U = \{a'\}$. Then $a \in X \setminus U$, so $f(a) \in f[X \setminus U] \subseteq Y \setminus f[U] = Y \setminus \{f(a')\}$, so $f(a) \neq f(a')$: contradiction. Hence $a = a'$.

S3.4 If n is even, $f(n) = n + 1$ is odd, so $f(f(n)) = (n + 1) - 1 = n$. If n is odd, $f(n) = n - 1$ is even, so $f(f(n)) = (n - 1) + 1 = n$. Thus $f \circ f = \text{id}_{\mathbb{Z}}$, so f is its own two-sided inverse, hence a bijection.

S3.5 $g = g \circ \text{id}_Y = g \circ (f \circ h) = (g \circ f) \circ h = \text{id}_X \circ h = h$.

S3.6 $(\Rightarrow)$ Suppose f surjective and $g \circ f = h \circ f$. Let $b \in Y$; fix a with $f(a) = b$. Then $g(b) = g(f(a)) = h(f(a)) = h(b)$. So $g = h$. $(\Leftarrow)$ Contrapositive. Suppose f is not surjective; fix $b \in Y \setminus f[X]$. Define $g, h : Y \rightarrow \{0, 1\}$ by $g(y) = 0$ for all y , and $h(b) = 1$, $h(y) = 0$ for $y \neq b$. For every $a \in X$, $f(a) \neq b$, so $h(f(a)) = 0 = g(f(a))$; thus $g \circ f = h \circ f$. But $g(b) = 0 \neq 1 = h(b)$, so $g \neq h$. Cancellation fails.

S3.7 Let $f : X \rightarrow \mathcal{P}(X)$ and $D = \{a \in X \mid a \notin f(a)\} \in \mathcal{P}(X)$. Suppose f is surjective; fix $d \in X$ with $f(d) = D$. Then $d \in D \Leftrightarrow d \notin f(d) \Leftrightarrow d \notin D$, which is impossible. So f is not surjective.

S3.8 Let $g(x) = a + (b - a)x$. If $0 < x < 1$ then $a < g(x) < b$ as $b - a > 0$, so $g : (0, 1) \rightarrow (a, b)$ is well-defined. Define $k : (a, b) \rightarrow (0, 1)$ by $k(y) = \frac{y-a}{b-a}$; for $a < y < b$ this lies in $(0, 1)$. Then $k(g(x)) = \frac{(b-a)x}{b-a} = x$ and $g(k(y)) = a + (y - a) = y$. So g is a bijection with inverse k .

B CHAPTER 4

S4.1 Base $n = 0$: empty sum $0 = 1! - 1$. Step: $\sum_{k=1}^{n+1} k \cdot k! = (n+1)! - 1 + (n+1)(n+1)! = (n+1)!(n+2) - 1 = (n+2)! - 1$.

S4.2 Base $n = 0$: $6 \mid 0$. Step: $(n+1)^3 - (n+1) = (n^3 - n) + 3n^2 + 3n = (n^3 - n) + 3n(n+1)$. By the IH $n^3 - n = 6k$; and $n(n+1)$ is even (the fact that $n(n+1)$ is even (one of $n, n+1$ is even)), say $2m$, so $3n(n+1) = 6m$. Hence $(n+1)^3 - (n+1) = 6(k+m)$.

S4.3 For $0 \leq k < n$: $\binom{n}{k} + \binom{n}{k+1} = \frac{n!}{k!(n-k)!} + \frac{n!}{(k+1)!(n-k-1)!} = \frac{n![(k+1)+(n-k)]}{(k+1)!(n-k)!} = \frac{(n+1)!}{(k+1)!(n-k)!} = \binom{n+1}{k+1}$. For $k = n$: $\binom{n}{n} + \binom{n}{n+1} = 1 + 0 = \binom{n+1}{n+1}$; for $k > n$ both sides are 0. Induction on n for $p(n)$: “ $\binom{n}{k} \in \mathbb{N}$ for all $k \in \mathbb{N}$ ”. $p(0)$: $\binom{0}{0} = 1$ and $\binom{0}{k} = 0$ for $k \geq 1$. Step: $\binom{n+1}{0} = 1$, and for $k \geq 0$, $\binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1} \in \mathbb{N}$ by the IH.

S4.4 Let $p(n)$ be the claim for $2^n \times 2^n$ boards. $p(1)$: a 2×2 board minus one square is exactly one L-piece. Step: given a $2^{n+1} \times 2^{n+1}$ board with one square removed, cut it into four $2^n \times 2^n$ quadrants. The removed square lies in one quadrant. Place one L-piece at the centre of the board covering one square from each of the other three quadrants. Now every quadrant is a $2^n \times 2^n$ board with exactly one square missing; tile each by $p(n)$.

S4.5 Let $p(n)$: no $d \geq 2$ divides both f_n and f_{n+1} . $p(0)$: $f_1 = 1$ has no divisor ≥ 2 . Step: suppose $d \geq 2$ divides f_{n+1} and f_{n+2} . Then $d \mid f_{n+2} - f_{n+1} = f_n$, so d divides both f_n and f_{n+1} , contradicting $p(n)$. Hence $p(n+1)$.

S4.6 Base $n = 1$: $1 \geq 1$. Step: by the IH the sum up to $n+1$ is at least $\sqrt{n} + \frac{1}{\sqrt{n+1}}$, so it suffices that $\sqrt{n} + \frac{1}{\sqrt{n+1}} \geq \sqrt{n+1}$. Multiplying by $\sqrt{n+1} > 0$ this is $\sqrt{n(n+1)} + 1 \geq n+1$, i.e. $\sqrt{n(n+1)} \geq n$, which holds since $n(n+1) \geq n^2$ and both sides are non-negative.

S4.7 Strong induction on $n \geq 1$. Since $2^0 \leq n$ and $2^n > n$, there is a largest $k \in \mathbb{N}$ with $2^k \leq n$. Then $n - 2^k < 2^k$ (otherwise $2^{k+1} \leq n$). If $n - 2^k = 0$, then $n = 2^k$. Otherwise $1 \leq n - 2^k < n$, so by the IH $n - 2^k$ is a sum of distinct powers of 2, each at most $n - 2^k < 2^k$; adding 2^k keeps the powers distinct.

S4.8 Base $n = 1$: $f_1 = 1 = f_3 - 1$. Step: $\sum_{k=1}^{n+1} f_k = (f_{n+2} - 1) + f_{n+1} = f_{n+3} - 1$.

S4.9 Base cases $8 = 3 + 5$, $9 = 3 \cdot 3$, $10 = 5 \cdot 2$. Step: let $n \geq 11$ and assume the claim for all $8 \leq k < n$. Then $8 \leq n - 3 < n$, so $n - 3 = 3a + 5b$ and $n = 3(a+1) + 5b$. Three base cases are needed because the step reaches back by 3.

C CHAPTER 5

S5.1 Reflexive: $a = 1 \cdot a$. Transitive: as for linear combinations of multiples: $b = ka$, $c = lb$ give $c = (lk)a$. Not symmetric: $1 \mid 2$ but $2 \nmid 1$. Not antisymmetric: $2 \mid -2$ and $-2 \mid 2$ but $2 \neq -2$.

S5.2 Let $a \in X$; fix b with aRb . By symmetry bRa . By transitivity applied to aRb and bRa , aRa .

S5.3 Reflexive: $x - x = 0 \in \mathbb{Q}$. Symmetric: $y - x = -(x - y) \in \mathbb{Q}$. Transitive: $x - z = (x - y) + (y - z) \in \mathbb{Q}$. $[0] = \{y \mid 0 - y \in \mathbb{Q}\} = \mathbb{Q}$. $[\sqrt{2}] = \{y \mid \sqrt{2} - y \in \mathbb{Q}\} = \{\sqrt{2} - r \mid r \in \mathbb{Q}\} = \{\sqrt{2} + r \mid r \in \mathbb{Q}\}$. If $[\sqrt{2}] = [\sqrt{3}]$ then $r = \sqrt{3} - \sqrt{2} \in \mathbb{Q}$, and $r \neq 0$; then $\sqrt{3} + \sqrt{2} = 1/r \in \mathbb{Q}$, contradicting the Chapter 0 book-level example. So $[\sqrt{2}] \neq [\sqrt{3}]$.

S5.4 Equivalence relations correspond to partitions. Partitions of a 4-element set by block sizes: one block of 4: 1; sizes 3 + 1: 4 (choose the singleton); 2 + 2: 3 (choose the partner of element 1); 2 + 1 + 1: 6 (choose the pair); 1 + 1 + 1 + 1: 1. Total 15.

S5.5 Reflexive: $U \triangle U = \emptyset$ is finite. Symmetric: $U \triangle V = V \triangle U$. Transitive: suppose $U \triangle V$ and $V \triangle W$ are finite. If $a \in U \triangle W$, say $a \in U$, $a \notin W$ (the other case is symmetric): if $a \in V$ then $a \in V \triangle W$; if $a \notin V$ then $a \in U \triangle V$. So $U \triangle W \subseteq (U \triangle V) \cup (V \triangle W)$, a subset of a union of two finite sets, hence finite.

S5.6 $g[X/\sim] = \{g([a]) \mid a \in X\} = \{f(a) \mid a \in X\} = f[X]$. For injectivity: g injective iff for all a, b , $g([a]) = g([b]) \Rightarrow [a] = [b]$, i.e. $f(a) = f(b) \Rightarrow a \sim b$, using $[a] = [b] \Leftrightarrow a \sim b$.

S5.7 $(\Leftarrow)$ Suppose $m \mid n$ and $[a]_n = [a']_n$. Then $n \mid a - a'$, and $m \mid n$, so $m \mid a - a'$ by transitivity of $\mid$; thus $[a]_m = [a']_m$. $(\Rightarrow)$ If the rule is well-defined, then since $[0]_n = [n]_n$ we must have $[0]_m = [n]_m$, i.e. $m \mid n - 0 = n$.

S5.8 $(\Rightarrow)$ An equivalence relation is reflexive; and if aRb and aRc , then bRa by symmetry, and bRa, aRc give bRc by transitivity. $(\Leftarrow)$ Suppose R is reflexive and has the stated property. Symmetric: let aRb . Also aRa . Applying the property to aRb and aRa gives bRa . Transitive: let aRb and bRc . By symmetry bRa . Applying the property to bRa and bRc gives aRc .

Appendix U

One-page proof toolbox

Contents

- A** Reading off the first line of a proof
Standard unfoldings

Planning

Use this page while doing the exercises: find the outermost symbol of what you must prove, copy the first line, unfold the definitions, repeat.

A READING OFF THE FIRST LINE OF A PROOF

goal looks like	first line(s)	last line
$\forall x \in X, p(x)$	Let $x \in X$.	... so $p(x)$. Since x was arbitrary, ...
$\exists x \in X, p(x)$	Put $x = \dots$ (then check $x \in X$).	... so $p(x)$.
$\exists! x$	Existence: as above. Uniqueness: suppose x, y both satisfy p .	... so $x = y$.
$P \Rightarrow Q$	Suppose P . (or: We prove the contrapositive; suppose $\neg Q$.)	... hence Q . (... hence $\neg P$.)
$P \wedge Q$	First we prove P Next we prove Q .	both done.
$P \vee Q$	Suppose $\neg P$.	... hence Q .
$P \Leftrightarrow Q$	$(\Rightarrow)$ Suppose P ... $(\Leftarrow)$ Suppose Q ...	both directions done.
$\neg P$, “there is no”, “irrational”, $X = \emptyset$	Suppose, for a contradiction, that P (that such an object exists; that $a \in X$).	... contradiction. Hence $\neg P$.
$X \subseteq Y$	Let $a \in X$.	... so $a \in Y$.
$X = Y$	$(\subseteq)$ Let $a \in X$... $(\supseteq)$ Let $a \in Y$... or a $\Leftrightarrow$ -chain.	both inclusions done.
f injective	Let $a, b \in X$ with $f(a) = f(b)$.	... so $a = b$.
f surjective	Let $b \in Y$. Put $a = \dots$	... so $f(a) = b$.
$\forall n \geq b, p(n)$	Base case $n = b$: ... Induction step: let $n \geq b$, suppose $p(n)$.	... so $p(n+1)$. By induction ...
R reflexive / symmetric / transitive	Let a . / Let a, b with aRb . / Let a, b, c with aRb, bRc .	aRa / bRa / aRc .
$F([a]) = h(a)$ well-defined	Let $a \sim b$.	... so $h(a) = h(b)$.

Using an assumption: $\exists$ or “ $b \mid a$ ” or “ $b \in f[U]$ ” $\rightarrow$ “Fix k with ...”. $P \vee Q$ or “ $a \in X \cup Y$ ” $\rightarrow$ cases. $P \wedge Q$ or “ $a \in X \cap Y$ ” $\rightarrow$ two facts. $\forall \rightarrow$ plug in what you need. $P \Rightarrow Q$ and $P \rightarrow Q$.

Standard unfoldings

$b \mid a$	$\exists q \in \mathbb{Z}, a = qb$	$a \in \bigcap_{i \in I} X_i$	$\forall i \in I, a \in X_i$
n even / odd	$n = 2k$ / $n = 2k + 1, k \in \mathbb{Z}$	$a \in \bigcup_{i \in I} X_i$	$\exists i \in I, a \in X_i$
$x \in \mathbb{Q}$	$x = \frac{a}{b}, a, b \in \mathbb{Z}, b \neq 0$	$(a, b) \in X \times Y$	$a \in X \wedge b \in Y$
$a \in \{x \in X \mid p(x)\}$	$a \in X \wedge p(a)$	$b \in f[U]$	$\exists a \in U, f(a) = b$
$b \in \{e(x) \mid x \in X\}$	$\exists x \in X, b = e(x)$	$a \in f^{-1}[V]$	$f(a) \in V$
$X \subseteq Y$	$\forall a \in X, a \in Y$	$g \circ f = \text{id}_X$	$\forall a \in X, g(f(a)) = a$
$U \subseteq \mathcal{P}(X)$	$U \subseteq X$	$b \in [a]_{\sim}$	$a \sim b$
$a \in X \cap Y$	$a \in X \wedge a \in Y$	$[a] = [b]$	$a \sim b$ (Lemma)
$a \in X \cup Y$	$a \in X \vee a \in Y$	$a \equiv b \pmod{n}$	$n \mid a - b$
$a \in X \setminus Y$	$a \in X \wedge a \notin Y$		